

Skeletal & Muscular Systems

Fikre Bayu

Skeletal system

- Provides our basic shape and support for the body and is what the muscular system acts on to produce movement
- Components of the skeletal system
 - ✓ Bones (skeleton)
 - ✓ Cartilages
 - ✓ Joints
 - ✓ Ligaments (connect bone to bone)
 - ✓ Tendons (connect bone to muscle)

Bones

- Bone, a living tissue, is a highly specialized, hard form of connective tissue
- Makes up most of the skeleton
- It is the chief supporting tissue of the body

Bones of the adult skeleton provide

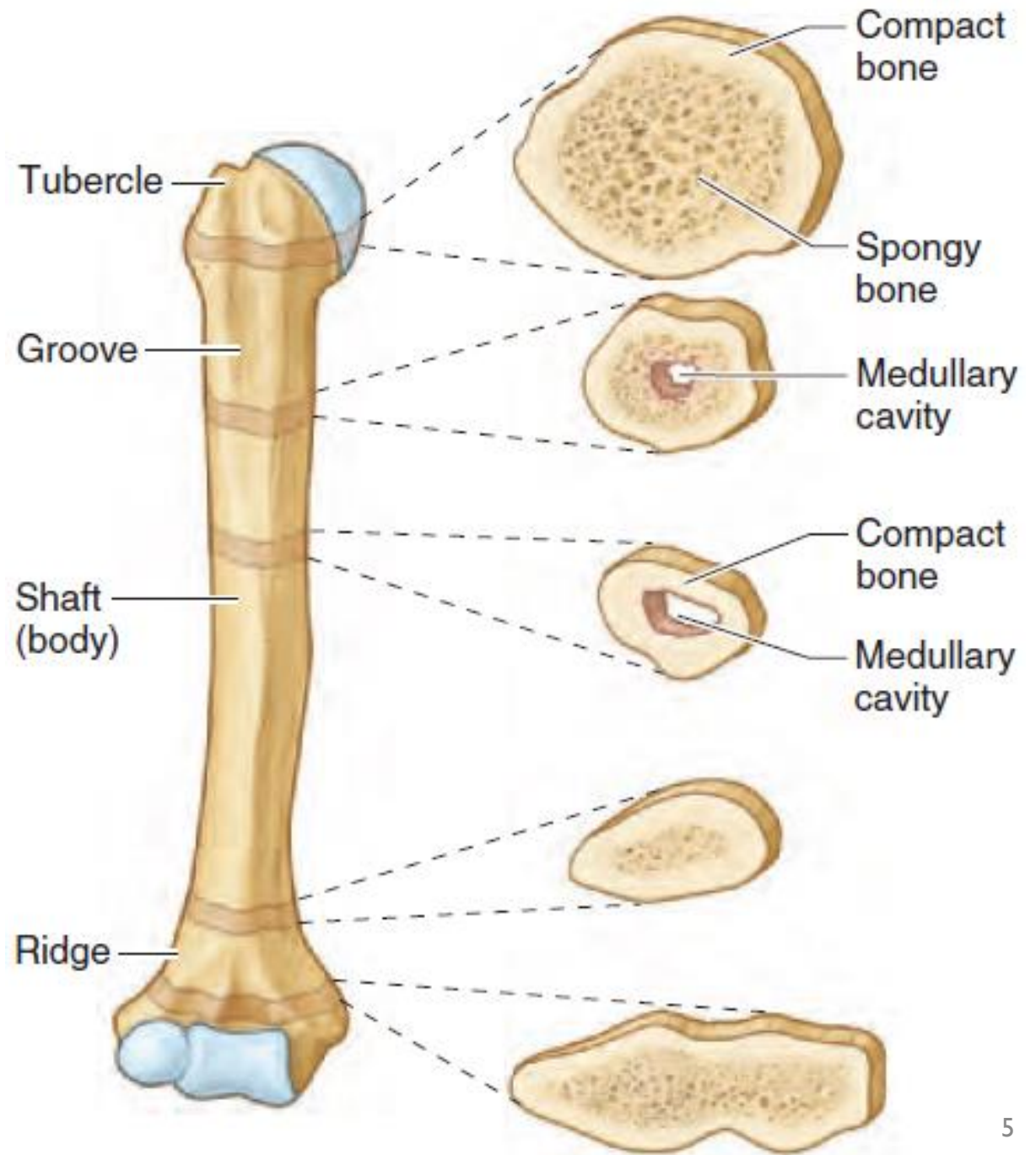
- ✓ Support for the body and its vital cavities
- ✓ Protection for vital structures
- ✓ The mechanical basis for movement (leverage)
- ✓ Storage for salts (e.g., calcium)
- ✓ A continuous supply of new blood cells (produced by the marrow contained within many bones)

Bones

- The two types of bone are
 - ✓ **Compact bone** and
 - ✓ **Spongy (trabecular or cancellous) bone**
- They are distinguished by
 - ✓ the relative amount of solid matter and
 - ✓ the number and size of the spaces they contain
- All bones have
 - ✓ a superficial thin layer of compact bone
 - ✓ a central mass of spongy bone, except the spongy bone is replaced by a *medullary (marrow) cavity*.
 - Their architecture & proportion are vary according to function
- Within the medullary cavity of adult bones, and between the spicules (trabeculae) of spongy bone,
 - ✓ *Yellow or red bone marrow* or a combination of both is found.

Fig. Transverse sections of the humerus.

The shaft of a living bone is a tube of compact bone that surrounds a medullary cavity.



Structure of Long Bone

- **Diaphysis**

- ✓ Shaft
- ✓ Composed of compact bone

- **Epiphysis**

- ✓ Ends of the bone
- ✓ Composed mostly of spongy bone

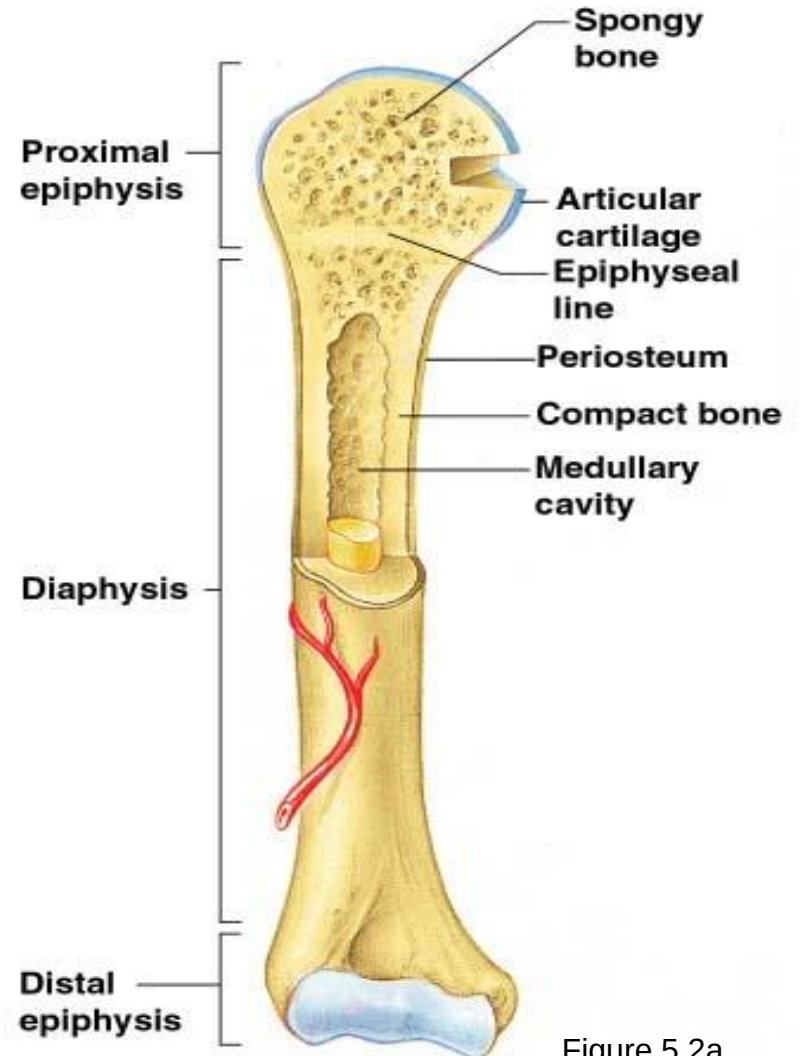


Figure 5.2a

Structures of a Long Bone

■ Periosteum

- ✓ Outside covering of the diaphysis
- ✓ Fibrous connective tissue membrane

■ Sharpey's fibers

- ✓ Secure periosteum to underlying bone

■ Arteries

- ✓ Supply bone cells with nutrients

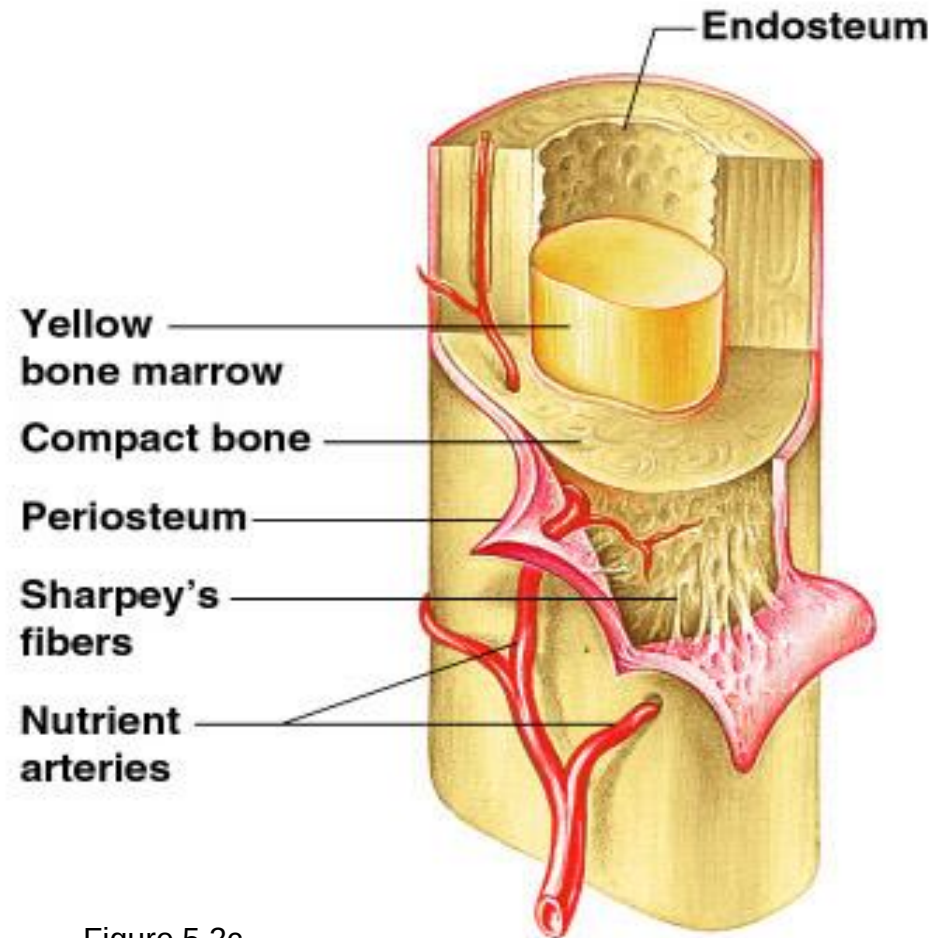
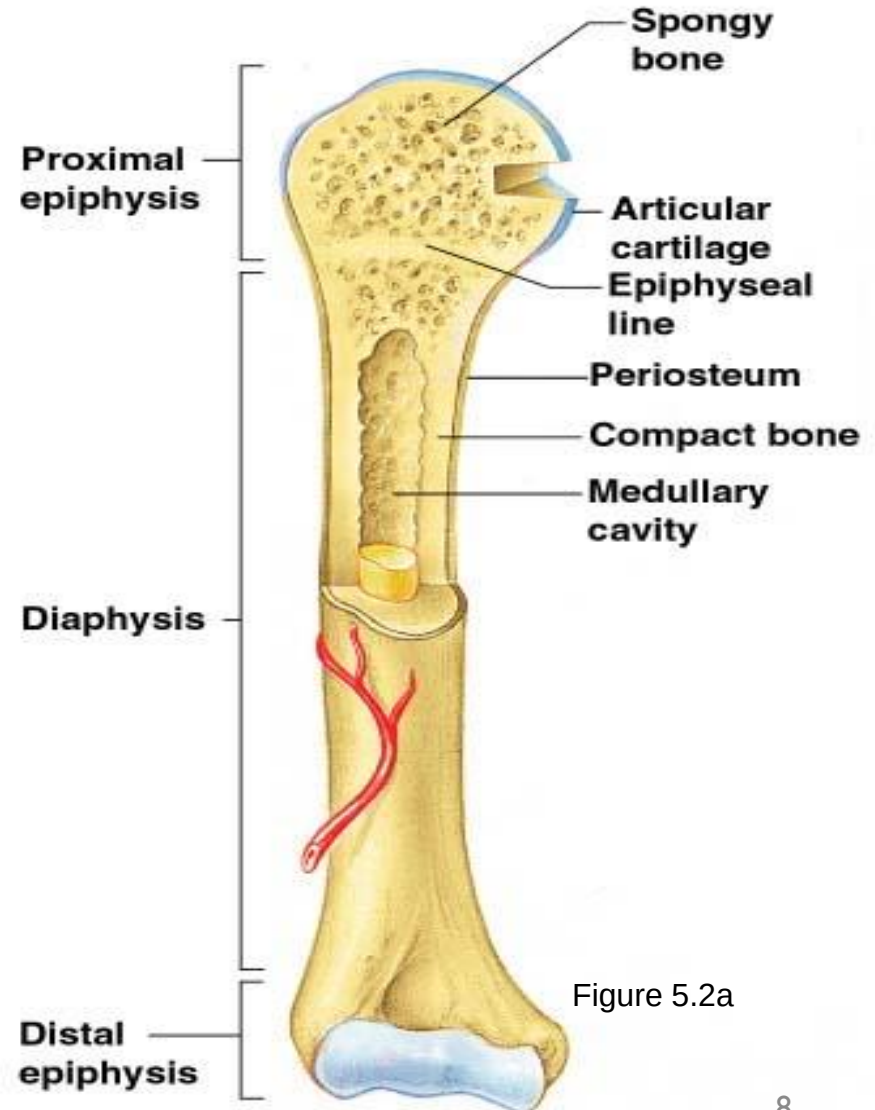


Figure 5.2c

Structures of a Long Bone.....

■ Articular cartilage

- ✓ Covers the external surface of the epiphyses
- ✓ Made of hyaline cartilage
- ✓ Decreases friction at joint surfaces



Structures of a Long Bone.....

- **Medullary cavity**
 - ✓ Cavity of the shaft
 - ✓ Contains yellow marrow (mostly fat) in adults
 - ✓ Contains red marrow (for blood cell formation) in infants

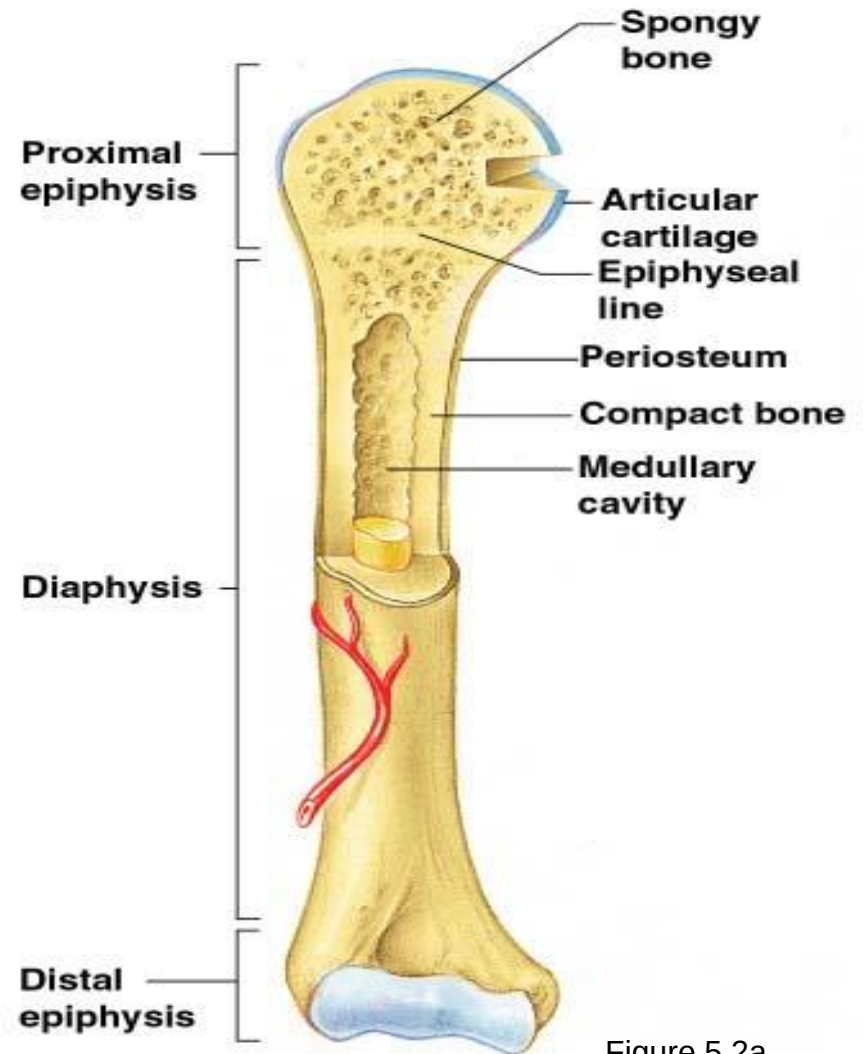


Figure 5.2a

Microscopic Anatomy of Bone

- Osteon (Haversian System)
 - ✓ A unit of bone
- Central (Haversian) canal
 - ✓ Opening in the center of an osteon
 - ✓ Carries blood vessels and nerves
- Perforating (Volkman's) canal
 - ✓ Canal perpendicular to the central canal
 - ✓ Carries blood vessels and nerves

Microscopic Anatomy of Bone...

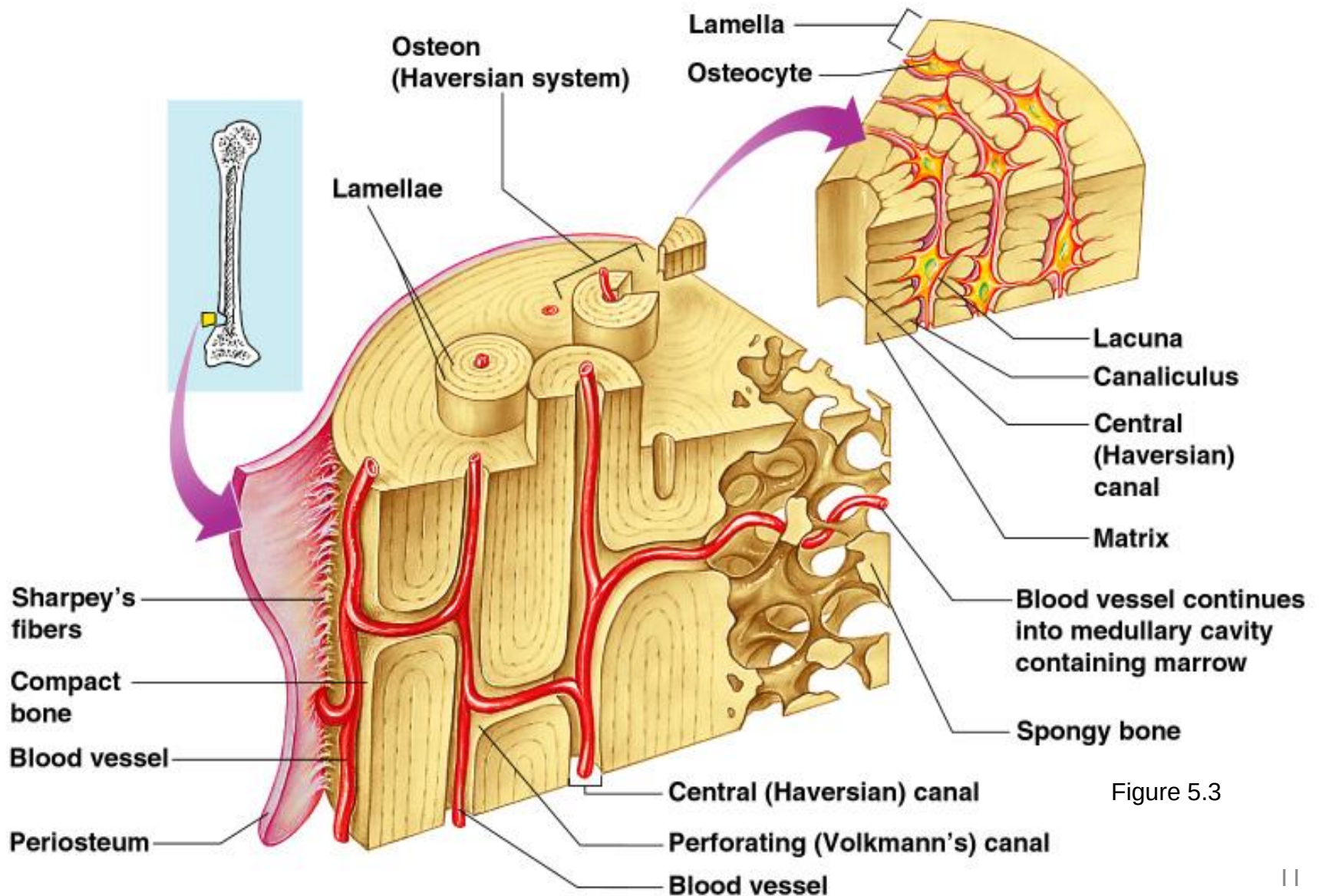
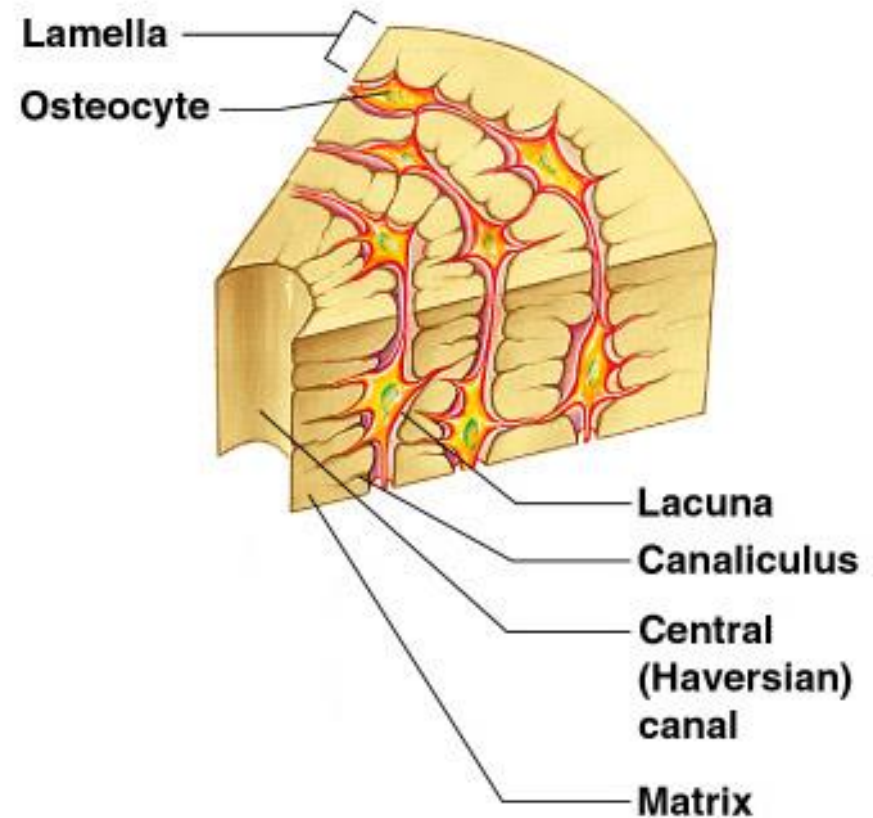


Figure 5.3

Microscopic Anatomy of Bone...

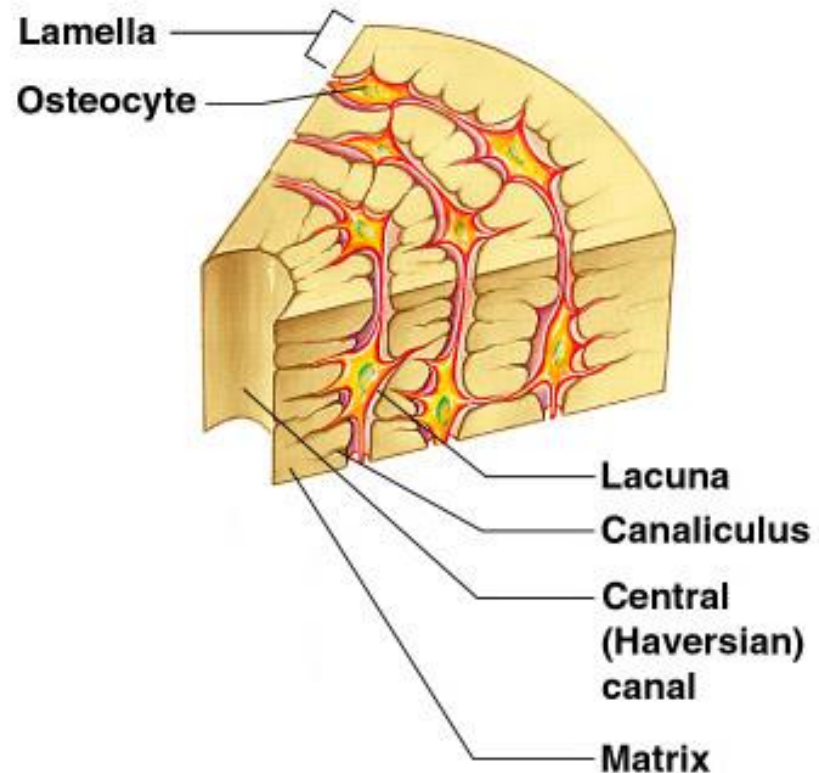
- Lacunae
 - ✓ Cavities containing bone cells (osteocytes)
 - ✓ Arranged in concentric rings
- Lamellae
 - ✓ Rings around the central canal
 - ✓ Sites of lacunae



Microscopic Anatomy of Bone...

■ Canaliculi

- ✓ Tiny canals
- ✓ Radiate from the central canal to lacunae
- ✓ Form a transport system



Cartilages

- **Cartilage** is a resilient, semirigid form of connective tissue that forms parts of the skeleton where more flexibility is required.
- The proportion of bone and cartilage in the skeleton changes as the body grows;
 - ✓ the younger a person is, the more cartilage he or she has
- The bones of a newborn are **soft and flexible** because they are mostly composed of cartilage

Cartilages

- Consists of cells (chondrocytes) in lacunae
- Matrix is gel-like
 - Contains collagenic and elastic fibers
- Has **no blood vessel**, its cells obtain oxygen and nutrients by diffusion.

I. Hyaline Cartilage

- Firm and somewhat flexible
- Matrix contains many **collagenic fibers**
- Found at end of long bones (**articular**), in nose, at **ends of ribs**

Cartilages.....

2. Fibrocartilage

- Matrix contains wide rows of **thick collagenous fibers**
- Able to withstand **tension and pressure**
- Found in **menisci in knee** and **discs** b/n vertebrae

3. Elastic Cartilage

- Matrix contains elastic fibers
- Very flexible
- Found in **epiglottis** and **ear** flaps

Bone Development

- In embryos, the skeleton is primarily **hyaline cartilage**
- During development, much of this cartilage is replaced by bone (the process is called ossification)
- Cartilage remains in isolated areas
 - ✓ Bridge of the nose
 - ✓ Parts of ribs
 - ✓ Joints

Bone Development...

- All bones derive from mesenchyme by two processes
 - I. **Intramembranous ossification:-** mesenchymal model of bones form during embryonic period & direct ossification of mesenchyme begins in the fetal period.
 - II. **Endochondral ossification:-** cartilage models of the bones form from mesenchyme during fetal period & bone subsequently replaces most of the cartilage.

Classification of the Bones

I. According to their Location

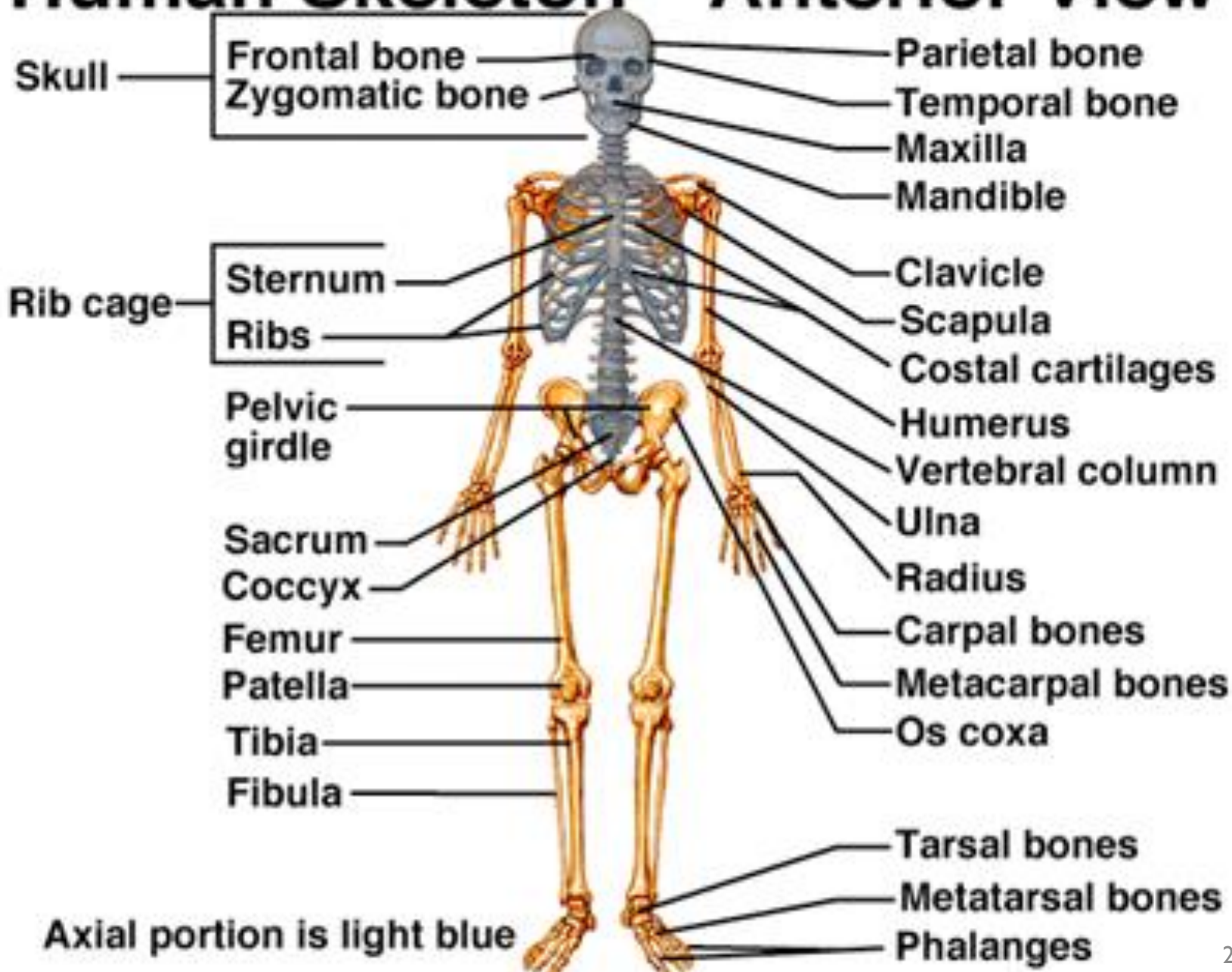
Axial Skeleton

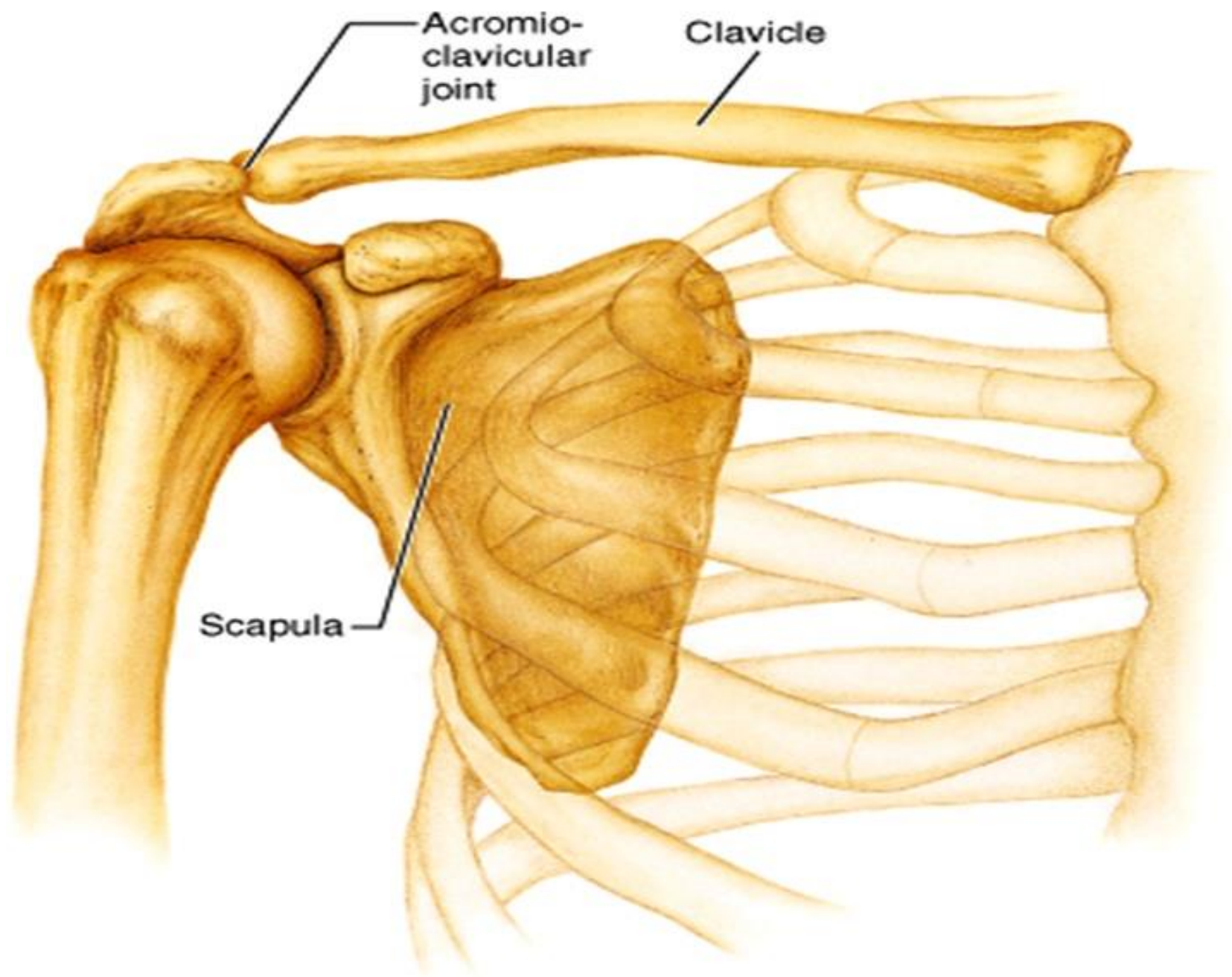
- Long axis of the body
- **80** bones in three major regions
 - Skull
 - Vertebral column
 - Bony thorax
 - Ribs
 - Sternum

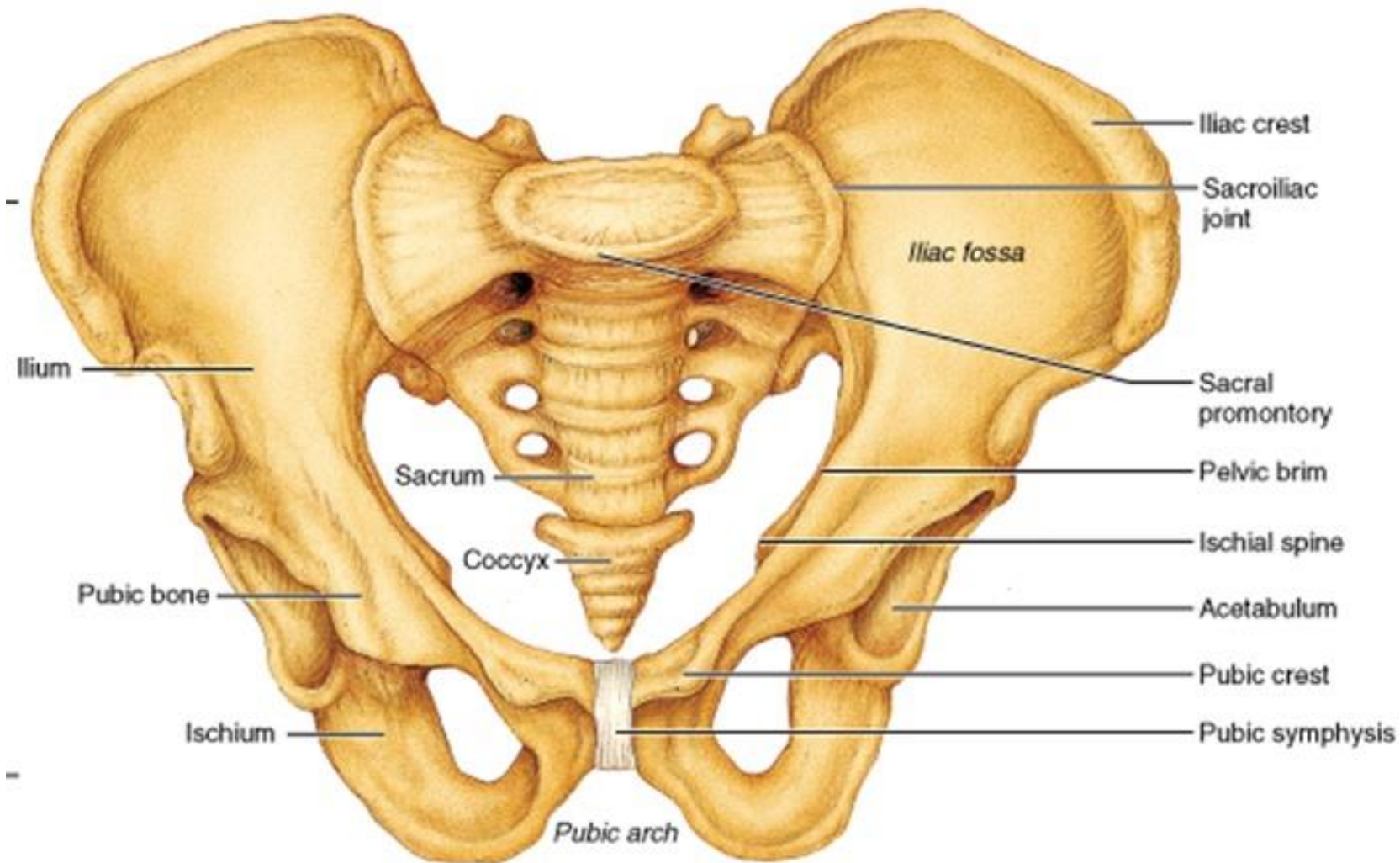
Appendicular Skeleton

- Bones of upper & lower extremities and girdles
- **126** bones in three major regions
 - Girdles
 - Shoulder girdle
 - Pelvic girdle
 - upper extremity
 - lower extremity

Human Skeleton – Anterior View

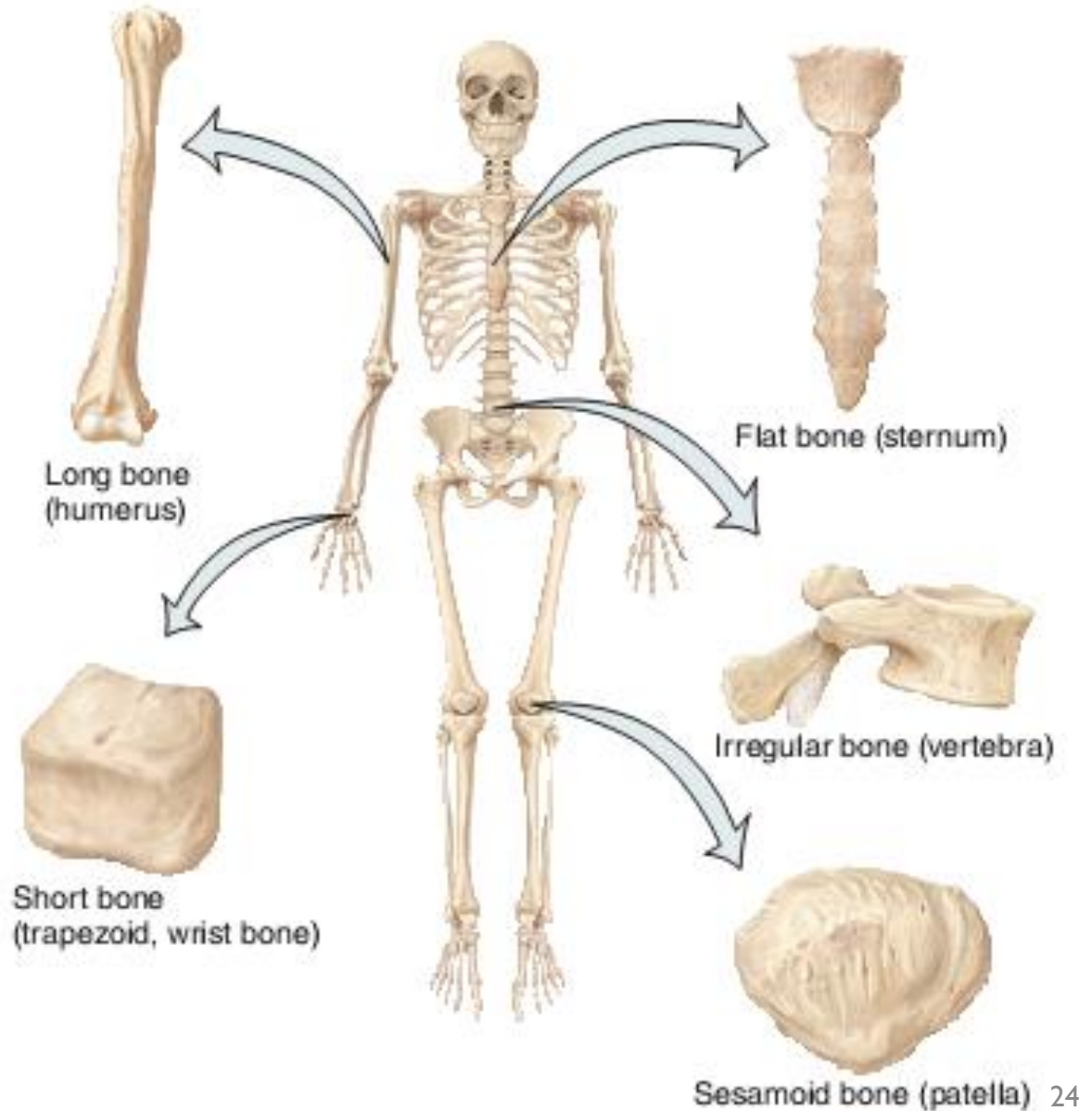






II. According to their Shape

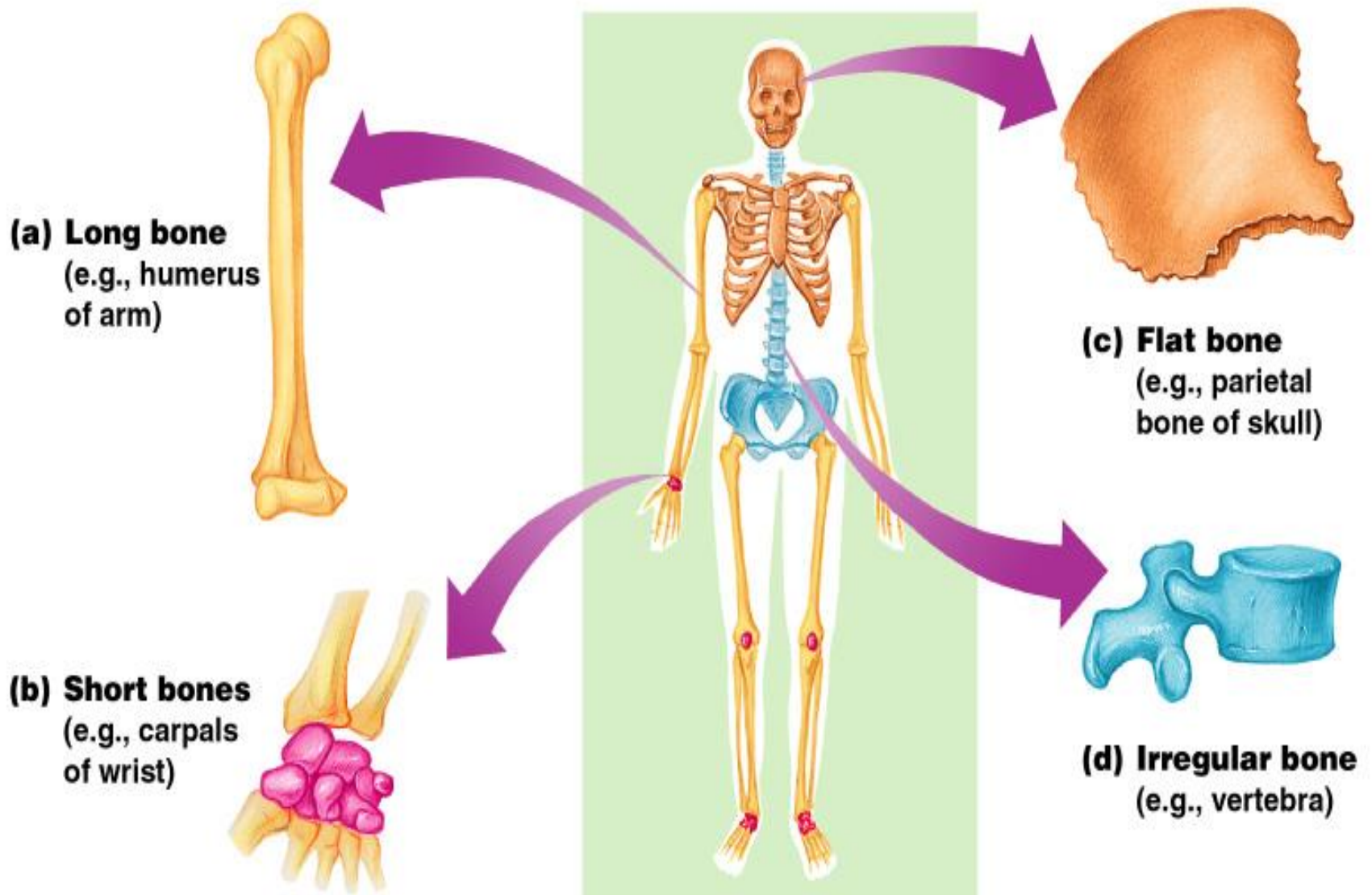
- Long
- Short
- Flat
- Irregular
- Sesamoid



II. According to their Shape

- **Long bones** are tubular (e.g., the humerus in the arm).
- **Short bones** are cuboidal and are found only in the ankle (tarsus) and wrist (carpus)
- **Flat bones** usually serve protective functions (e.g., those forming the cranium protect the brain).

Classification of Bones on the Basis of Shape



II. According to their Shape

- **Irregular bones** (e.g., in the face, vertebrae) have complex shapes and cannot be grouped into any of the previous categories
- **Sesamoid bones**
 - ✓ Bones develop in certain tendons where there is considerable friction, tension, and physical stress, such as the palms and soles.
 - ✓ They may vary in number from person to person, are not always completely ossified.
 - ✓ Protect tendons from excessive wear and tear.
 - Eg:- The patellae, the largest sesamoid bones

III. Classification of bone according to mode of development

Membranous bones.

- These bones develop through intramembranous ossification.
- Intra membranous ossification is a process of conversion of **embryonic mesenchyme** directly into bone.
 - Eg:- **Bones of the vault of the skull, Bones of the face, Clavicle**

Cartilaginous bones.

- These develop by the process of intracartilaginous ossification.
- Here a cartilage model of future bone is formed first which changes into bone.

Eg:- **Humerus, radius etc...**

Membrano-cartilaginous bones.

- These develop through both of the above processes. These include bones at the junction of vault and base of skull.
 - Eg:- **Occipital bone**

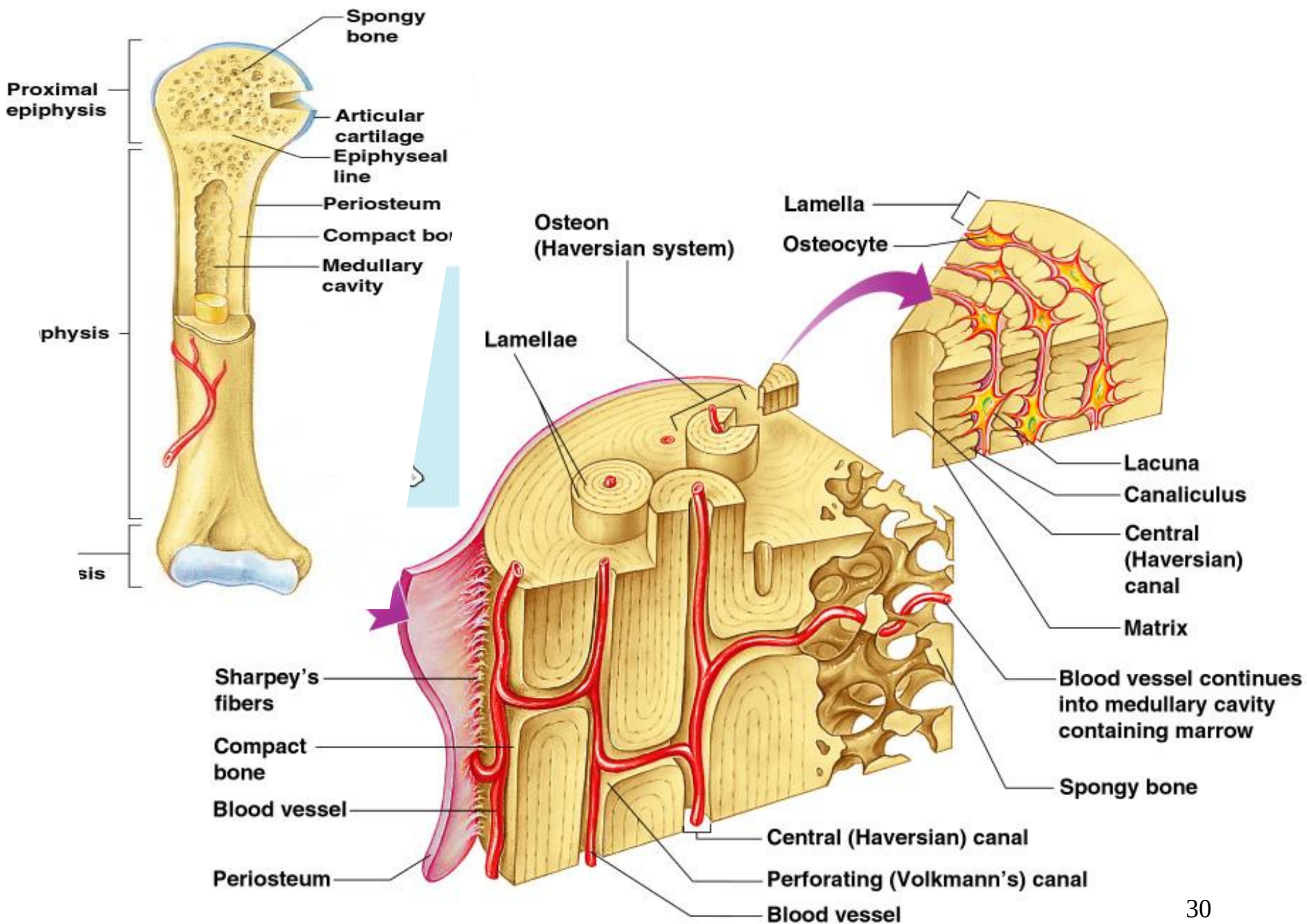
IV. Classification of bone According to structure

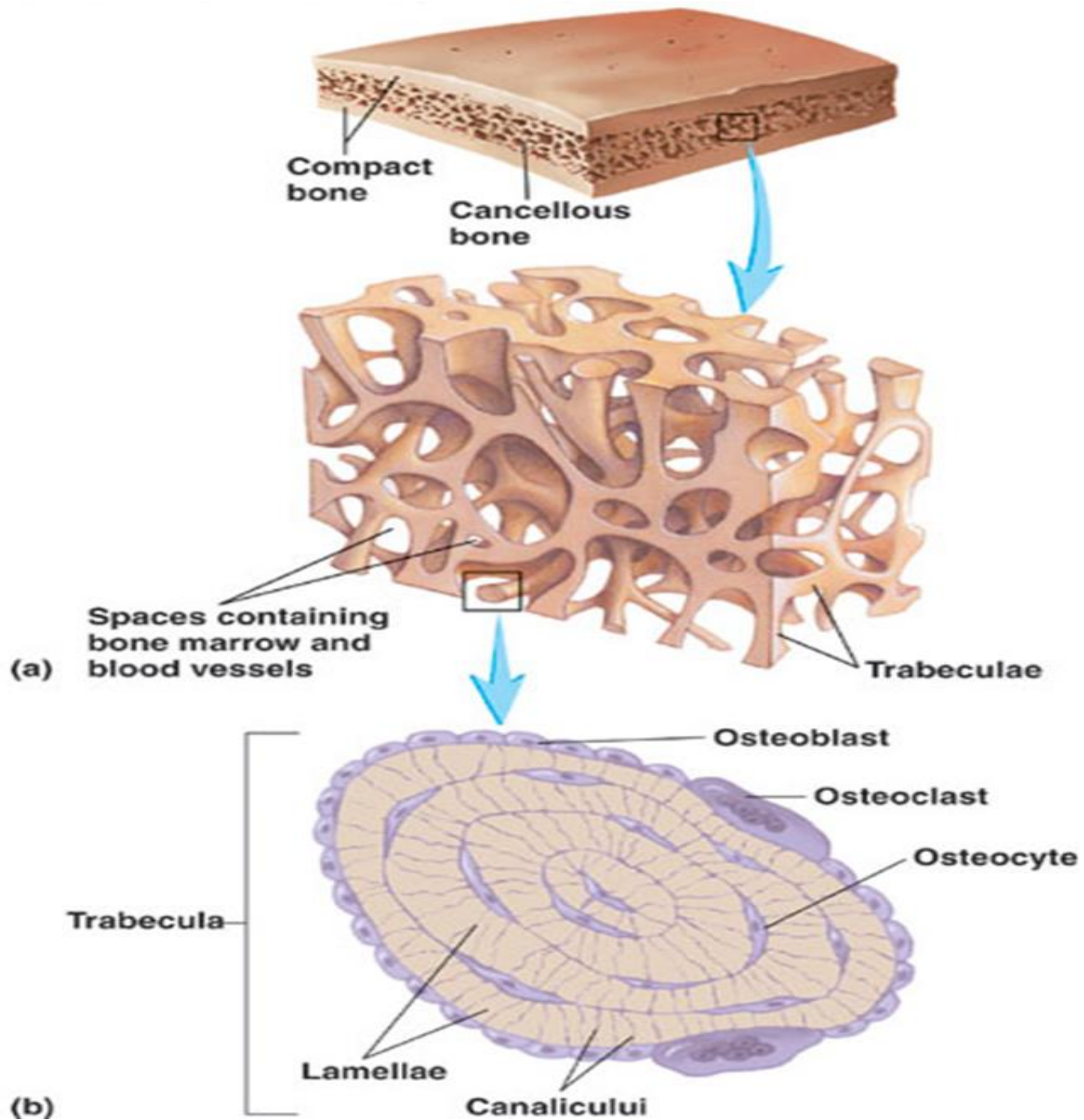
Compact bone

- Compact bone tissue is compactly arranged & contains few spaces and is the strongest form of bone tissue.
- Compact bone tissue provides protection and support and resists the stresses produced by weight and movement.

Cancellous bone or spongy bone

- It is spongy in nature
- It is seen internal to compact bones and at the ends of long bones.
- In contrast to compact bone tissue, spongy bone tissue does not contain osteons but it contains lamellae of bone called trabeculae.



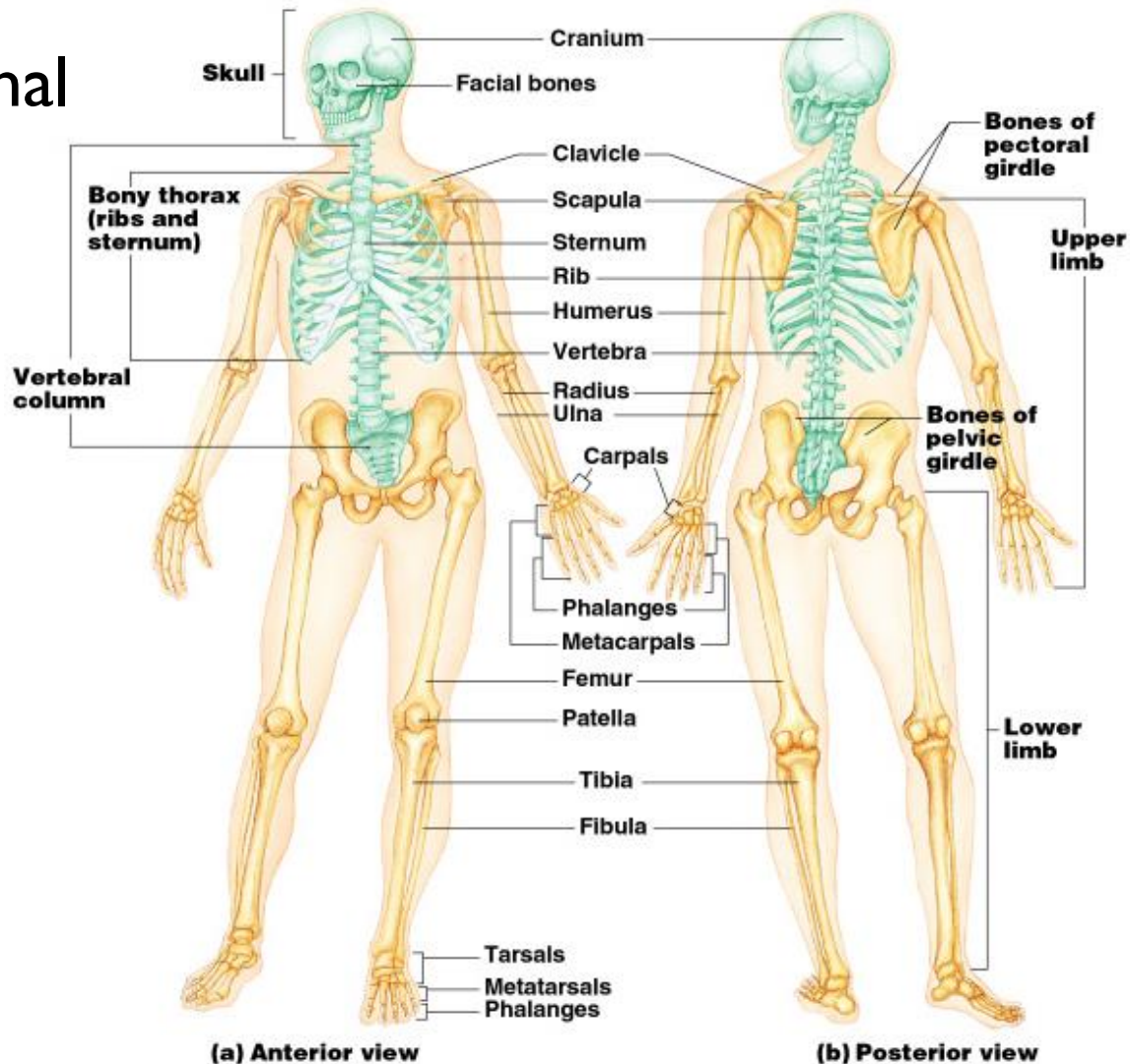


Bone Markings and Formations

- Bone markings appear wherever
 - ✓ tendons, ligaments, and fascia are attached.
 - ✓ arteries, nerves lie adjacent to or enter bones.
 - ✓ in relation to the passage of a tendon.
 - ✓ to control the type of movement occurring at a joint.
- **Reading Assignment**
 - ✓ Bone markings and features

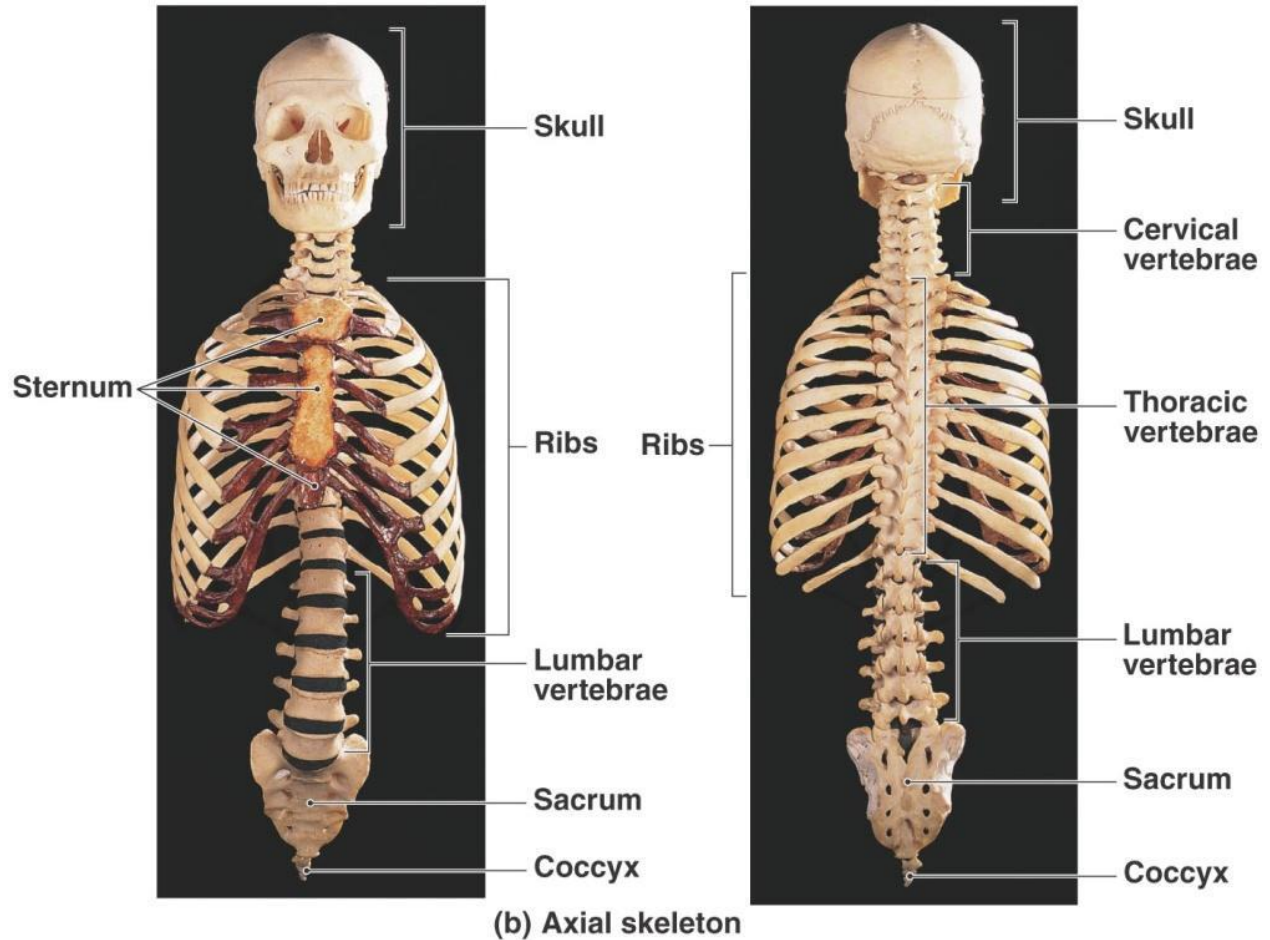
The Axial Skeleton

- Forms the longitudinal part of the body
- Divided into three parts
 - ✓ Skull
 - ✓ Vertebral column
 - ✓ Bony thorax



Axial Skeleton

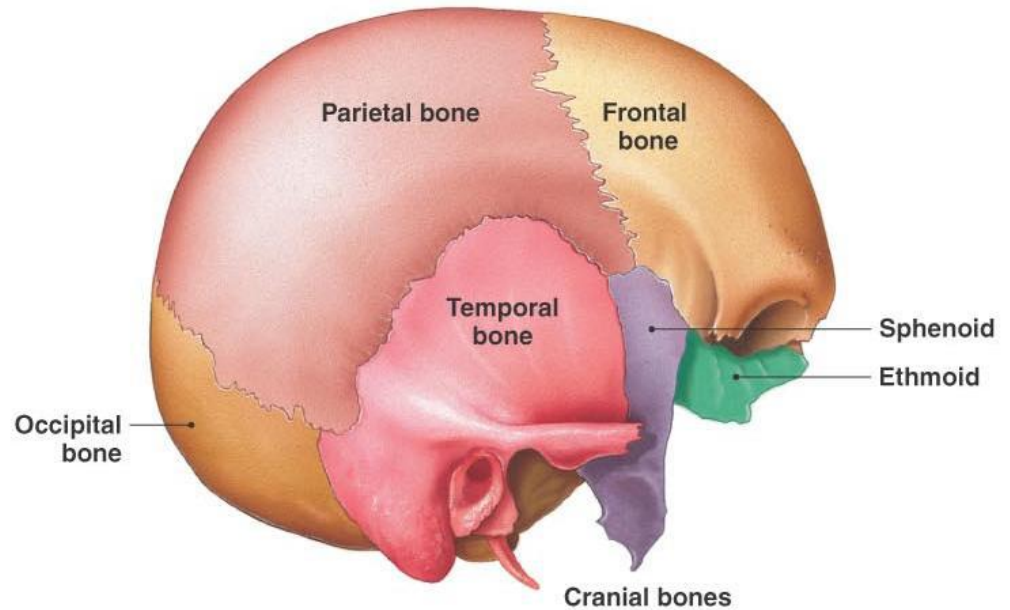
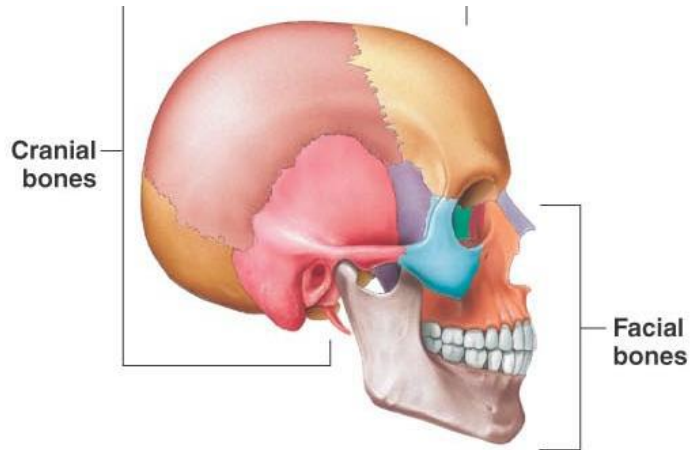
Skull	22
Auditory ossicles	6
Hyoid	1
Vertebrae	26
Ribs	24
<u>Sternum</u>	<u>1</u>
Total = 80	



The Skull

- Two sets of bones
 - ✓ Cranium (8)
 - ✓ Facial bones (14)
- Bones are joined by sutures

The Skull



SKULL

ASSOCIATED BONES 7

Auditory ossicles enclosed in temporal bones (detailed in Chapter 18) 6

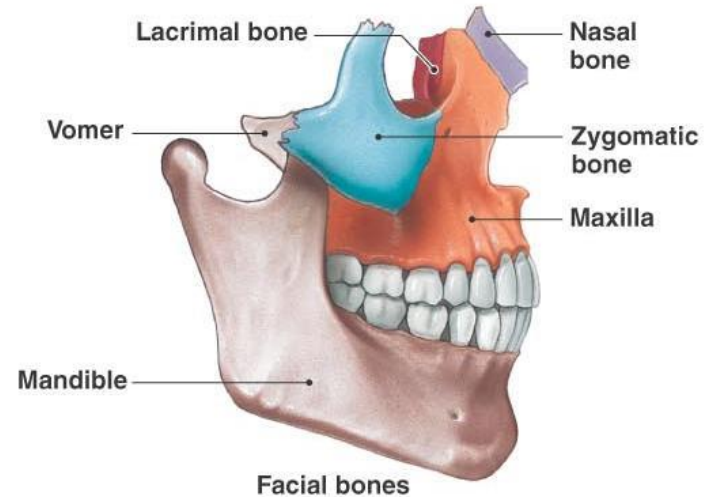
Hyoid bone 1

CRANIUM 8

Occipital bone	1
Parietal bones	2
Frontal bone	1
Temporal bones	2
Sphenoid	1
Ethmoid	1

FACE 14

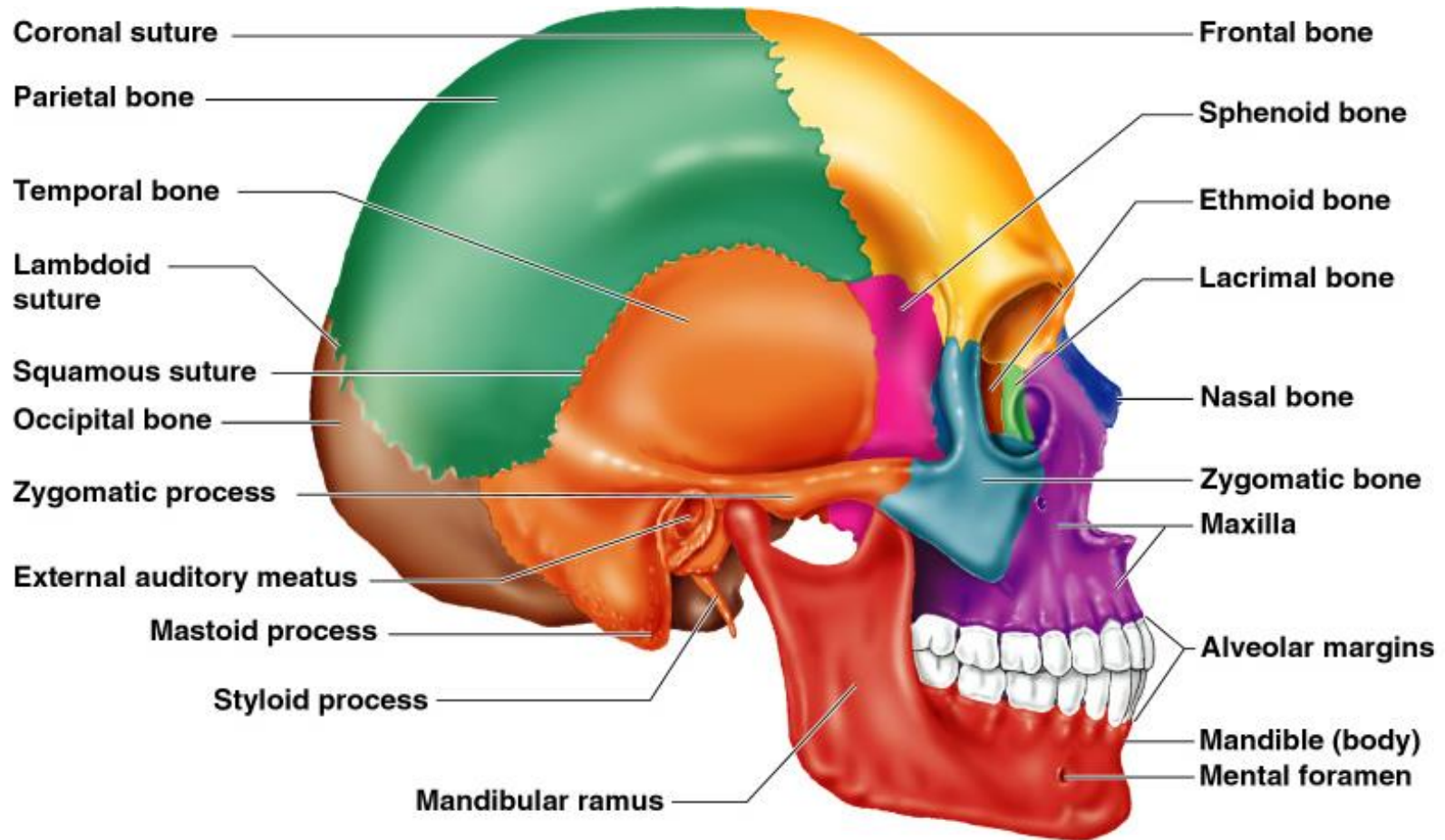
Maxillae	2
Palatine bones	2
Nasal bones	2
Inferior nasal conchae	2
Zygomatic bones	2
Lacrimal bones	2
Vomer	1
Mandible	1



Cranial Bones

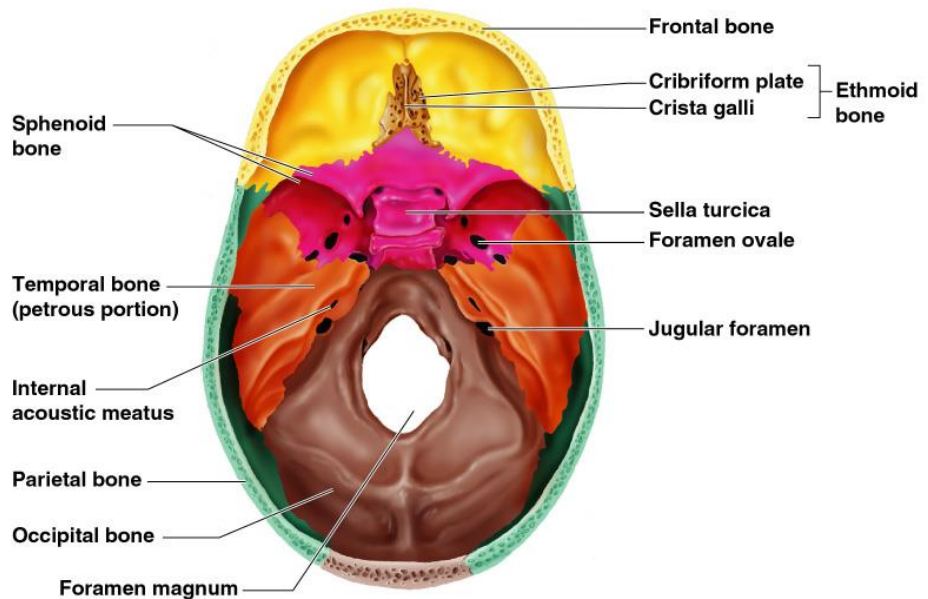
- Protects brain
- Composed of 8 flat bones (fibrous joints in adults; fontanelles in new born)
 - ✓ Frontal bone (1):- forms anterior roof of cranium, nasal cavity & orbit
 - ✓ Parietal bone(2):- separated from frontal bone by coronal suture
 - ✓ Occipital bone (1):- forms the back & much of the base of the skull
 - ✓ Temporal (2):- forms lower sides of the cranium & separated from parietal bone by squamous suture

Cranial Bones



Cranial Bones...

- **Sphenoid bone (I):-**
keystone bone (all other bones articulate with it)
 - ✓ shaped like bat with outstretched wings
 - ✓ completes sides of skull and contributes to orbits
- **Ethmoid bone (I):-**
lies in front of sphenoid
 - ✓ forms orbits and nasal septum



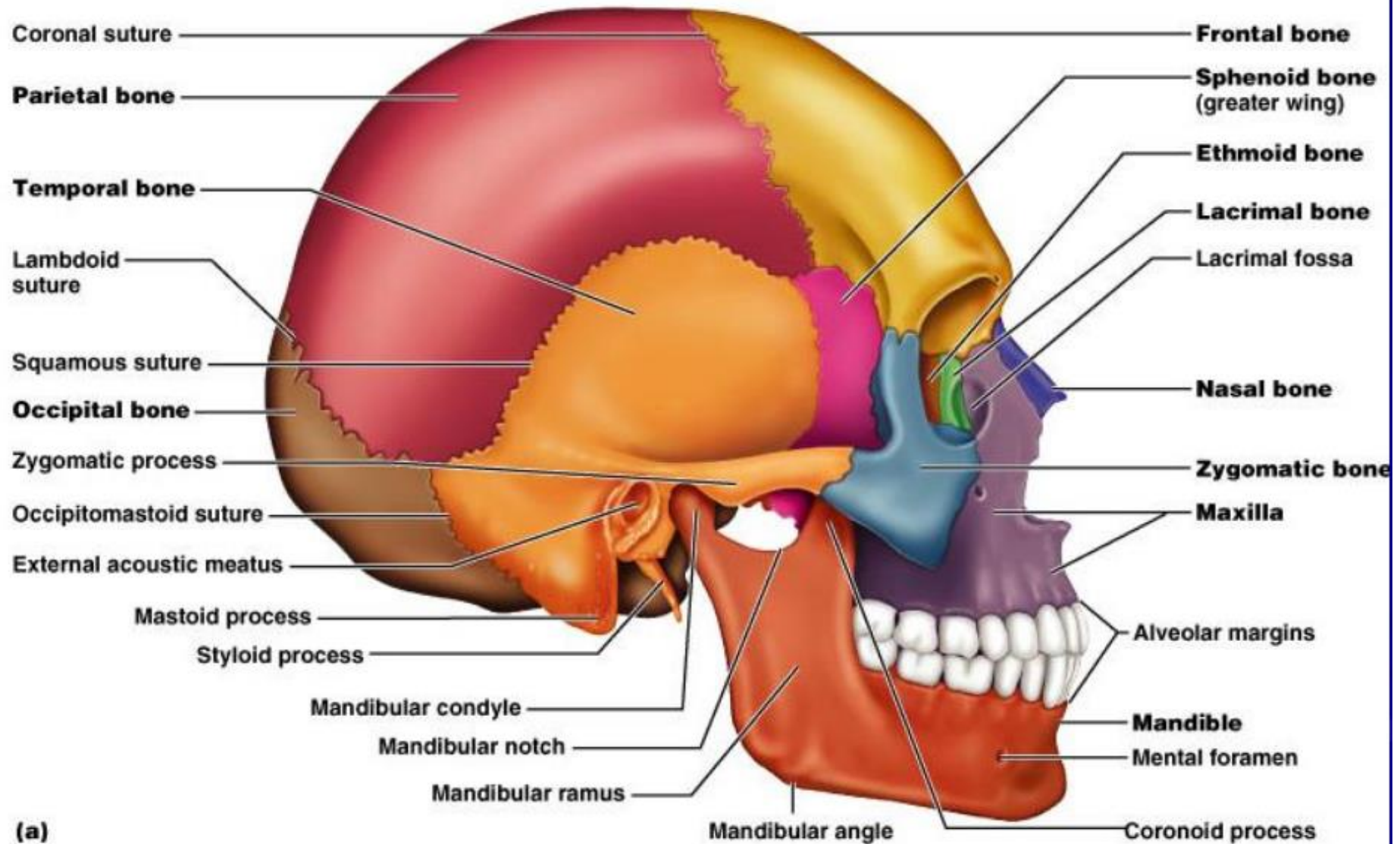
Facial Bones

- Provide the basic shape of the face
- All are paired except vomer & mandible
- Consist of about 14 bones

Mandible (1):- forms the lower jaw bone

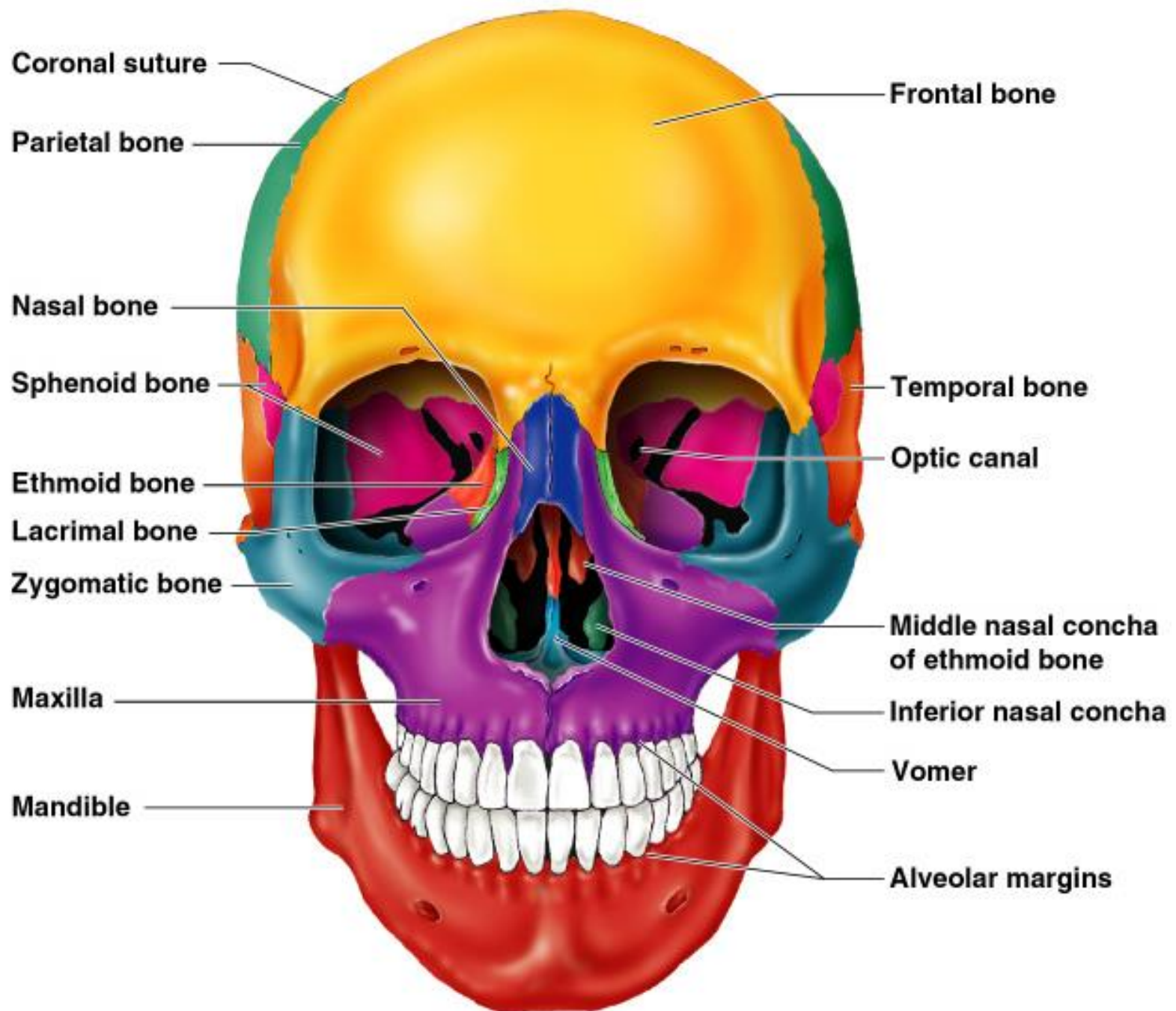
- only movable bone of skull
- attached to the cranium by TMJ
- has 3 parts: body, ramus, and condyloid & coronoid processes

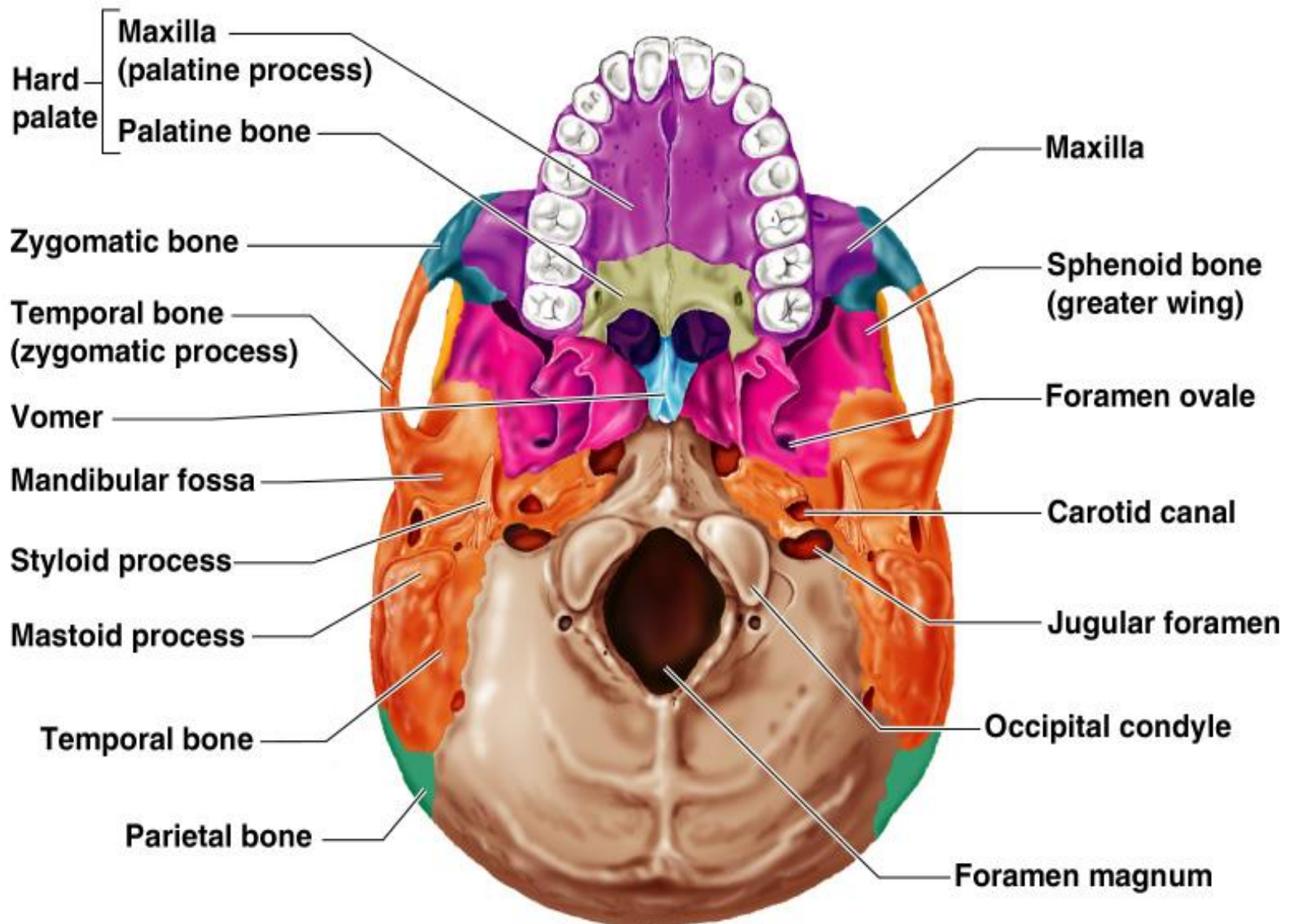
Facial Bones



Facial Bones...

- **Maxilla** (2):- holds upper teeth, forms hard palate
- **Zygomatic bones** (2):- forms cheek bone of face
- **Nasal bone** (2):- helps to support the flexible cartilaginous plates
- **Nasal concha** (2):- superior, middle & inferior
- **Vomer** (1):- forms the lower part of nasal septum
- **Lacrimal bone** (2):- forms medial wall of orbit
- **Palatine bone** (2):-forms posterior part of hard palate, part of orbit & nasal cavity.



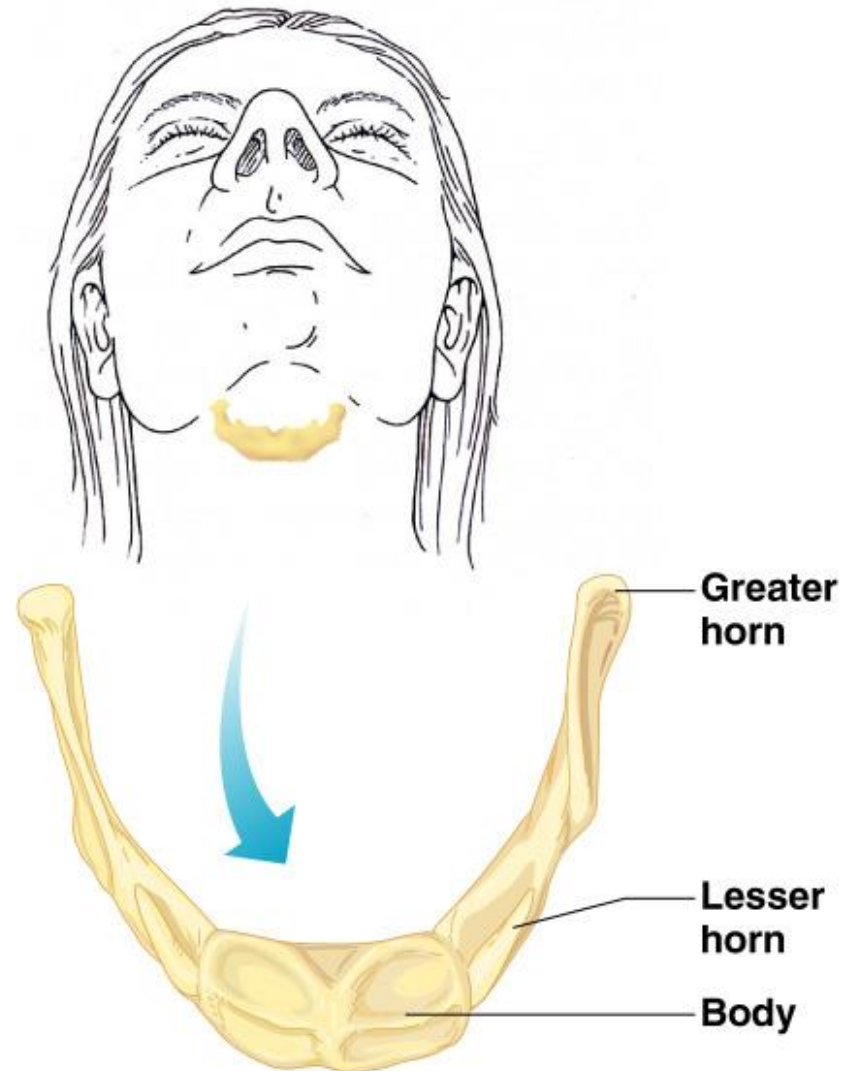


Auditory Ossicles

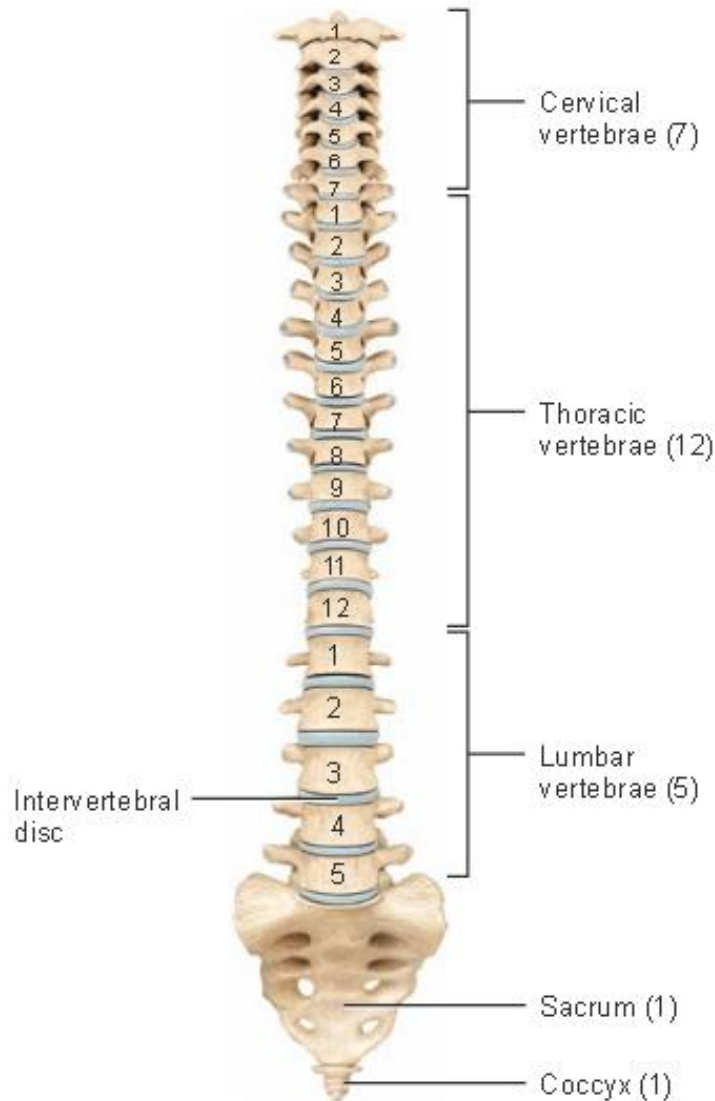
- Three auditory ossicles in the middle ear of each ear
 - ✓ malleus
 - ✓ incus
 - ✓ stapes
- Helps to transmit sound impulses

The Hyoid Bone

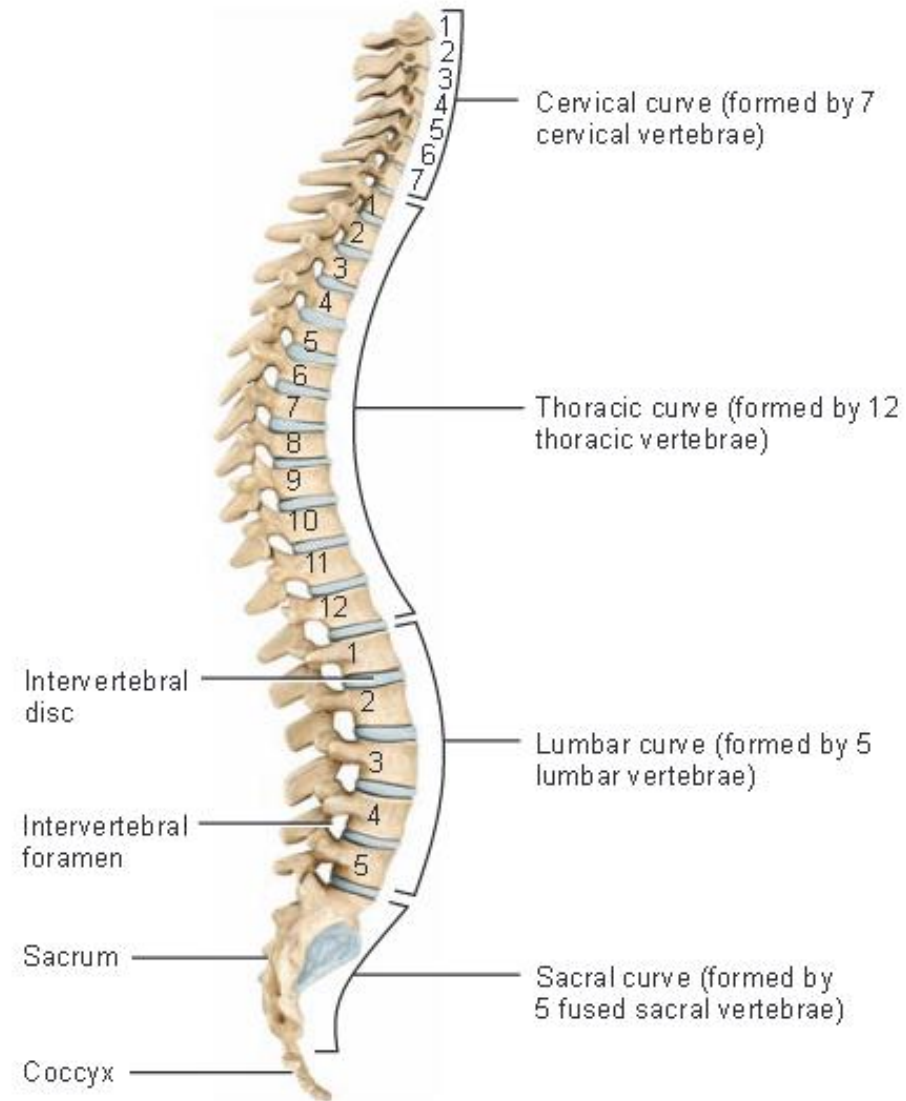
- The only bone that does not articulate with another bone
- Serves as a moveable base for the tongue



The Vertebral Column



(a) Anterior view showing regions of the vertebral column



(b) Right lateral view showing four normal curves

The Vertebral Column...

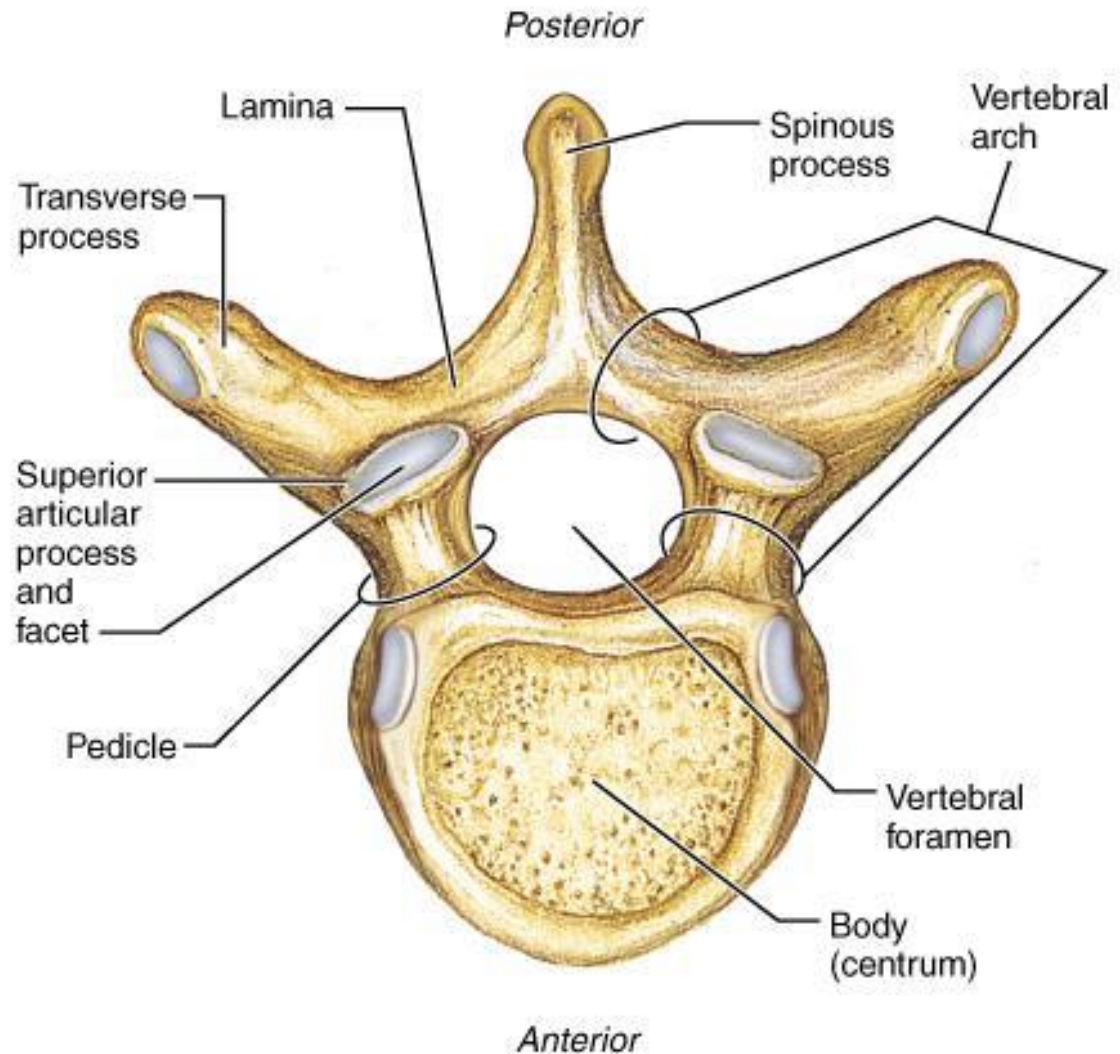
- The vertebral column performs several functions:
 - Encloses and protects the spinal cord
 - Supports the weight of the head, neck, and trunk
 - Transfers weight to the lower limbs
 - Helps to maintain the upright position of the body

Composition of vertebral column

- The vertebral column in an adult typically consists of 26 vertebrae
 - become larger as the vertebral column descends to the sacrum and then become progressively smaller toward the apex of the coccyx
- Intervertebral discs are pads of fibrocartilage that bind together the vertebral bodies of adjacent vertebrae.
- Divided in to five regions:
 - ✓ Cervical7
 - ✓ Thoracic12
 - ✓ Lumbar5
 - ✓ Sacral 1 (five fused vertebrae)
 - ✓ Coccygeal..... 1 (four fused vertebrae)

Characteristics of Typical Vertebra

- **Consists of:-**
 - I. **Vertebral body,**
 - II. **Vertebral arch**
and
 - III. **Seven processes**



Characteristics of Typical Vertebra

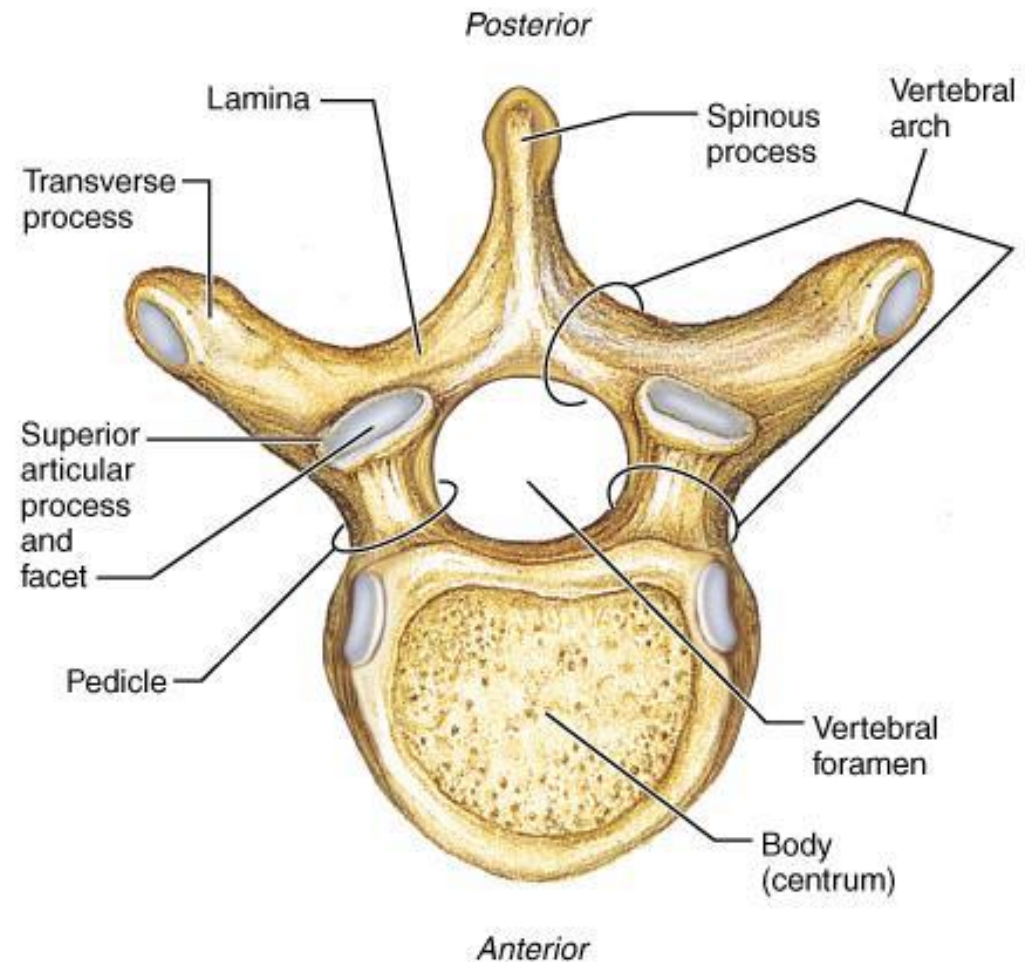
I. Vertebral body

- More massive & roughly cylindrical; weight bearing part of the vertebra
- linked to adjacent vertebral bodies by **IVD** and **ligaments**, forming the **cartilaginous joints**.
- Thoracic vertebrae has **costal facets**, which articulate with the heads of the corresponding and subjacent (just below) ribs.
- The size of vertebral bodies increases inferiorly as the amount of weight supported increases.
 - most markedly from T4 inferiorly

Characteristics of Typical Vertebra

II. Vertebral arch

- is posterior to the vertebral body and
- forms the lateral and posterior parts of the **vertebral foramen**.
- consists of two (R&L) pedicles and laminae
- has seven processes



Characteristics of Typical Vertebra...

II. Vertebral arch...

- The vertebral foramina of all the vertebrae together form the **vertebral canal**, which contains
 - Spinal cord, spinal nerve roots and spinal meninges
 - Blood vessels
 - Continuous superiorly with the cranial cavity, through the **foramen magnum** of the skull.
- Consists of pedicles and laminae:
 - two **pedicles**; bony pillars that attach the vertebral arch to the vertebral body;
 - two **laminae**; flat sheets of bone that extend from each pedicle to meet in the midline and form the roof of the vertebral arch.

Characteristics of Typical Vertebra

7 Processes from the vertebral arch of a typical vertebra

- **One median spinous process**
 - ✓ Projects posteriorly (and usually inferiorly, typically overlapping the vertebra below) from the vertebral arch at the junction of the laminae
- **Two transverse processes project posterolaterally from the junctions of the pedicles and laminae**
- ✓ The spinous and transverse processes, afford attachments for deep back muscles and serve as levers, facilitating the muscles that fix or change the position of the vertebrae

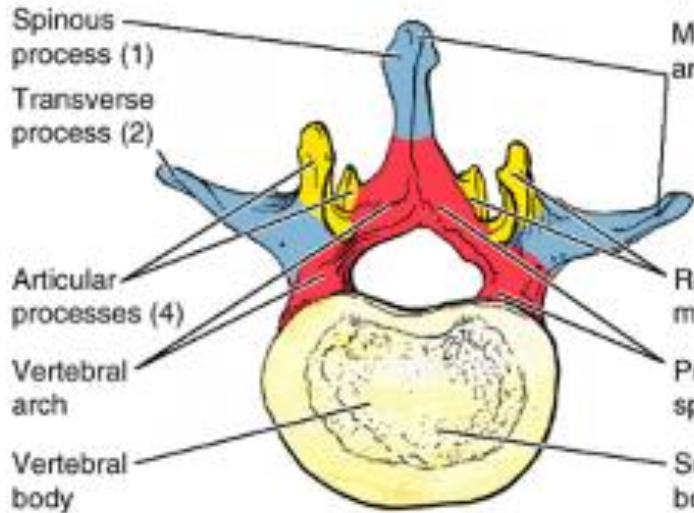
Characteristics of Typical Vertebra

7 Processes from the vertebral arch of a typical vertebra...

- **Four articular processes (G. zygapophyses)**
 - ✓ two superior and two inferior also arise from the junctions of the pedicles and laminae
 - ✓ each bearing an articular surface (facet)
- **Vertebral notches (sup. & inf.)** are indentations on each pedicle superiorly and inferiorly
 - ✓ b/n the superior and inferior articular processes posteriorly and the corresponding projections of the body anteriorly

Characteristics of Typical Vertebra

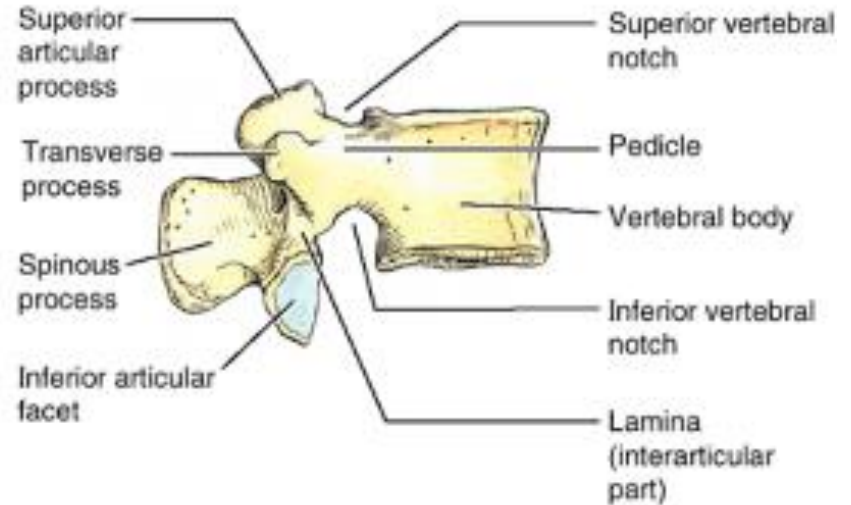
Parts:



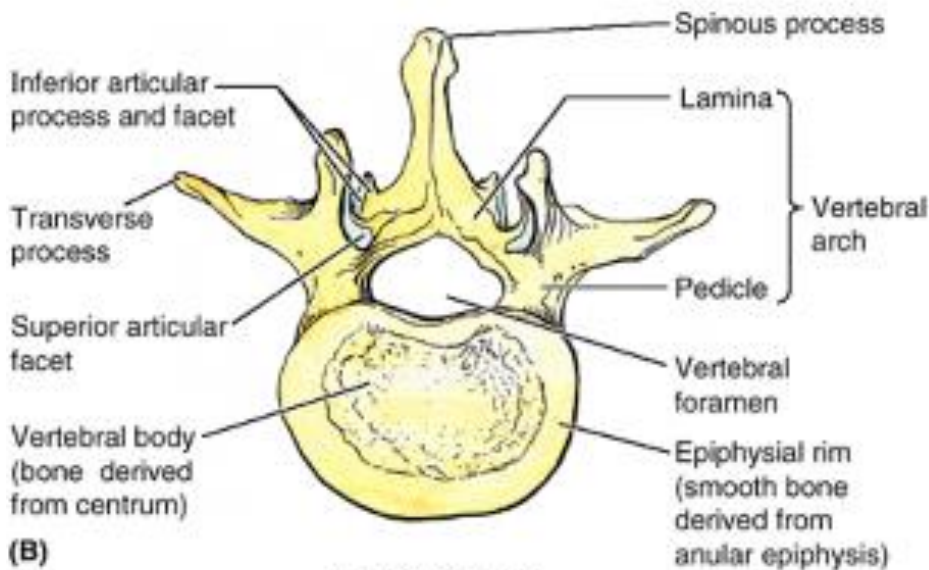
(A)

Functions:

Muscle attachment and movement
Restriction of movement
Protection of spinal cord
Support of body weight

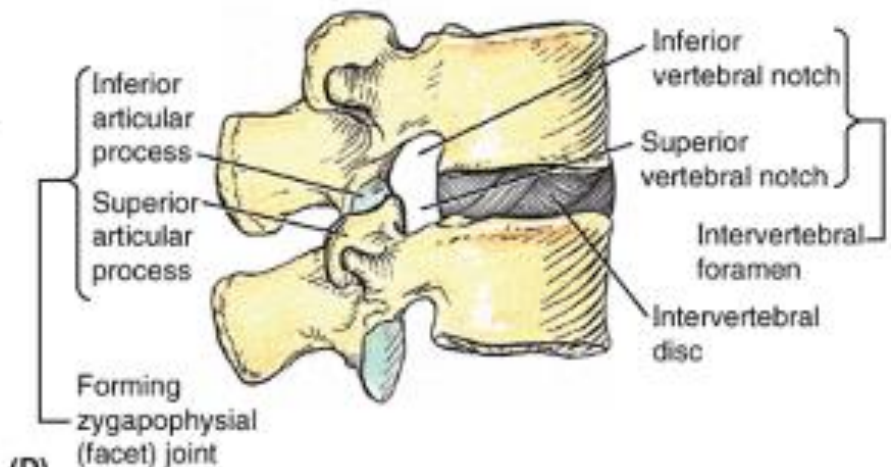


(C)



(B)

Superior views

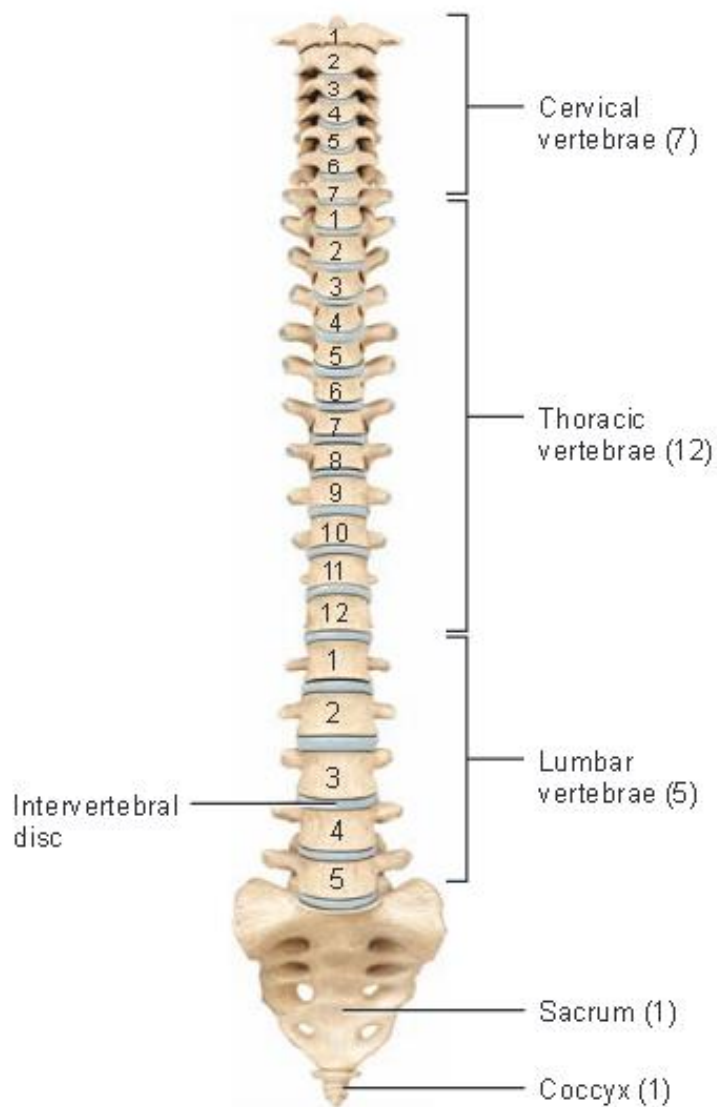


(D)

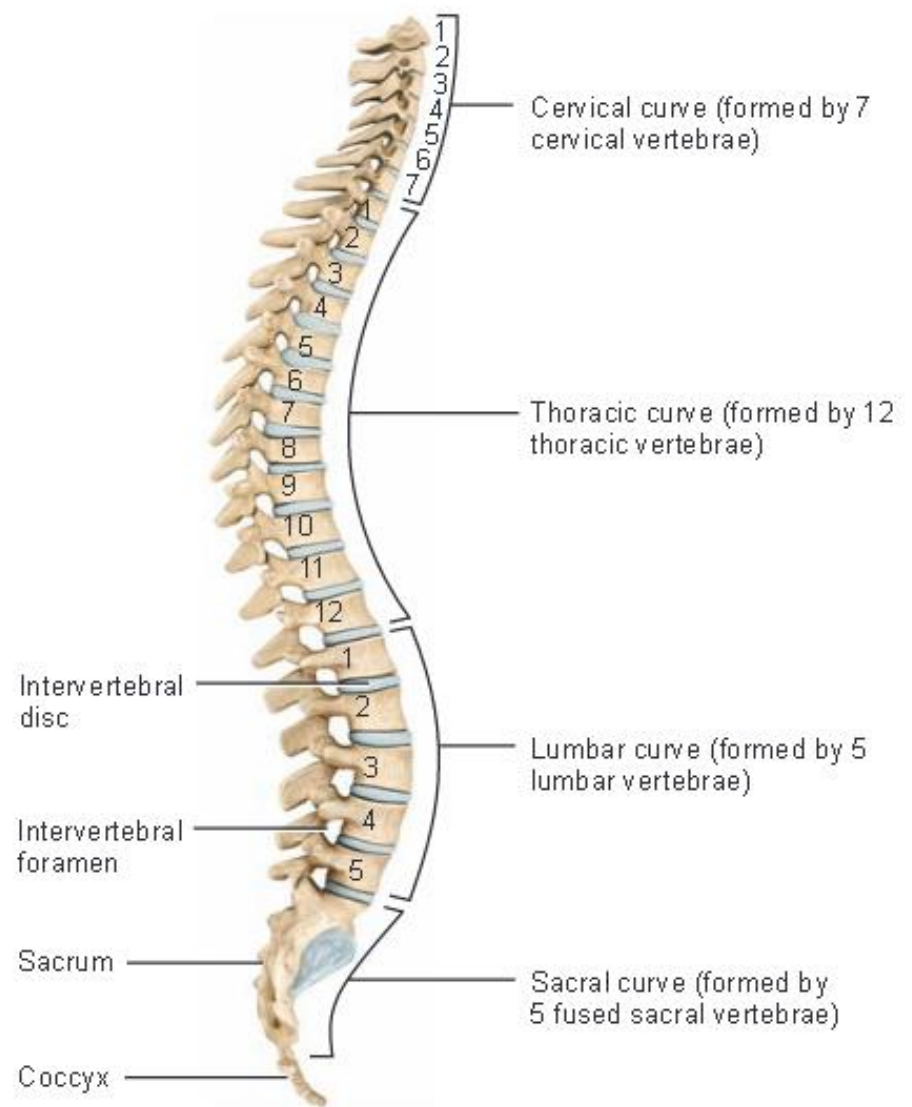
Lateral views

The Vertebral Column...

- The spine has a normal curvature
- Has four curvatures in adults (cervical, thoracic, lumbar & sacral)
 - ✓ **thoracic & sacral**: are primary curves
 - ✓ **cervical & lumbar**: are secondary curves
- The curves function in:- increasing the strength
 - ✓ maintaining balance



(a) Anterior view showing regions of the vertebral column



(b) Right lateral view showing four normal curves

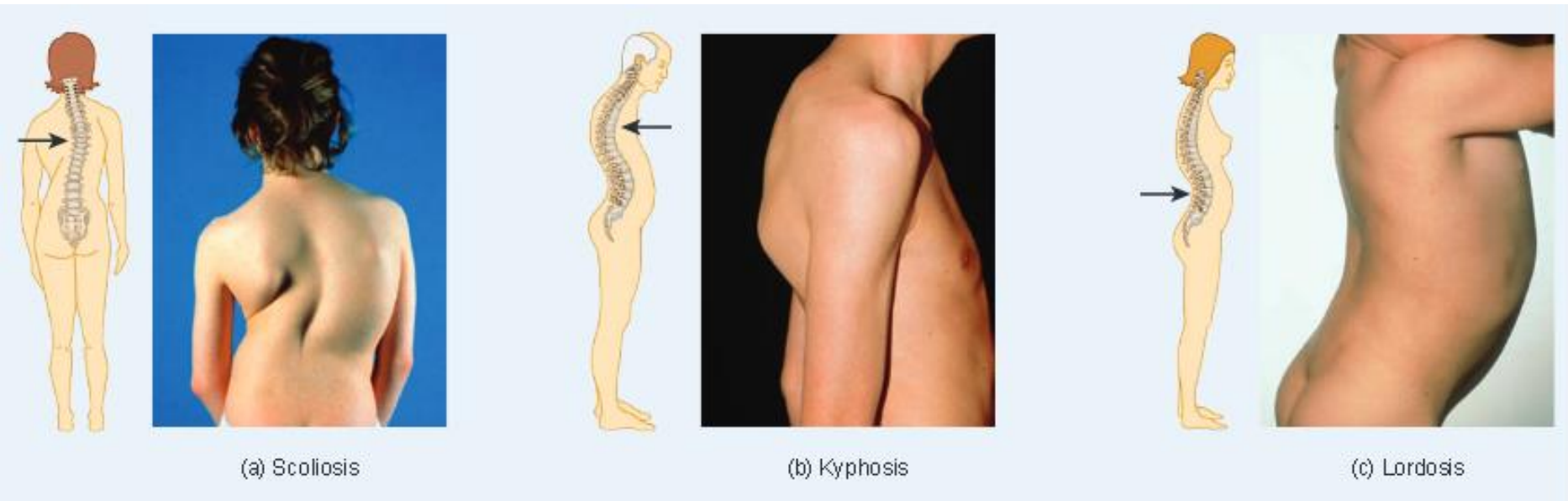
The Vertebral Column...

- Primary curves/**accommodation curves** are the posteriorly sweeping curves of the thoracic and sacral regions.
 - These curves **develop before birth to allow the abdominopelvic viscera more room.**
- Secondary curves/ **compensation curves** develop in the infant and toddler as anteriorly sweeping curves of the **cervical and lumbar regions.**
 - These curves develop as the infant learns to hold up his or her head (cervical) and begins to **walk (lumbar).**

The Vertebral Column...

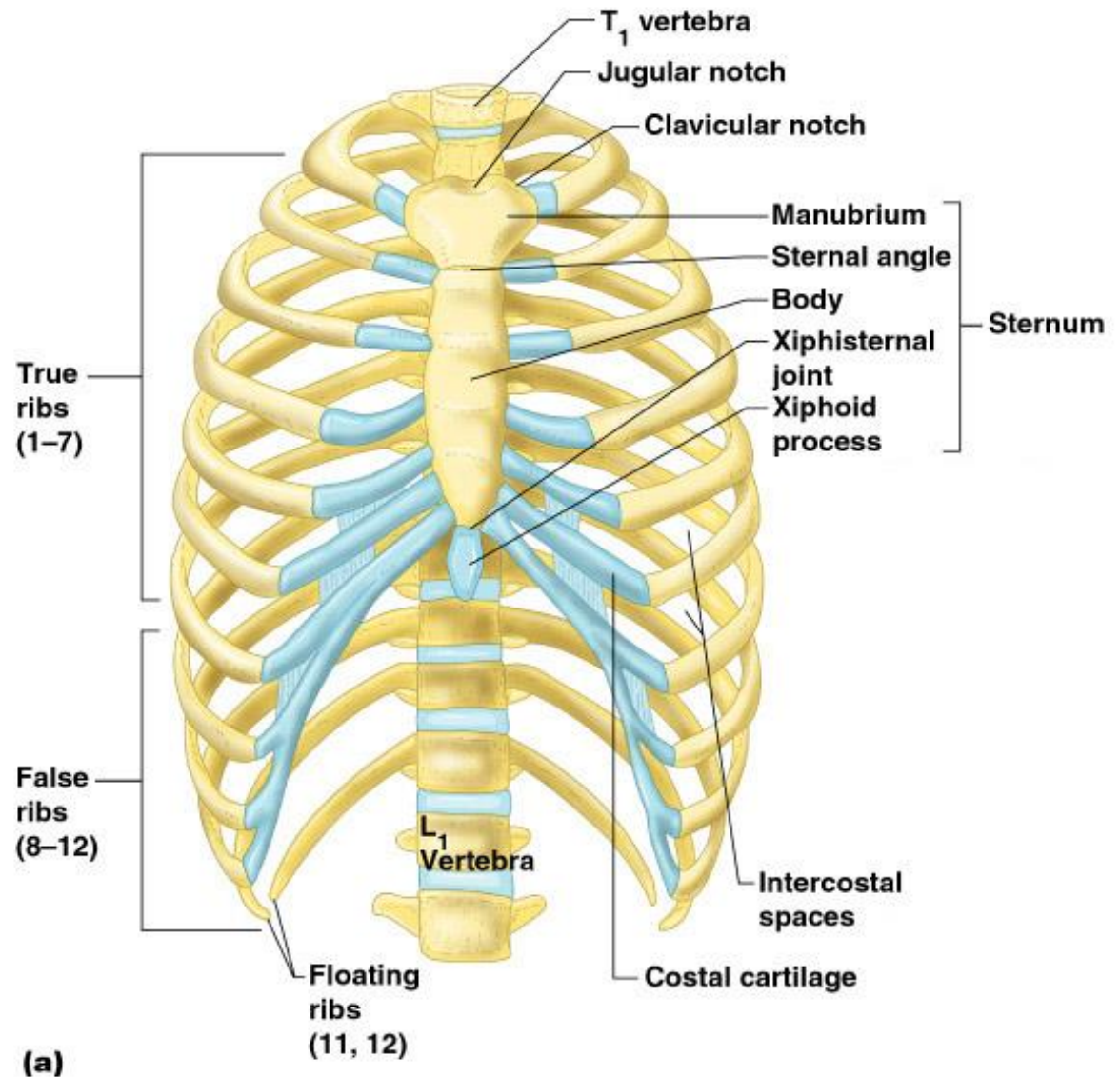
Abnormal curvatures

- Result from exaggeration of a normal curve include;
 - ✓ **Scoliosis** (abnormal lateral curvature)
 - ✓ **Kyphosis** (abnormal posterior curvature - hunchback)
 - ✓ **Lordosis** (abnormal anterior curvature – swayback)



The Bony Thorax

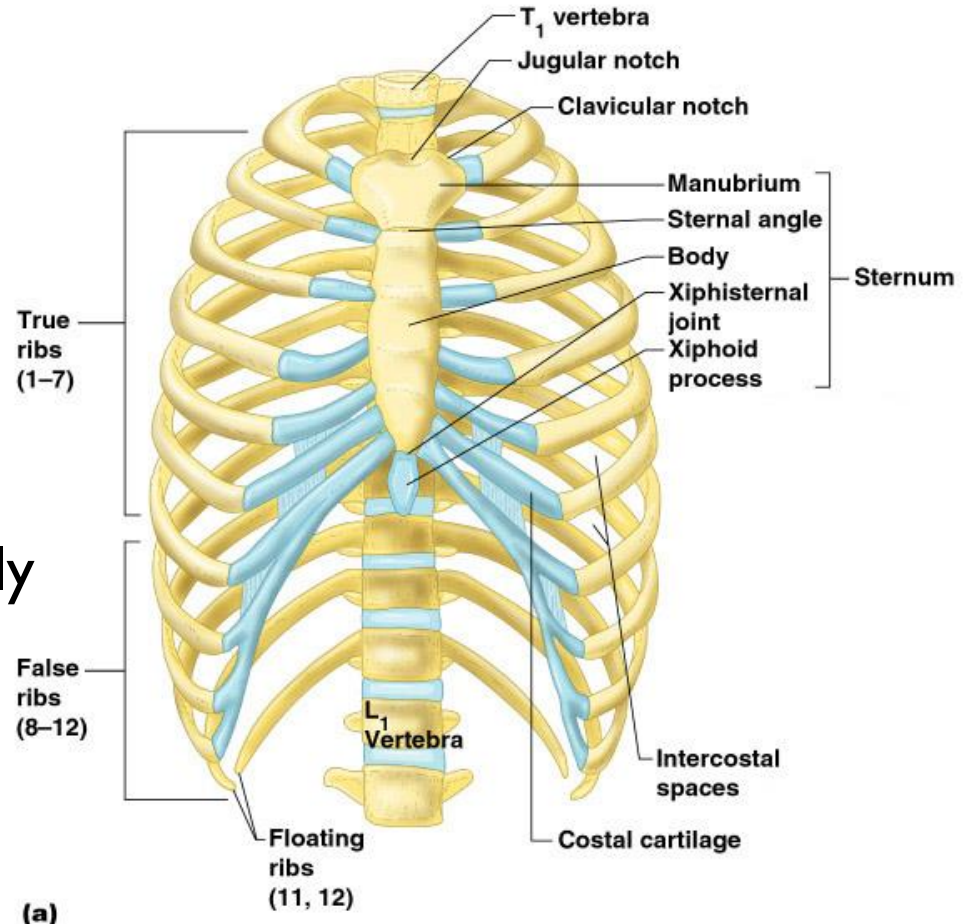
- Forms a cage to protect major organs
- Made-up of three parts
 - ✓ Sternum
 - ✓ Ribs
 - ✓ Thoracic vertebrae



The Bony Thorax...

Ribs

- 12 pairs (connect directly to thoracic vertebrae in the back)
- 7 pairs “true ribs” connect directly to sternum anteriorly
- the remaining 5 pairs are called “false ribs”
 - ✓ last 2 pairs “floating ribs” do not attach to sternum at all



The Bony Thorax...

Sternum

- Flat bone located anteriorly
- Has 3 fused bones:
 - ✓ manubrium, body & xiphoid process
- On its side, it has costal & clavicular notches and superiorly jugular notch
- Sternal angle is palpated between manubrium & body

The Appendicular Skeleton

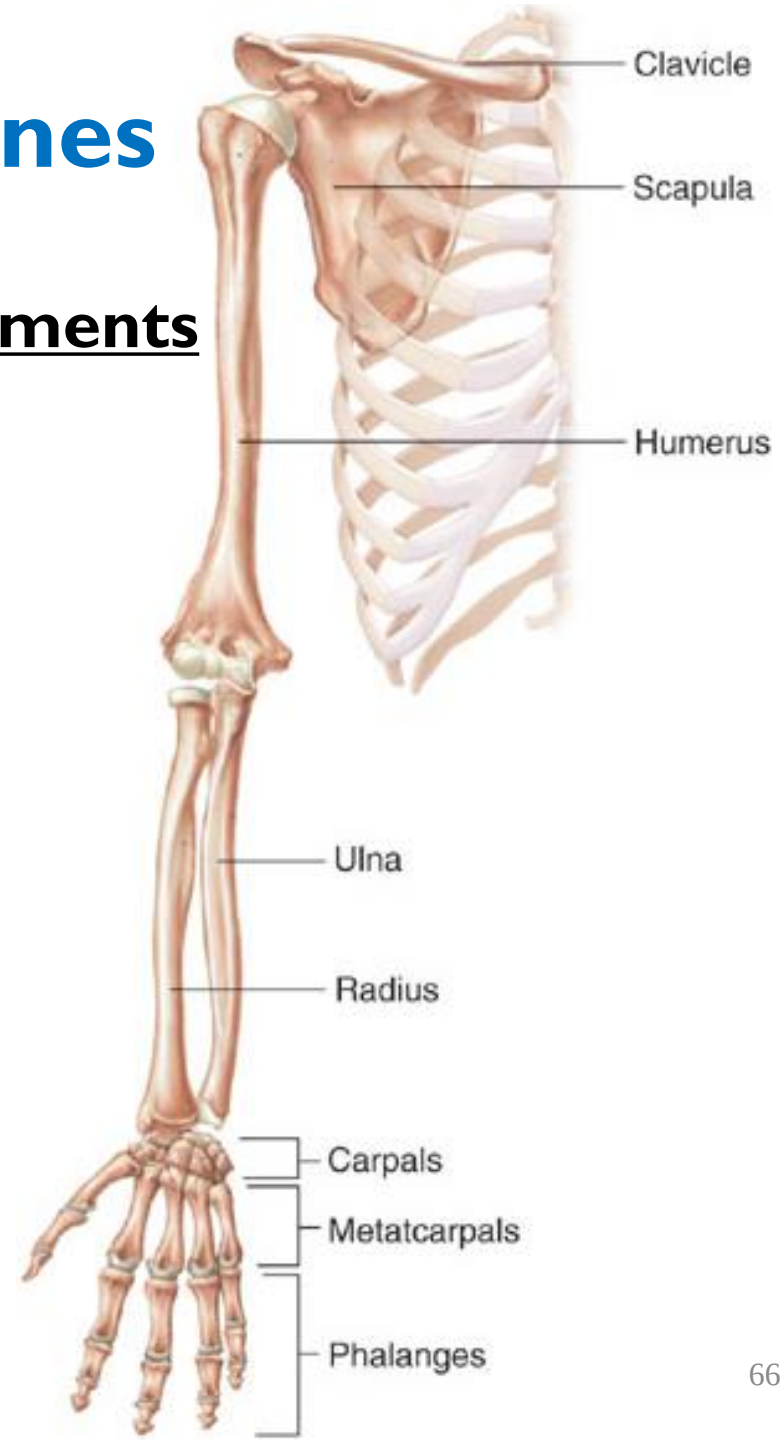
The Appendicular Skeleton

- 126 bones of the:
 - ✓ Limbs (appendages)
 - ✓ Pectoral girdle
 - ✓ Pelvic girdle

The Upper Limb bones

Upper limb consists of 4 segments

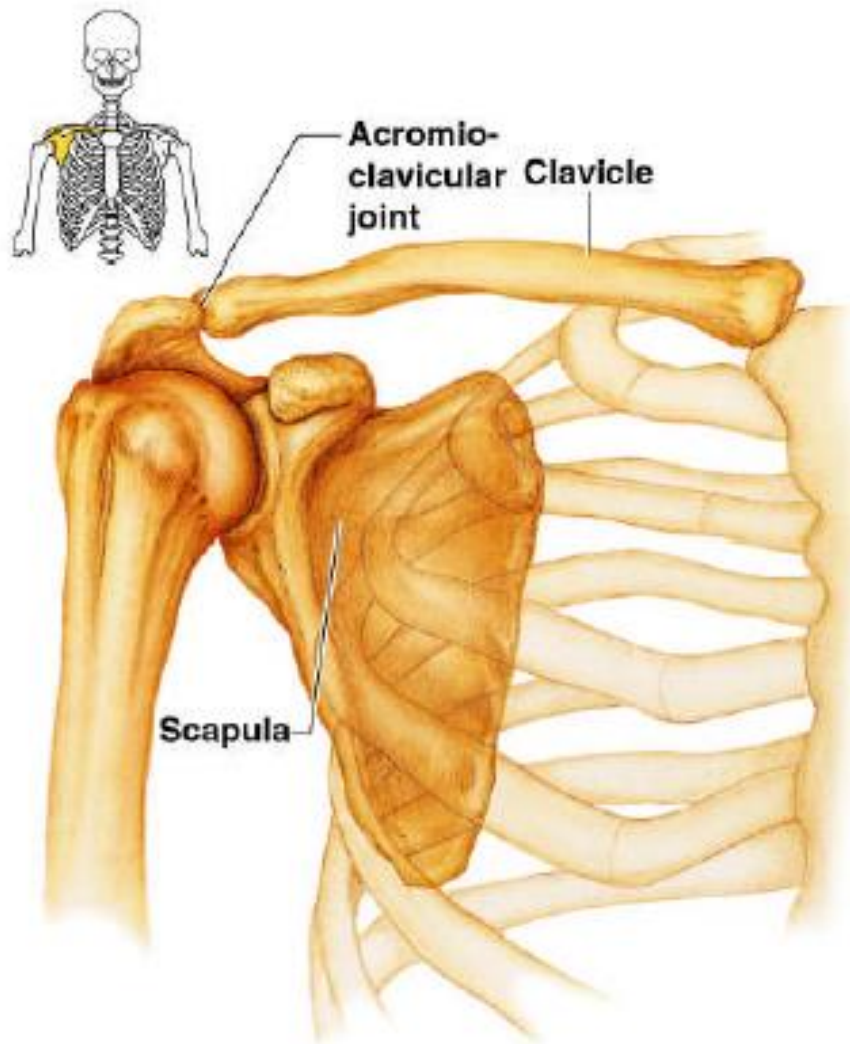
- **Shoulder**; pectoral girdle
- **Arm**
 - Humerus
- **Forearm**
 - Ulna
 - Radius
- **Hand**
 - Wrist (**carpal bones**)
 - Palm (**metacarpal bones**)
 - Fingers (**phalanges**)



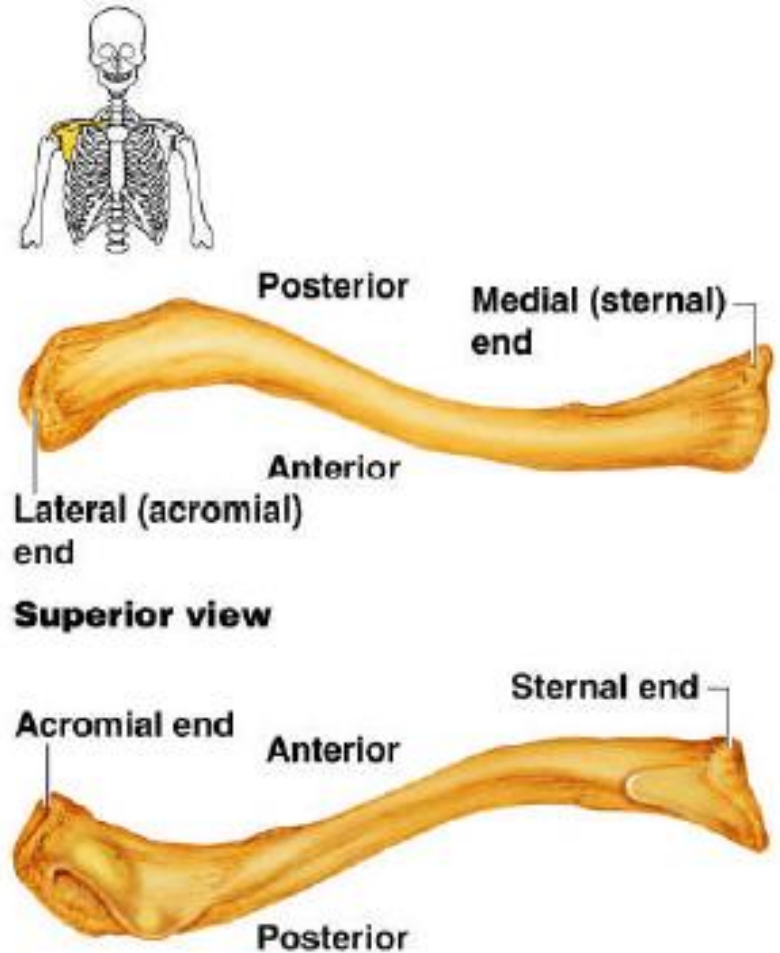
The Pectoral (Shoulder) Girdle

- Composed of two bones
 - ✓ Clavicle (2) – collarbone
 - ✓ Scapula (2) – shoulder blade
- These bones allow the upper limb to have **exceptionally free movement**
 - Attached only anteriorly to the **axial skeleton on sternum** by sternoclavicular joint
 - Loosely attached, held in place largely by musculature

Bones of the Shoulder Girdle



(a) Articulated pectoral girdle



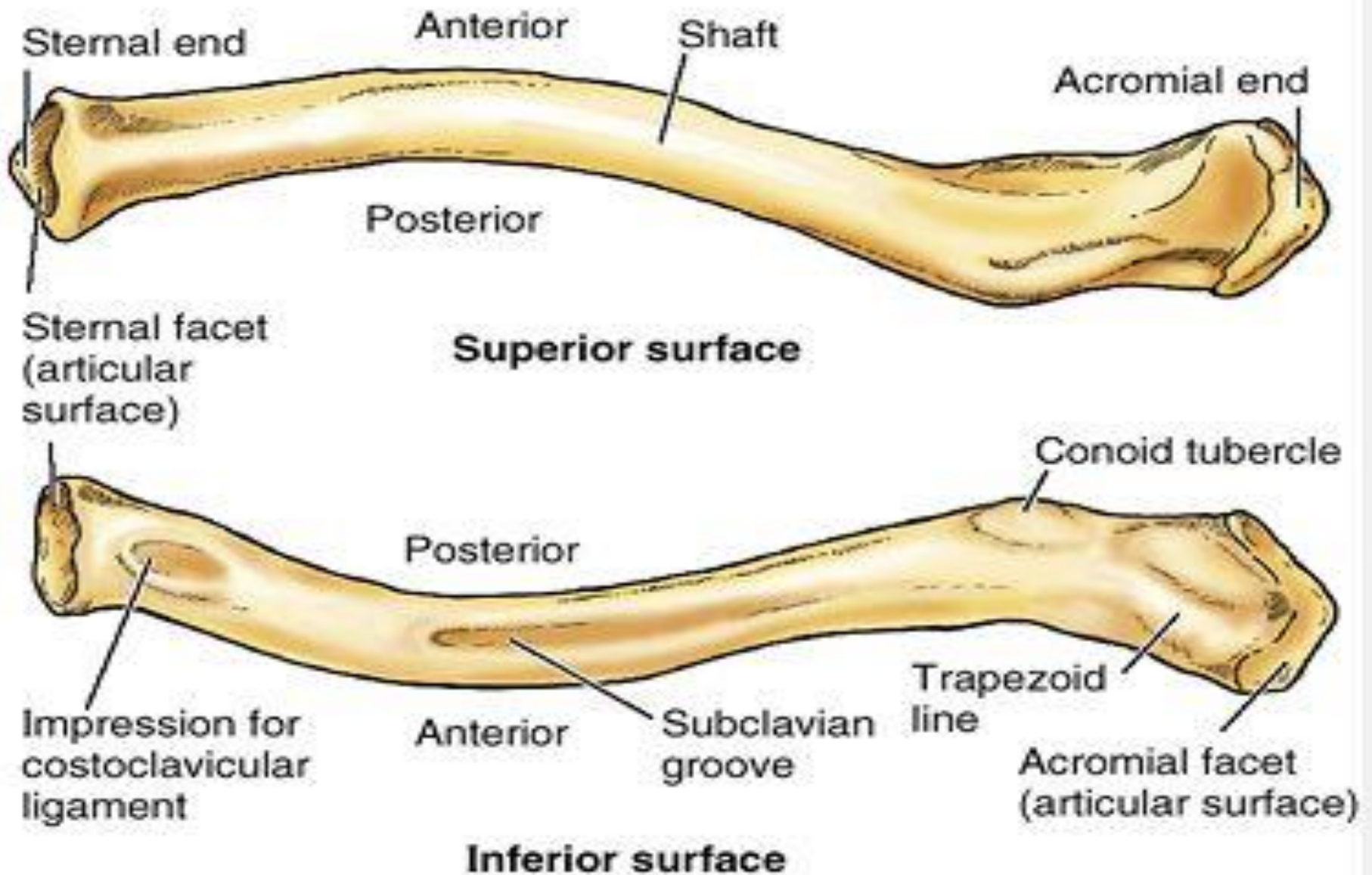
Inferior view

(b) Right clavicle

Clavicle (collar bone)

- Double curved bones extending along the superior aspect thorax
- The medial two thirds of the shaft of the clavicle are convex anteriorly, whereas the lateral third is flattened and concave anteriorly
 - ✓ gives S-shaped slender collar bone
 - ✓ **curvatures increase the resilience** of the clavicle
- **sternal end**: articulates with manubrium of the sternum at the sternoclavicular (SC) joint
- **acromial end**: articulates with acromion of scapula at the acromioclavicular (AC) joint
- ***It has two surface superior & inferior surface***

Right clavicle



Clavicle: Function

- **Connects** the upper limb to the trunk
- Attachment site for muscles of the thorax and shoulder
- Position scapula away from thorax so that the arm **has maximum freedom of motion**
- Afford **protection to the neurovascular** bundle supplying the upper limb
- **Transmits shocks** (traumatic impacts) from the upper limb to the axial skeleton

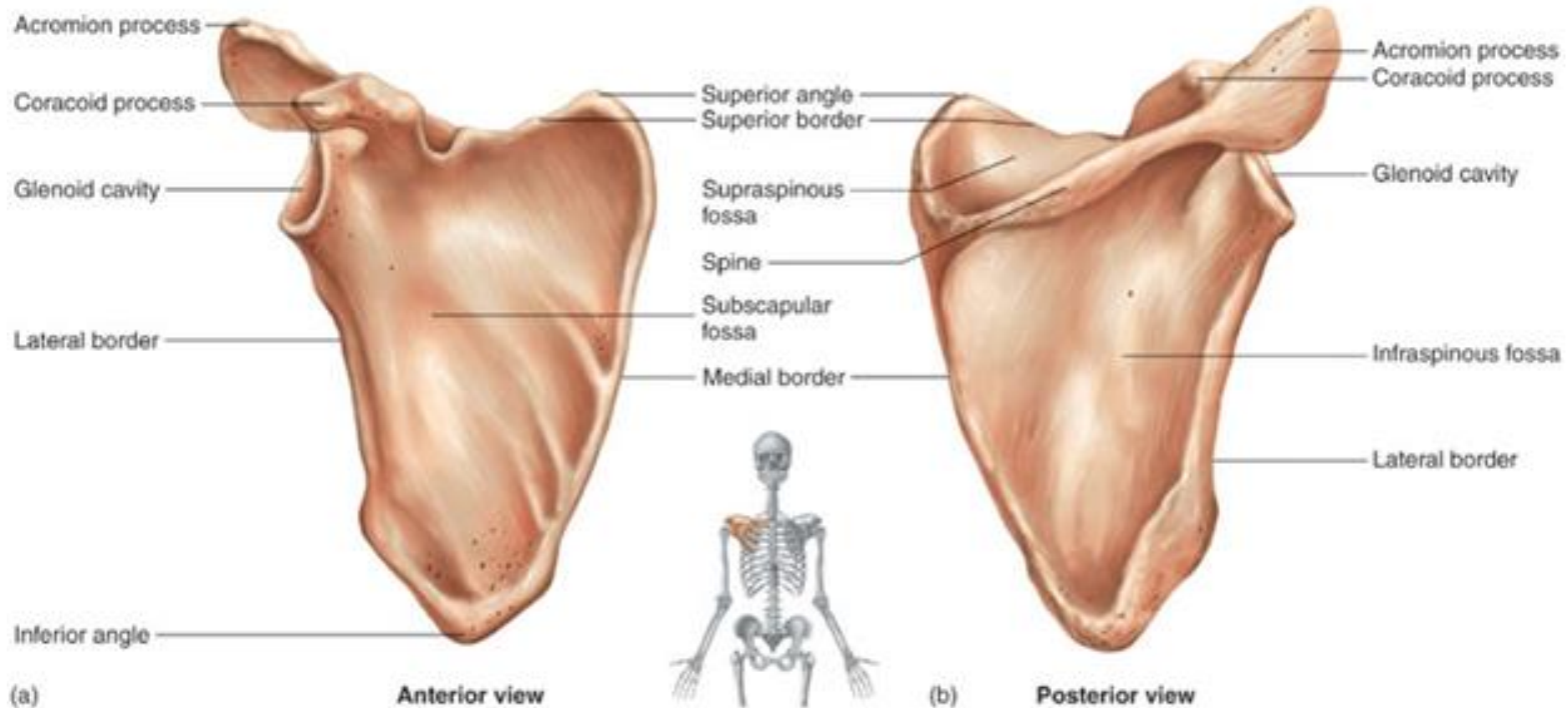
Scapula (shoulder blade)

- Large triangular flat bone
- Lies on the posterolateral aspect of the thorax, overlying the 2nd -7th ribs
- Has 3 borders: superior, medial & lateral
- Has 3 angles: superior, inferior & lateral

Posterior surface

- ◆ Convex
- ◆ Divided by spine into supraspinous and infraspinous fossa
- ◆ Spine continues laterally as flat expanded acromion which articulates with the acromial end of the clavicle

Scapula



- **Anterior surface**

- ✓ forms subscapular fossa

- **Lateral surface**

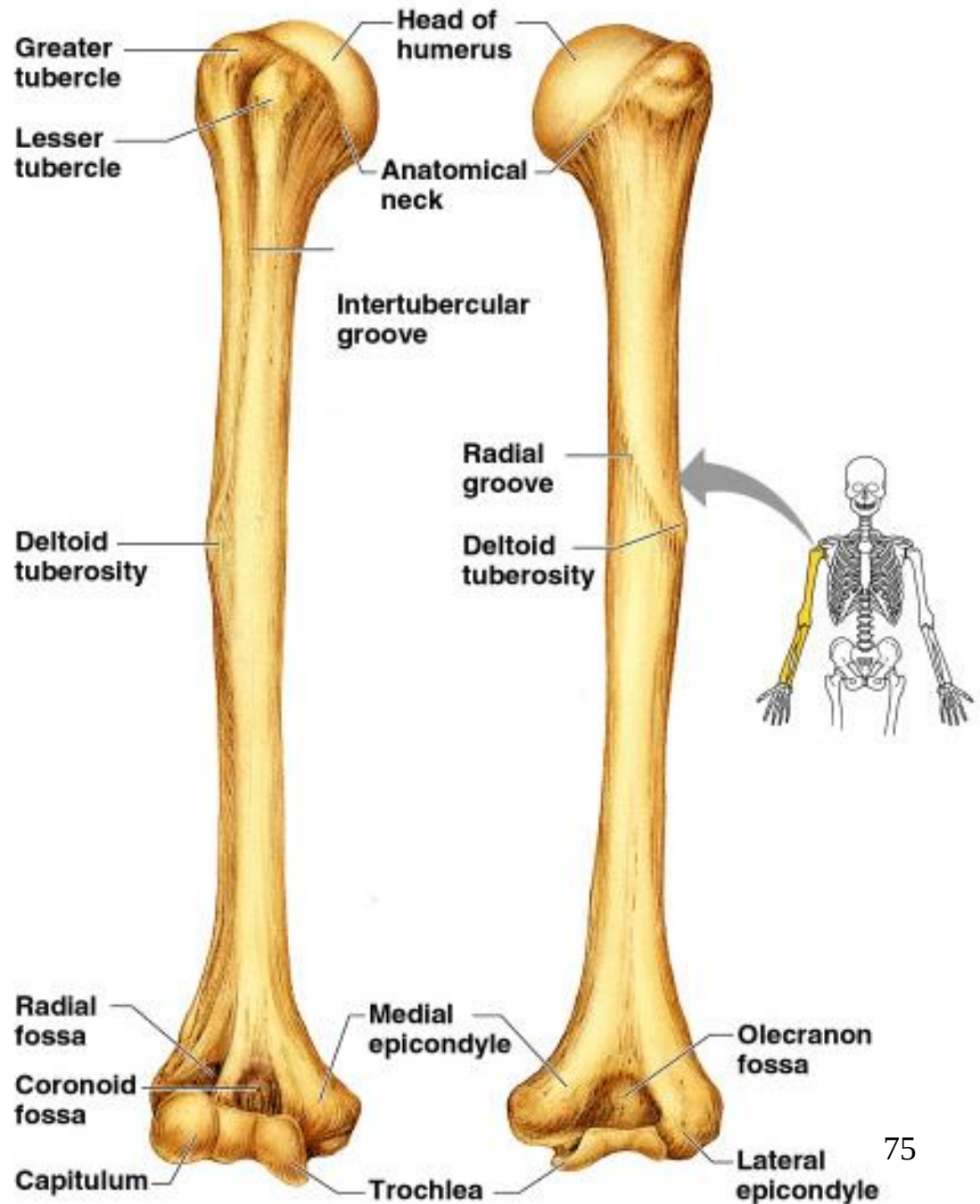
- ✓ has a glenoid cavity
 - shallow, concave, oval fossa
 - directed anterolaterally and slightly superiorly
 - articulates with the head of the humerus - glenohumeral (shoulder) joint
 - considerably smaller than the head of the humerus

- **Coracoid process**

- ✓ superior to glenoid cavity and projects anterolaterally

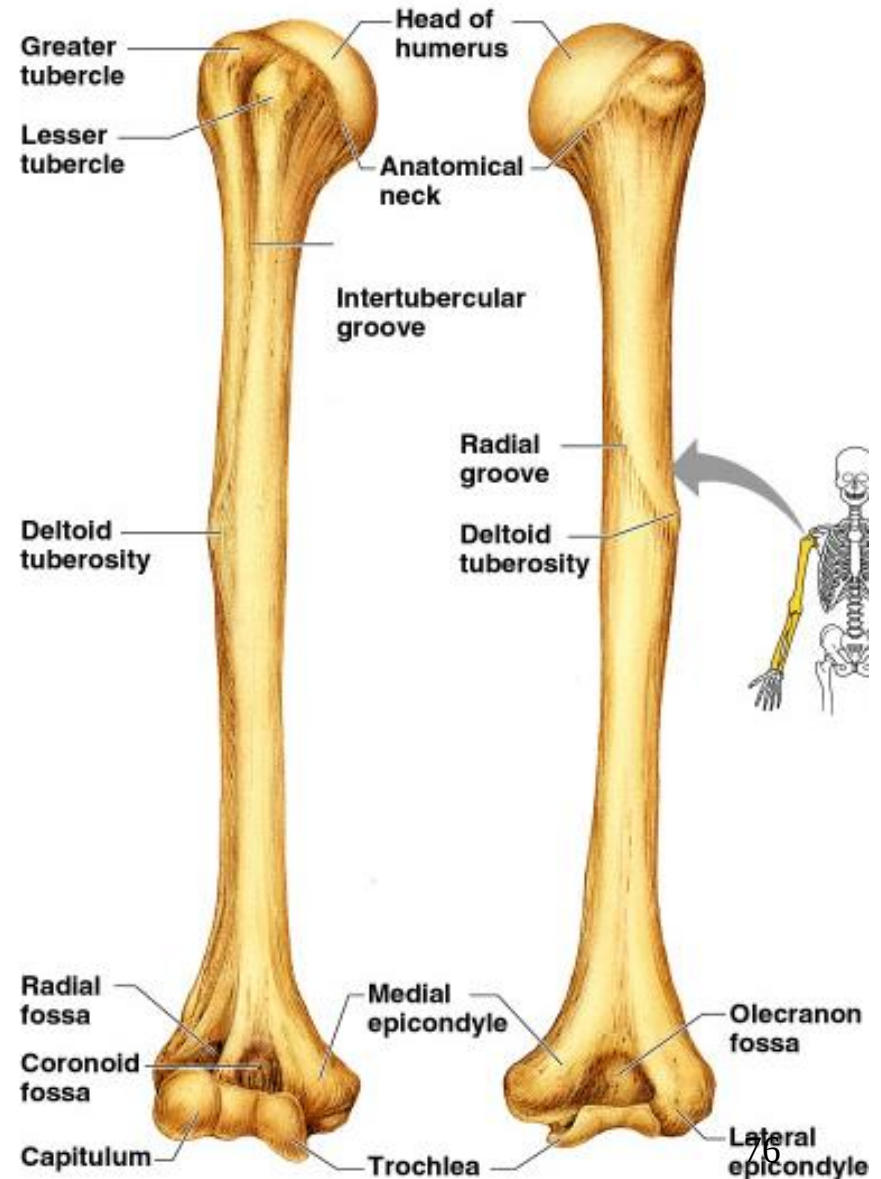
Bones of the Upper Limb

- The arm is formed by a single bone
 - ✓ Humerus



Humerus

- The longest bone of upper extremity
- **Consist of**
 - ✓ Head
 - ✓ Anatomical & surgical necks,
 - ✓ Tubercles(greater & lesser)
 - ✓ Intertubercular groove &
 - ✓ Shaft (body)



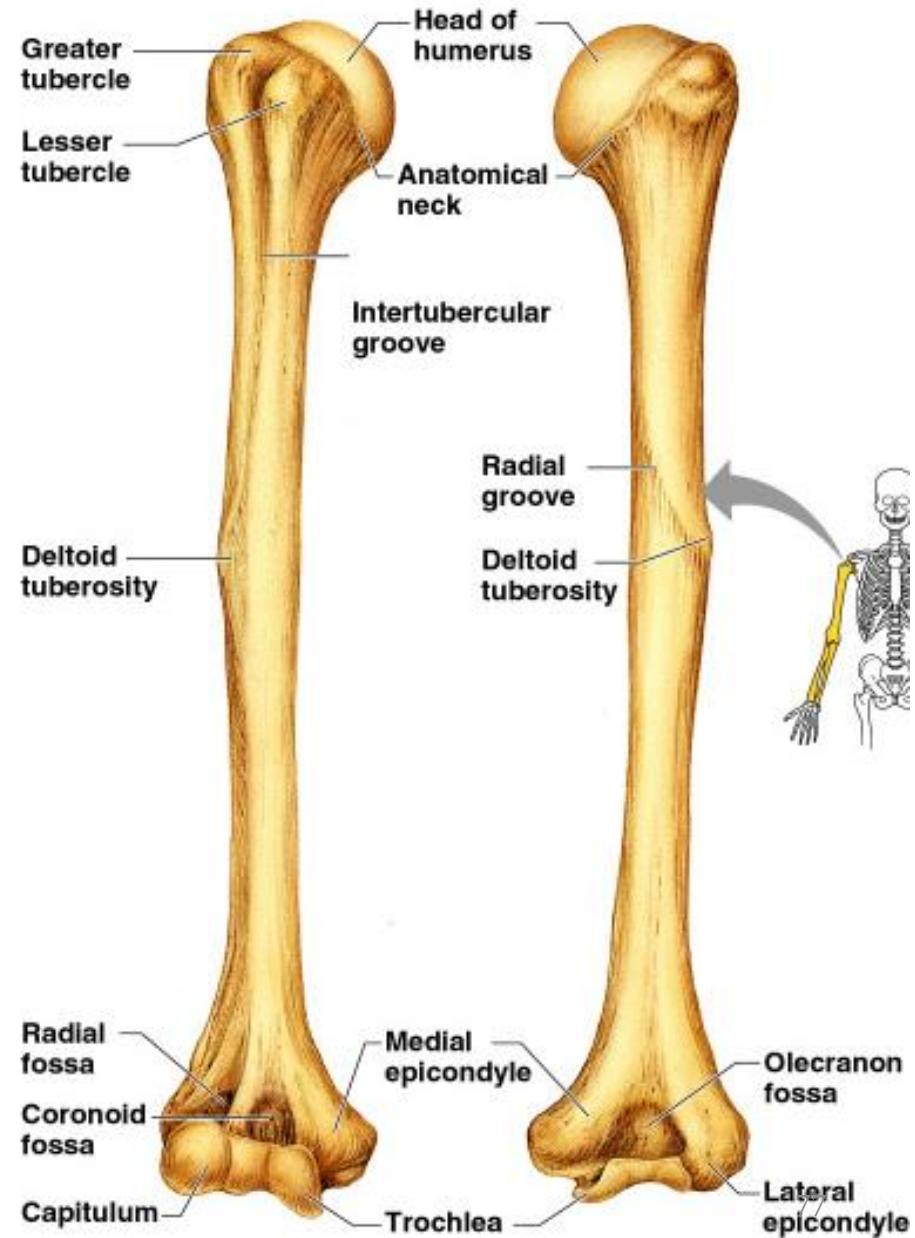
Humerus

■ Head

- ✓ ball-shaped
- ✓ articulate with the glenoid cavity of the scapula

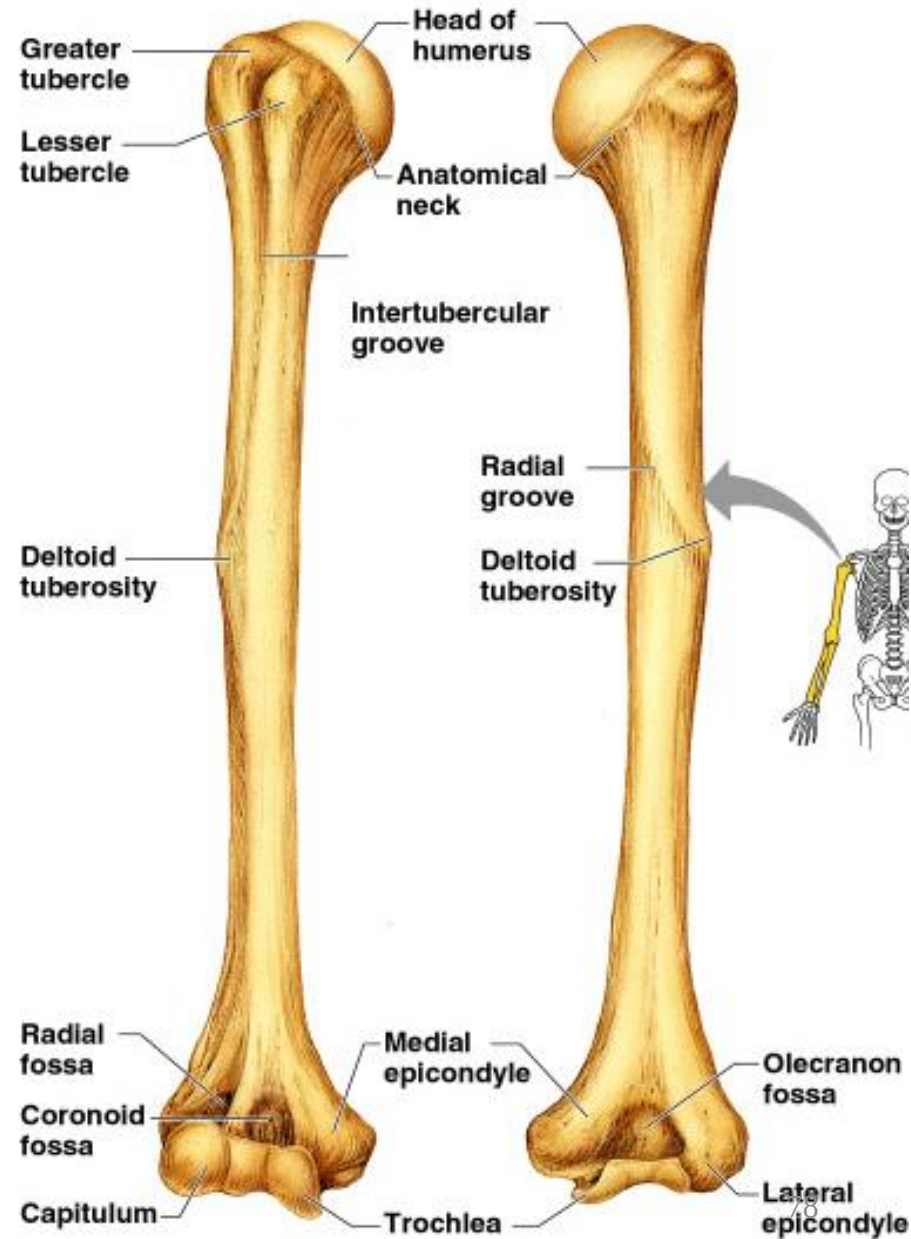
■ Tubercles

- ✓ lesser and greater tubercle
- ✓ *intertubercular groove* (bicipital groove) separates the two



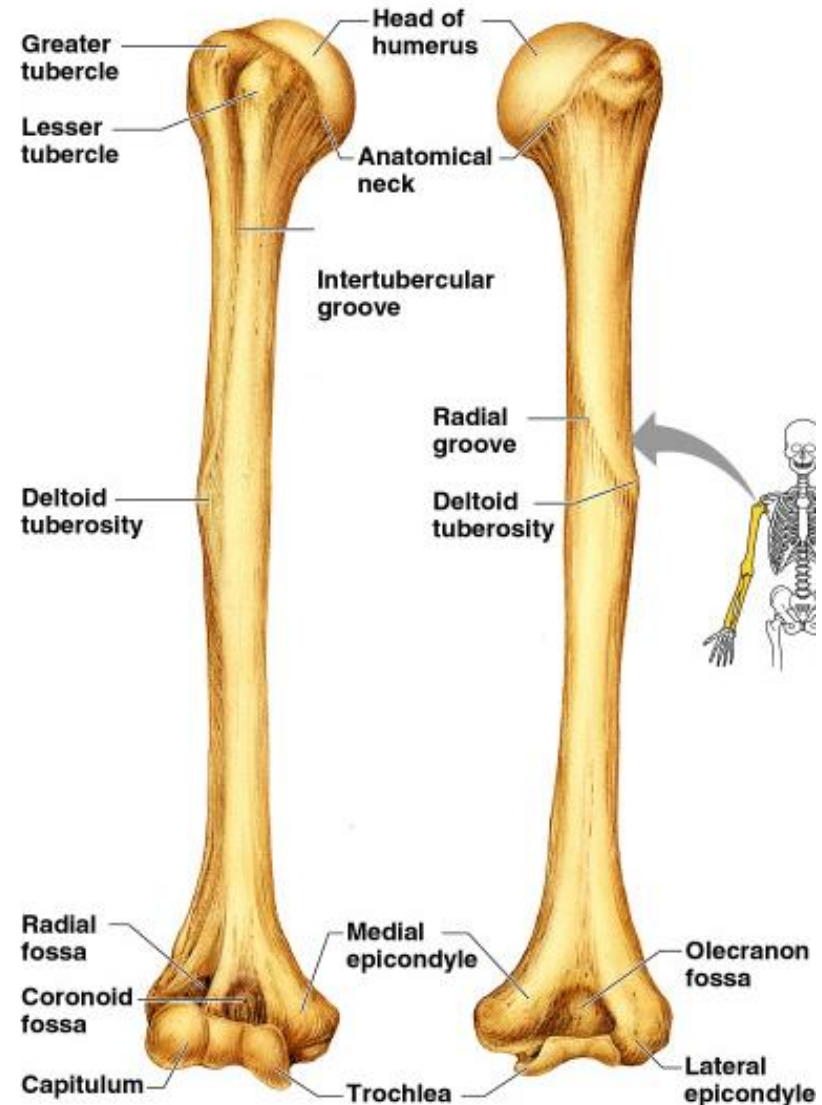
Humerus

- **Anatomical neck**
 - ✓ just distal to the humeral head
 - ✓ ***separates*** the ***head*** from the ***tubercles***
- **Surgical neck**; commonly fractured site of humerus
 - ✓ narrow part just distal to the tubercles



Humerus...

- **Shaft (body)**
 - ✓ Two prominent features:
 - **Deltoid tuberosity**
laterally for attachment of deltoid m.
 - **Radial (spiral) groove**
posteriorly for **radial nerve** and **deep artery of the arm**
- **Inferior end**
 - widens as sharp medial and lateral supracondylar ridges
 - ends distally as medial and lateral epicondyle



Humerus...

▪ Condyles of the humerus

- ✓ distal end of the humerus
- ✓ include trochlea, capitulum, olecranon, coronoid, and radial fossae
- ✓ Distal end has two articular surfaces
 - **Capitulum**: for articulation with the **head of the radius**
 - **Trochlea**: for articulation with the **trochlear notch of ulna**
- ✓ **Coronoid fossa**
 - superior to the trochlea anteriorly
 - receives the coronoid process of the ulna during full flexion of the elbow

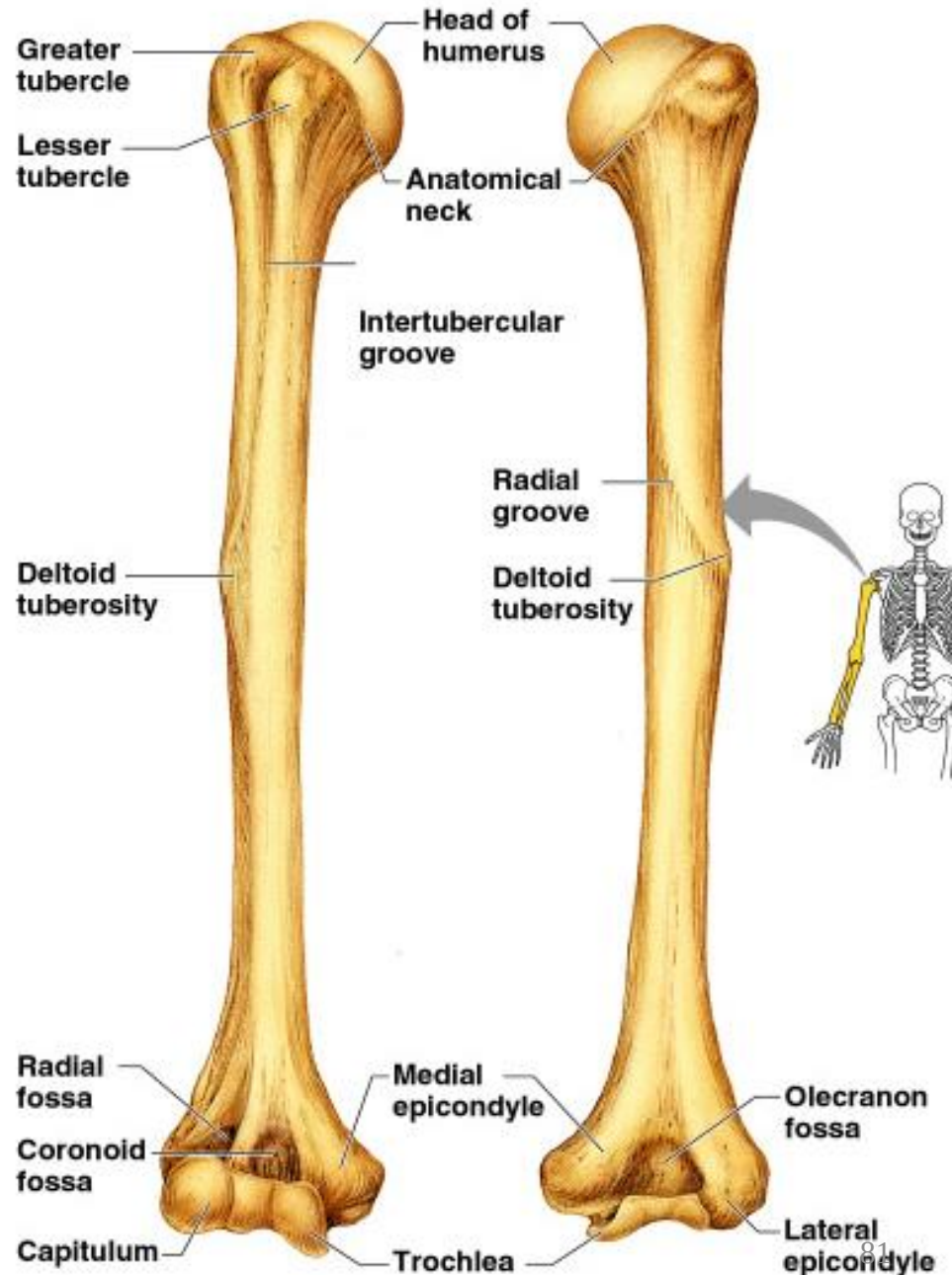
Humerus...

Olecranon fossa

- ✓ Postly, accommodates the olecranon of the ulna during extension of the elbow

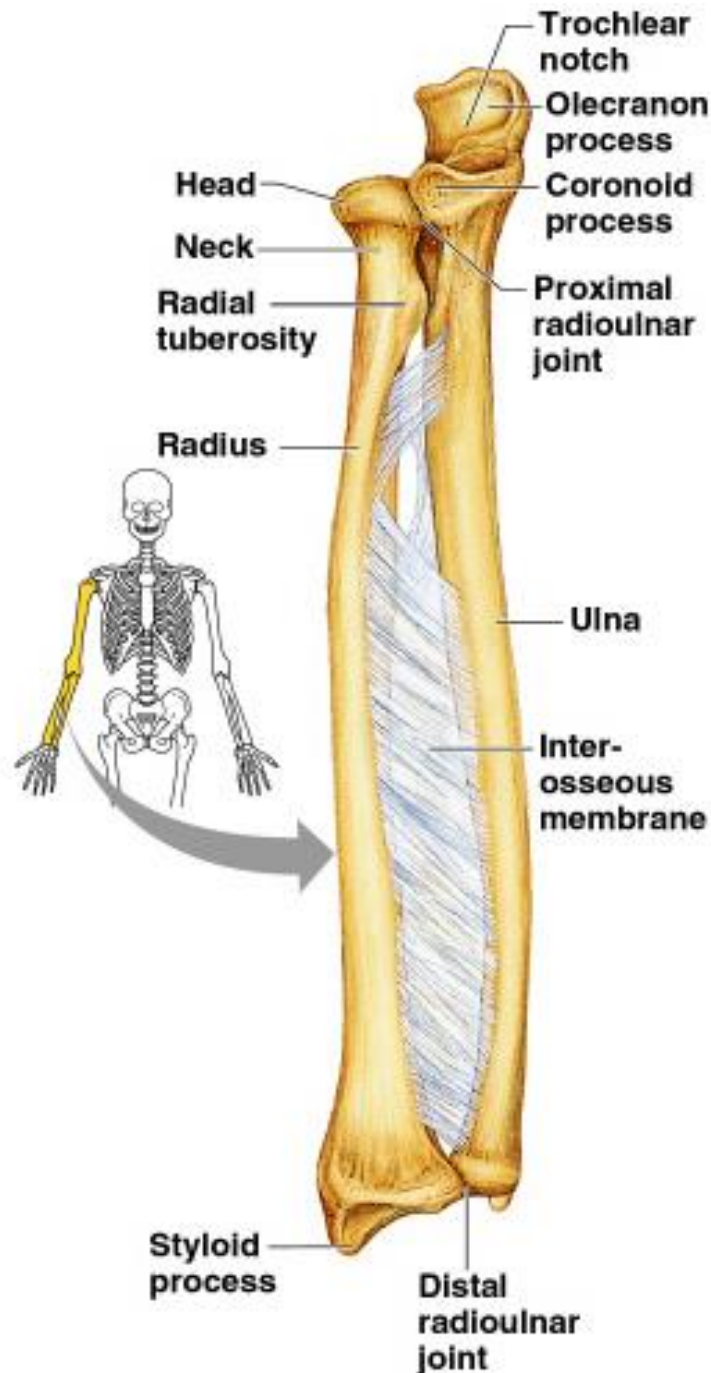
Radial fossa

- ✓ superior to the capitulum anteriorly
- ✓ accommodates the edge of the head of the radius when the elbow is fully flexed



Bones of the Upper Limb.

- The forearm has two bones
 - ✓ Ulna
 - ✓ Radius

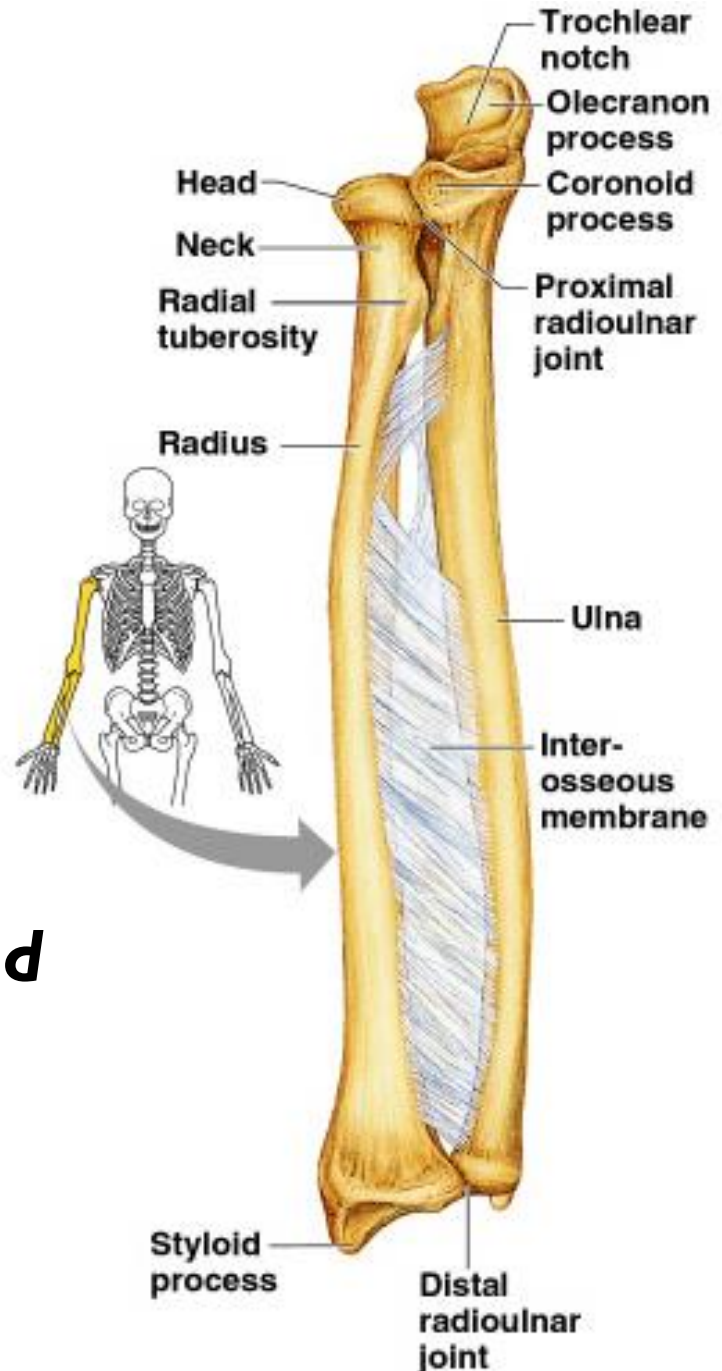


The Ulna

- The medial and longer of the two forearm bones
- Its proximal end has two prominent projections
 - ✓ olecranon posteriorly
 - ✓ coronoid process anteriorly
 - The two form the walls of the trochlear notch
- **Trochlear notch**
 - ✓ articulates with the trochlea of the humerus
- **Radial notch**
 - ✓ on the lateral side of the coronoid process
 - ✓ articulates with the head of radius

The Ulna...

- **Shaft (body)**
 - ✓ proximally it is thick, but it tapers distally
- **Head**
 - ✓ narrow distal end
 - ✓ rounded with small, conical ulnar styloid process
- ***The ulna does not reach, and therefore does not participate in the wrist (radiocarpal) joint***



The Radius

- The lateral and shorter of the two forearm bones
- Its proximal end consists of a cylindrical head, a short neck, a projection from the medial surface (radial tuberosity)

The Radius...

Head

- ✓ superior surface is concave for articulation with **capitulum** of humerus
- ✓ medially articulate with **radial notch** of ulna

Neck

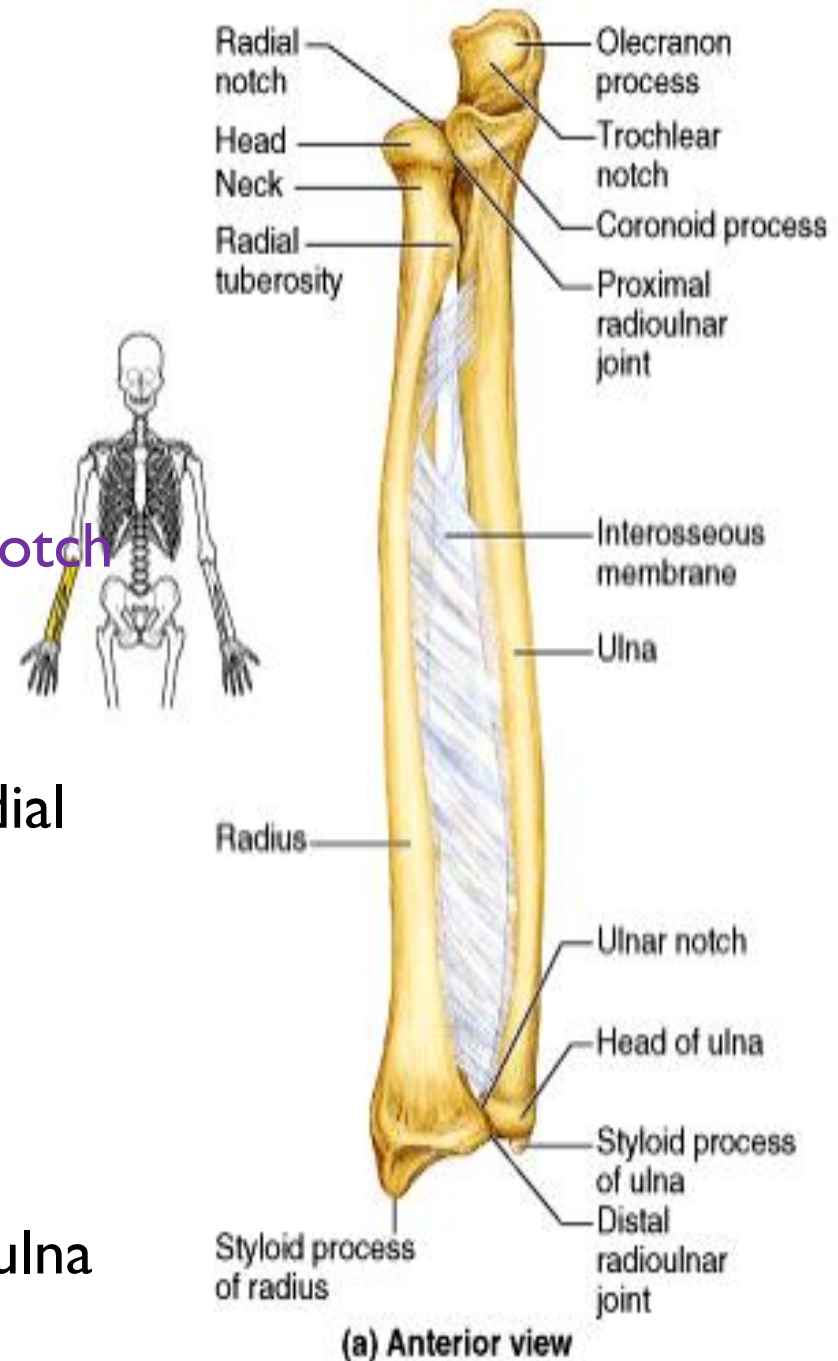
- ✓ narrow part between head and radial tuberosity

Shaft (body)

- ✓ gradually enlarges distally

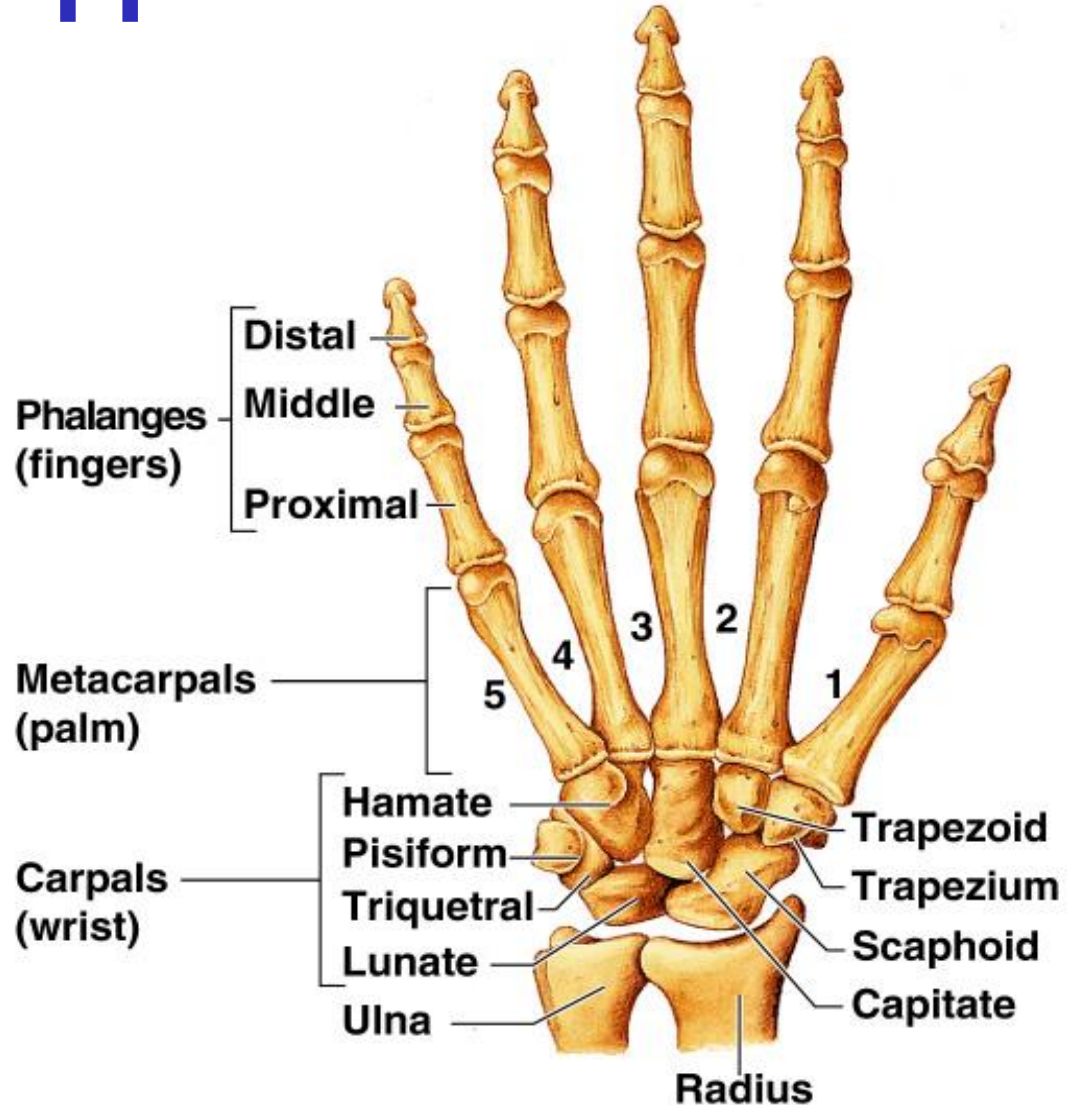
Distal end

- ✓ medially - ulnar notch for head of ulna
- ✓ laterally - radial styloid process



Bones of the Upper Limb...

- The hand
 - ✓ Carpals – wrist
 - ✓ Metacarpals – palm
 - ✓ Phalanges – fingers



The Carpal Bones

- 8 bones, form the wrist (carpus)
- Arranged in two transverse rows of 4 bones

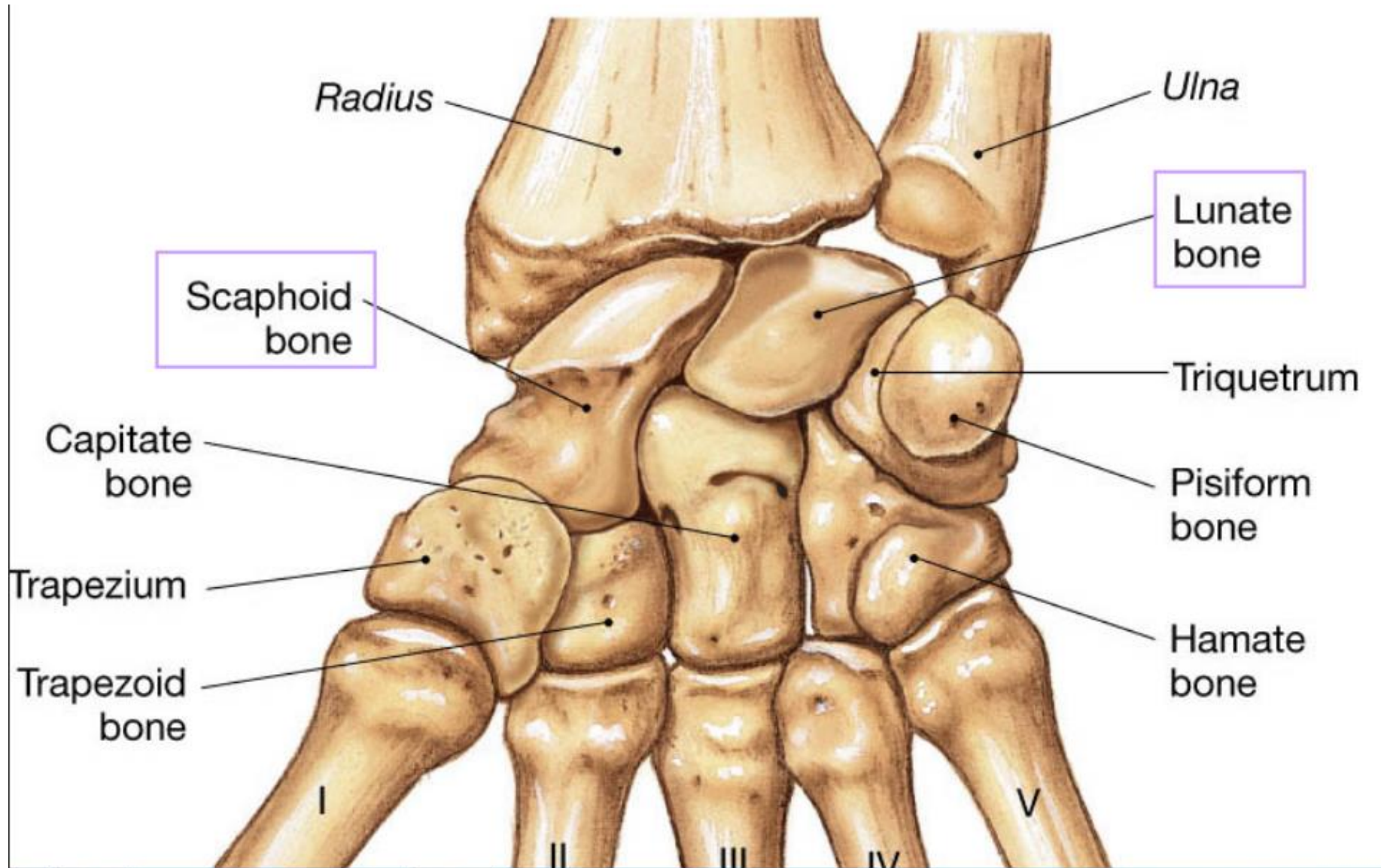
Proximal row

- articulate with radius and articular disc of wrist joint proximally and distal row of carpals distally
- Proximal row from lateral to medial: “**She Like To Play**”
 - ✓ **Scaphoid**: boat-shaped with scaphoid tubercle
 - ✓ **Lunate**: moon-shaped
 - ✓ **Triquetrum**: pyramidal
 - ✓ **Pisiform**: pea-shaped; lies on palmar surface of triquetrum

The Carpal Bones...

▪ Distal row

- Articulate with proximal row of carpals proximally and metacarpals distally
- **Distal row from lateral to medial: “TTCH”**
 - ✓ **Trapezium**: four sided
 - ✓ **Trapezoid**: wedge-shaped
 - ✓ **Capitate**: head like; largest
 - ✓ **Hamate**: wedge-shaped, has hooked process (hook of hamate)

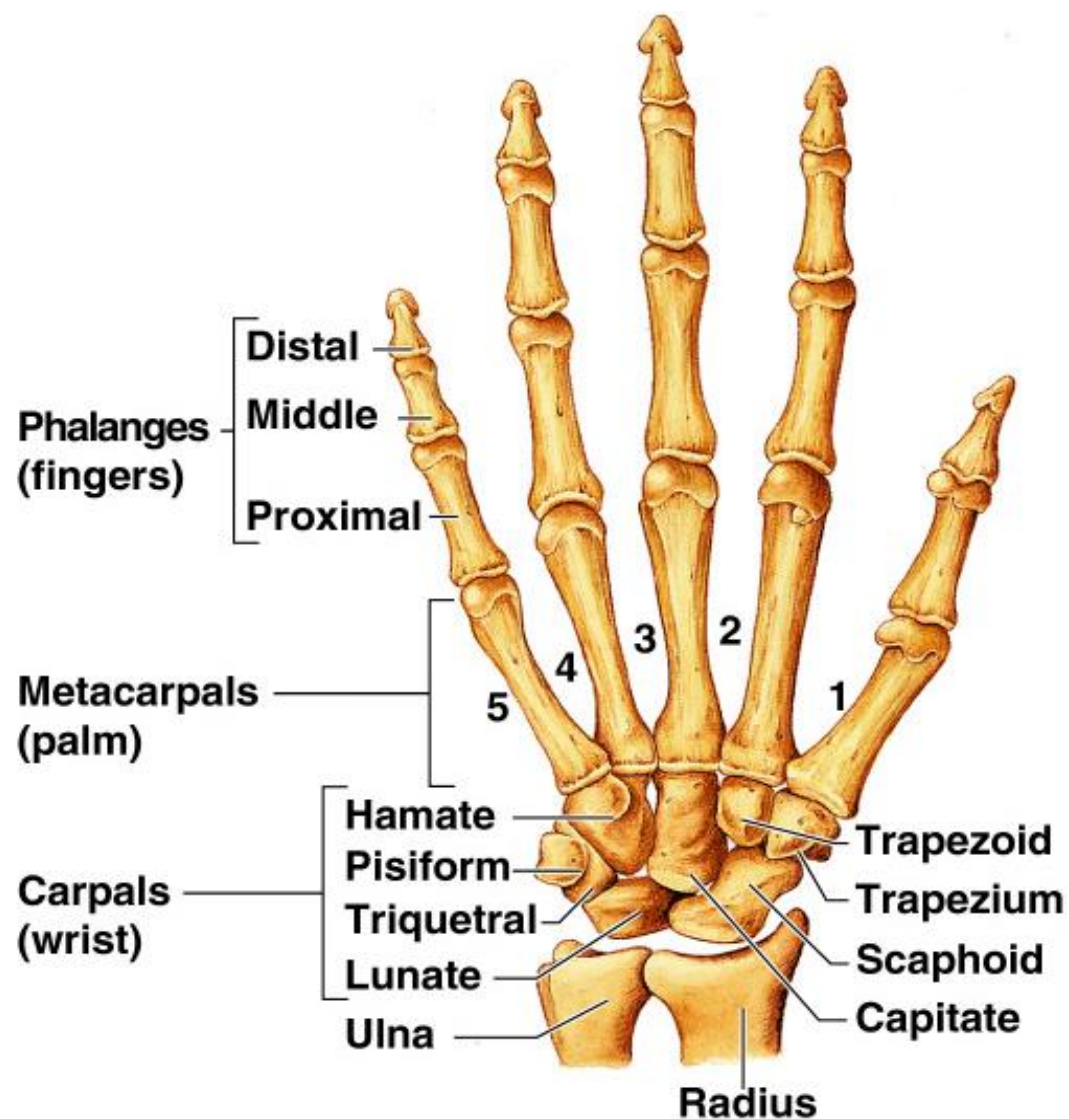


Metacarpals

- Forms the skeleton of the palm
- Composed of five metacarpal bones
- Each metacarpal consists of a base, shaft, and head
- The proximal bases of the metacarpals articulate with the carpal bones
- The distal heads of the metacarpals articulate with the proximal phalanges and form the knuckles

Phalanges

- These are 14 bones
 - Medial 4 digits have proximal, middle & distal phalanx
 - Thumb has only proximal & distal phalanx
 - Each phalanx has base, body & head

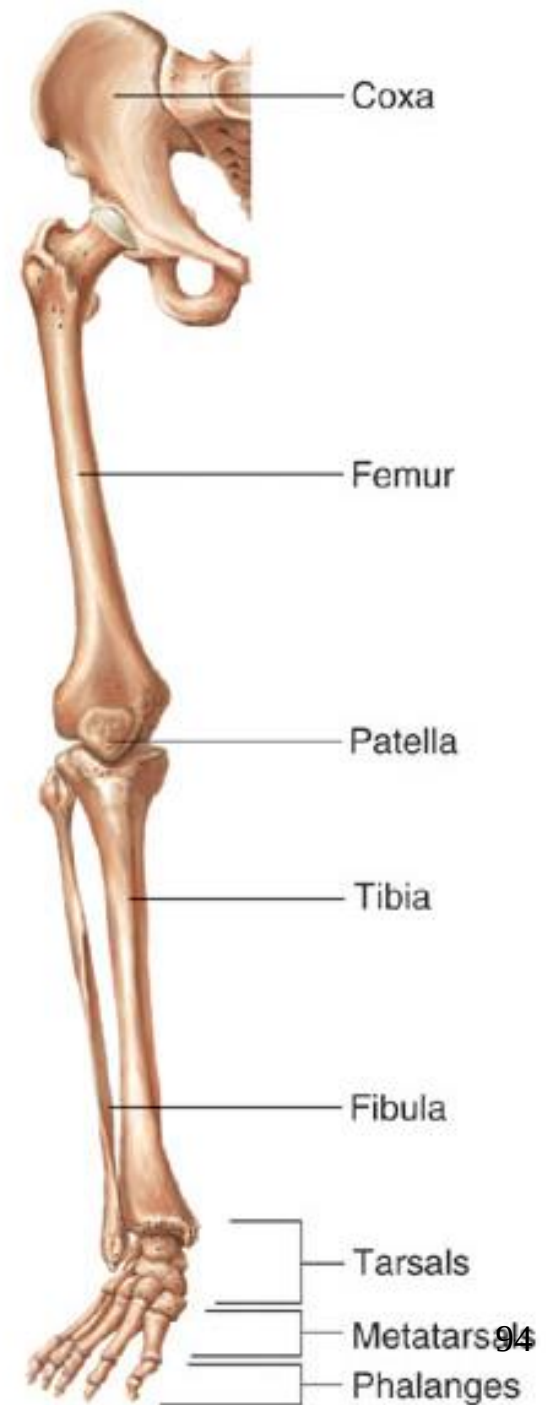


- The distal phalanges are flattened and expanded at their distal ends, which underlie the nail beds

Bones of the lower limbs

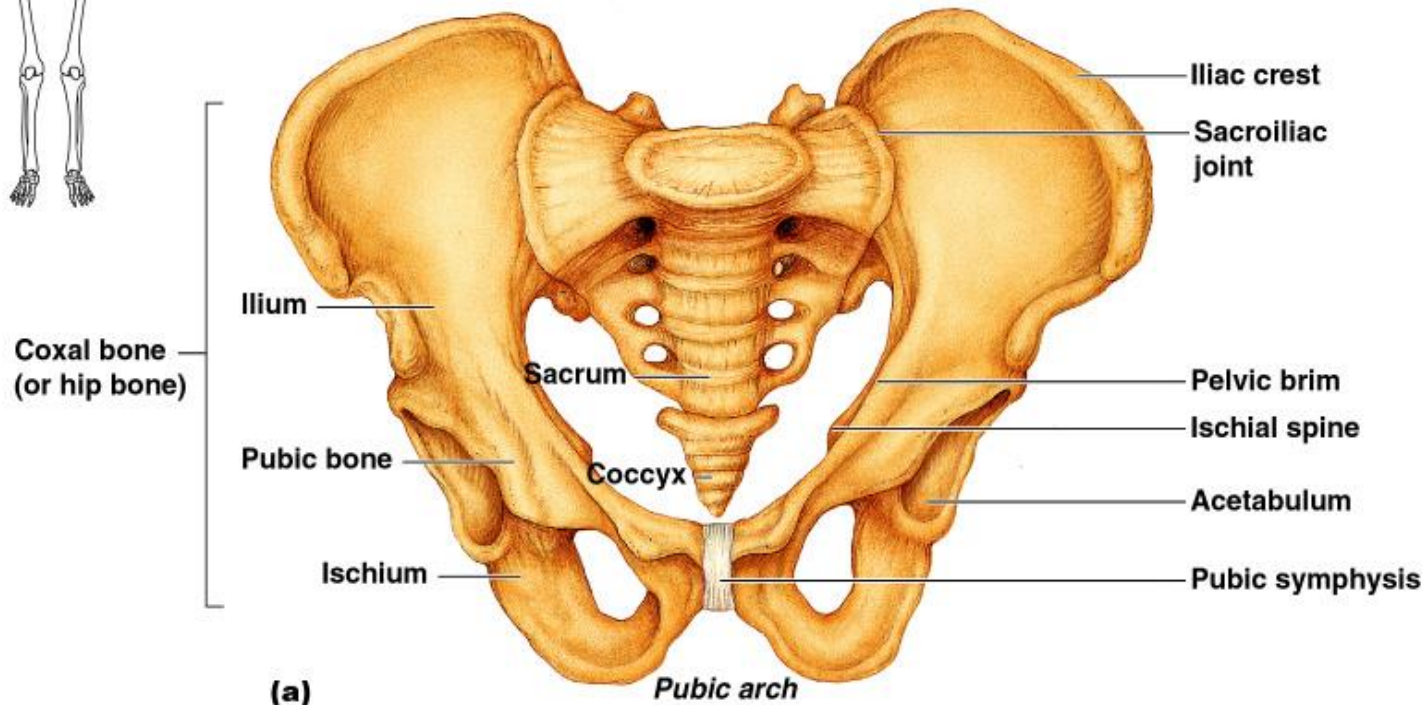
The Lower Limb

- Responsible for transferring the body weight to the ground
- Consists of the following structures:
 - ✓ The hip bones
 - ✓ The femur (thigh)
 - ✓ The patella (kneecap)
 - ✓ The tibia (leg)
 - ✓ The fibula (leg)
 - ✓ Tarsal bones of the ankle
 - ✓ Metatarsal bones and phalanges of the foot



Bones of the Pelvic Girdle

- **Pelvic girdle** is a basin-shaped ring bones → connects vertebral column to the two femurs.
- **Formed by:**
 - ✓ Bones: 2 hip bones, sacrum and coccyx
 - ✓ Joints: 2 sacroiliac joints, pubic symphysis & sacrococcygeal J



Bones of the Pelvic Girdle

Functions

- Bear the weight of the upper body when sitting and standing.
- Transfer it to the lower appendicular skeleton for standing and walking
- Provide attachment for the erectile bodies of the external genitalia and muscles
- Protects several organs
 - ✓ Reproductive organs
 - ✓ Urinary bladder
 - ✓ Part of the large intestine

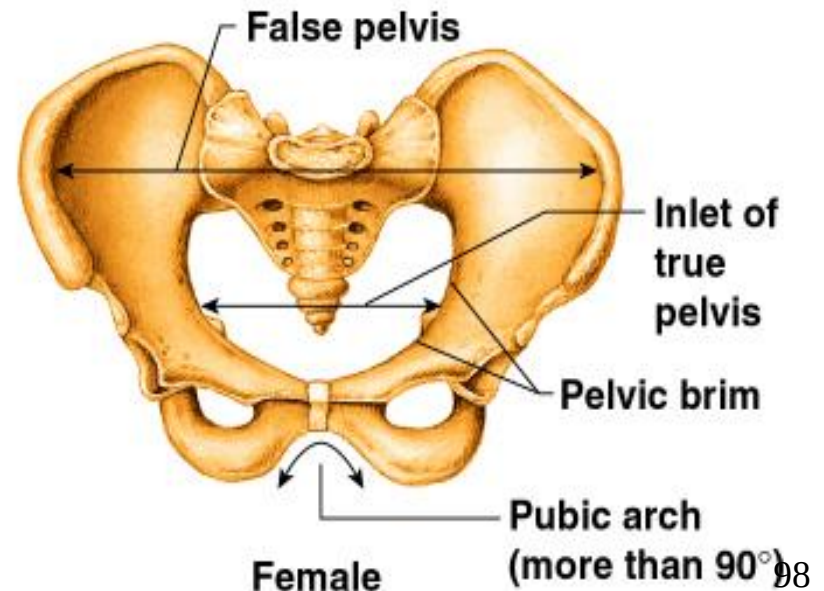
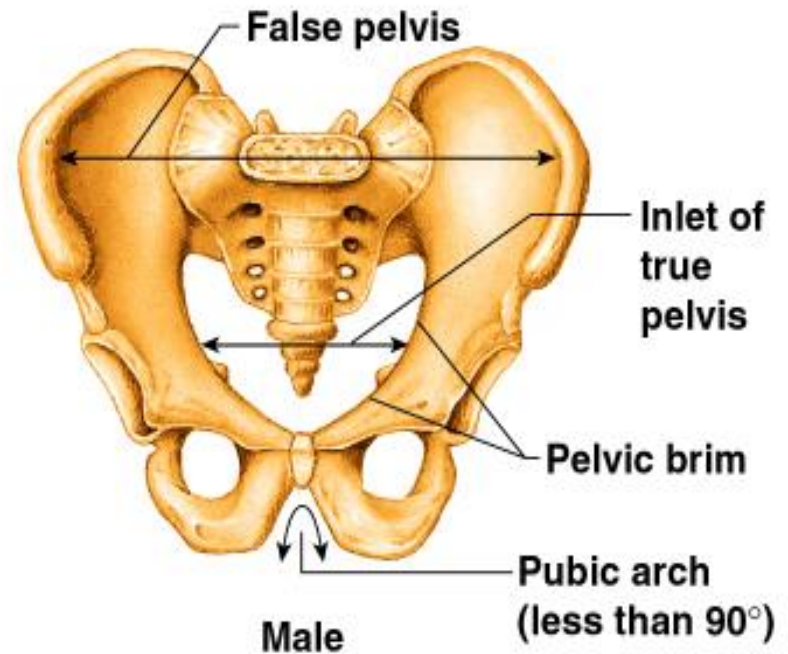
Gender Differences of the Pelvis

Bony Pelvis	Male (♂)	Female (♀)
General structure	Thick and heavy	Thin and light
Greater pelvis (false pelvis)	Deep	Shallow
Lesser pelvis (true pelvis)	Narrow and deep, tapering	Wide and shallow, cylindrical
Pelvic inlet (superior pelvic aperture)	Heart-shaped, narrow	Oval and rounded; wide
Pelvic outlet (inferior pelvic aperture)	Comparatively small	Comparatively large
Pubic arch and subpubic angle	Narrow (<70°)	Wide (>80°)
Obturator foramen	Round	Oval
Acetabulum	Large	Small

... continued

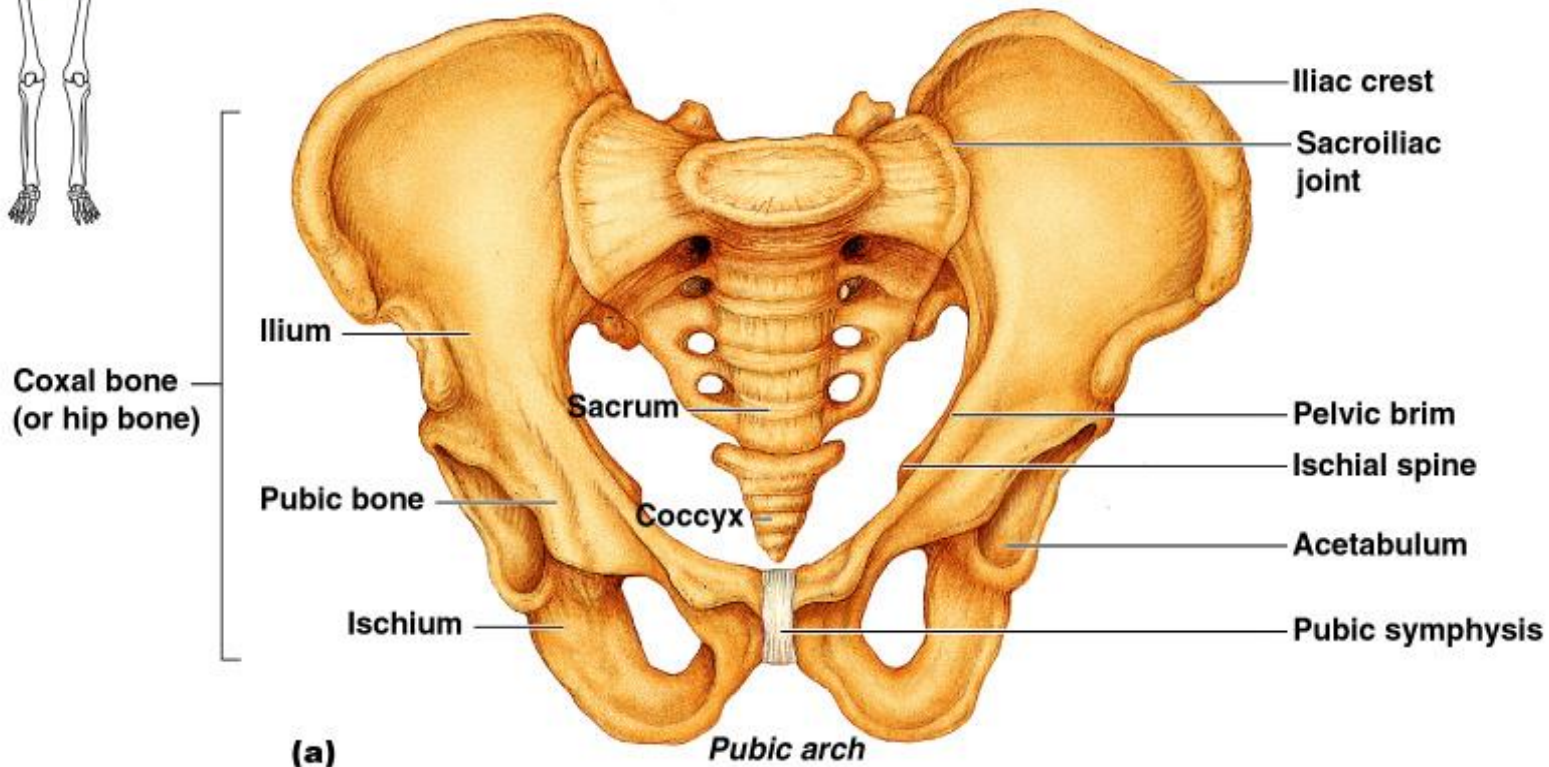


Bony Pelvis
General structure
Greater pelvis (false pelvis)
Lesser pelvis (true pelvis)
Pelvic inlet (superior pelvic aperture)
Pelvic outlet (inferior pelvic aperture)
Pubic arch and subpubic angle
Obturator foramen
Acetabulum



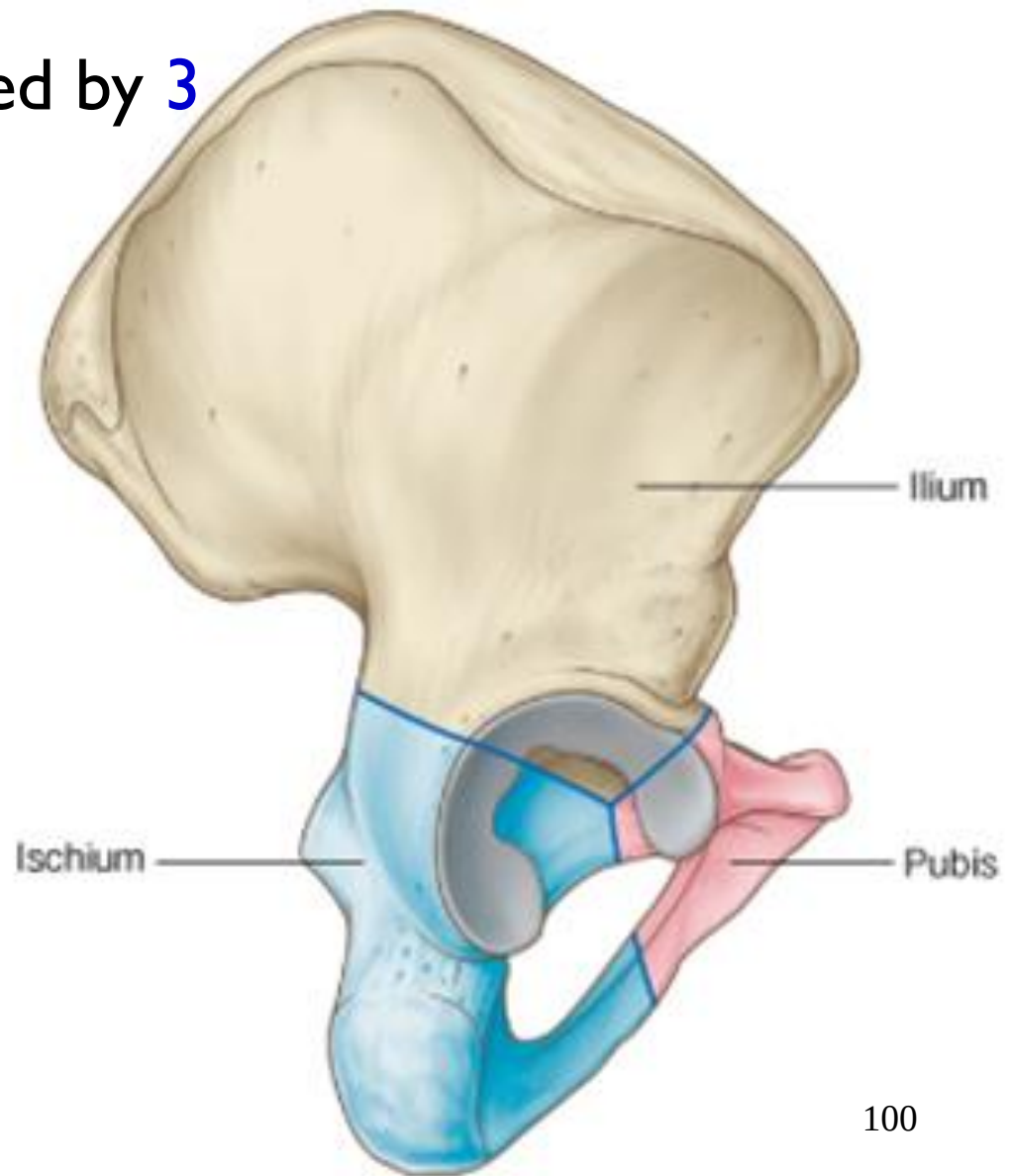
HIP BONES

- Two in number; Rt & Lt
- Joined at the **pubic symphysis** anteriorly and to the sacrum posteriorly at the **sacroiliac joints** to form a bony ring, the pelvic girdle



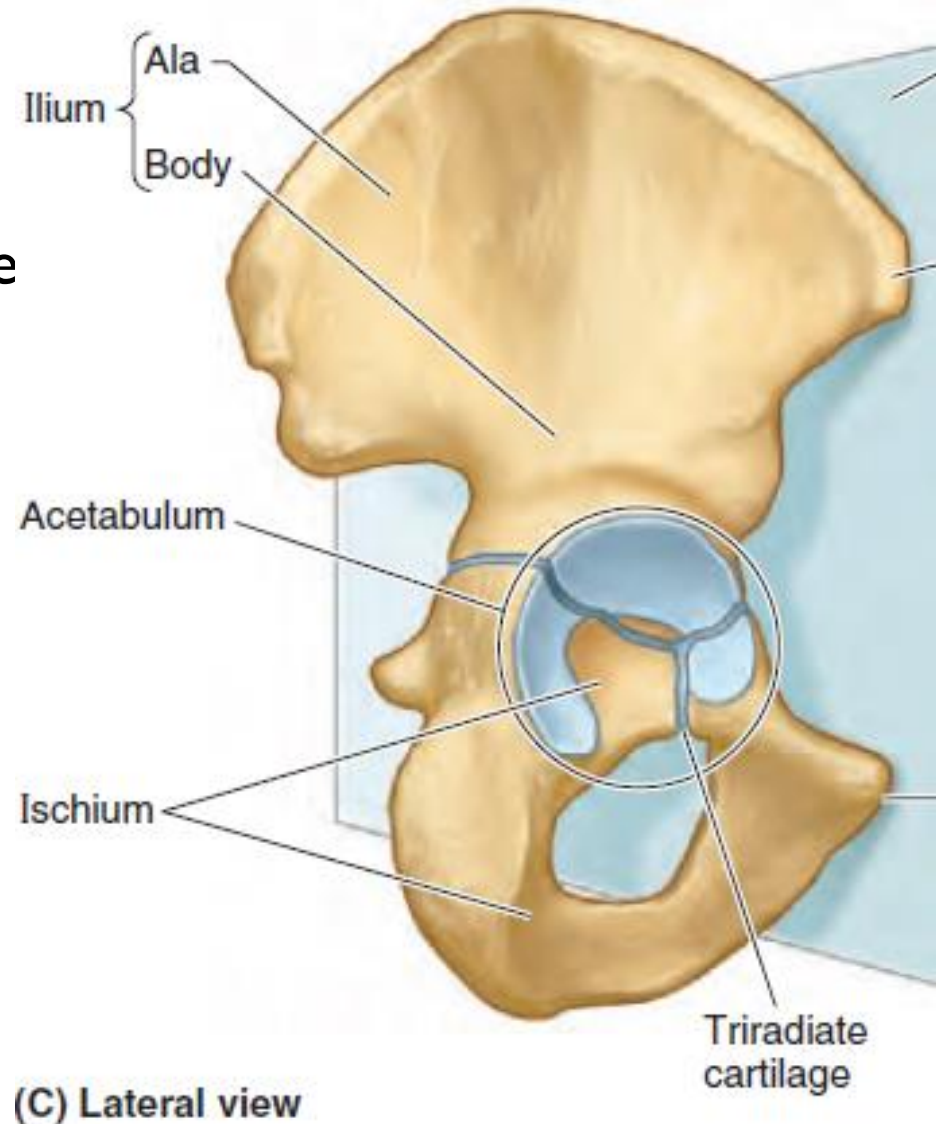
HIP BONES

- Each hip bone is formed by **3 bones fusing at the acetabulum**
- The 3 bones are:
 - Ilium,
 - Ischium &
 - Pubis



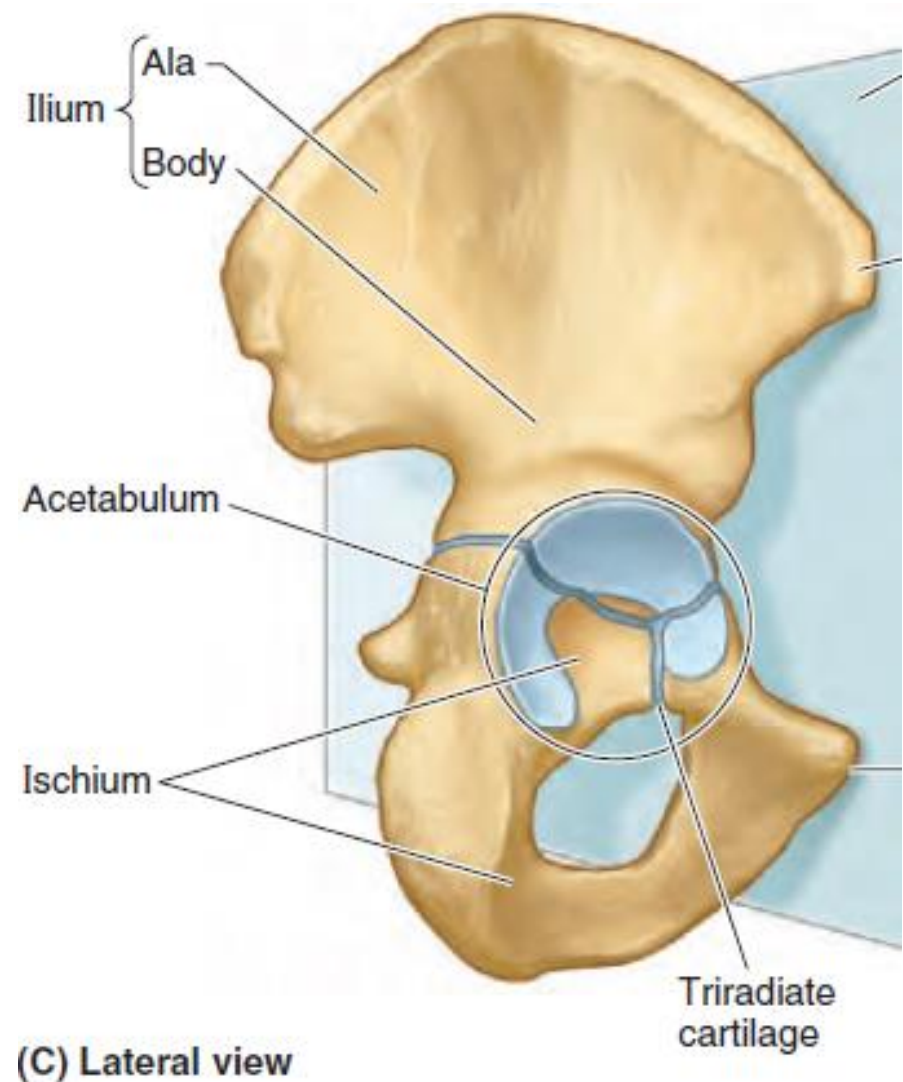
Ilium

- The largest & uppermost of the three
- Flattened, fan-shaped part of the hip bone
- Located superior to the acetabulum
- **Parts**
 - ✓ Body
 - ✓ Ala (wing)
 - ✓ Iliac crest

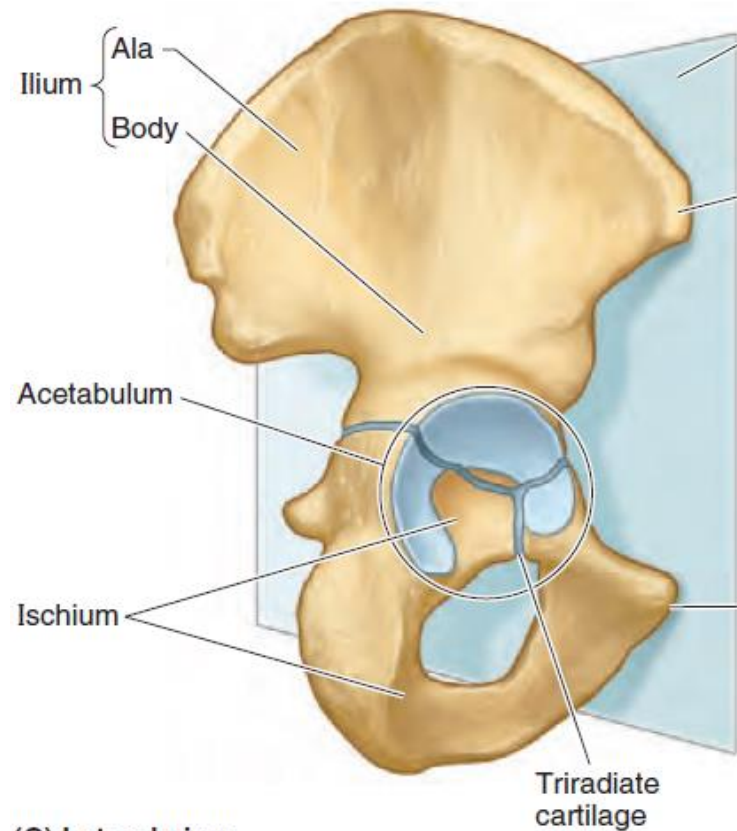


Landmarks:

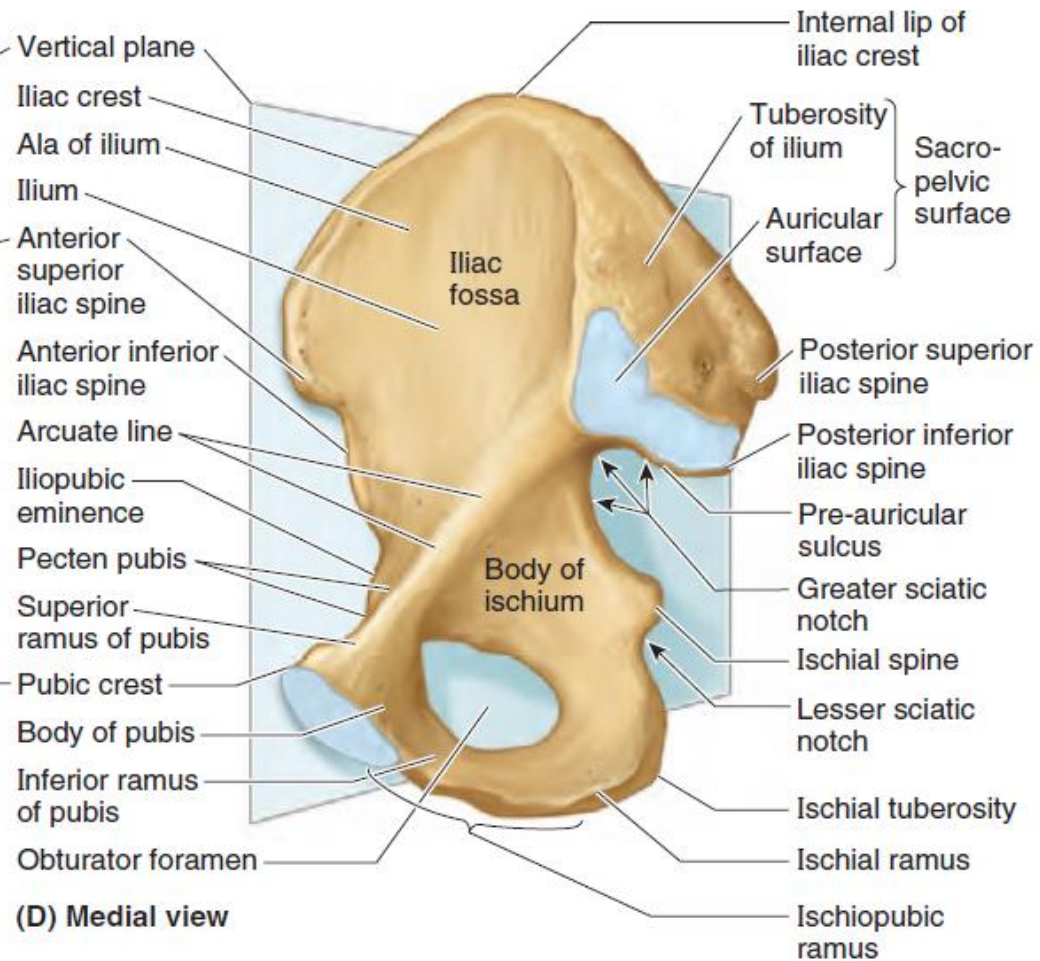
- has 4 spines and notches
 - ASIS
 - AIIS
 - PSIS
 - PIIS
 - Greater sciatic notch



(A) Anterior view



(C) Lateral view



(D) Medial view

Ishium

- ✓ lies posteroinferior
- ✓ ischial spine inferior to greater sciatic notch
- ✓ Ischial tuberosity, the part we sit on
- Has a body and a ramus
 - **Body**
 - ✓ forms the posterior part of the acetabulum
 - **Ramus**
 - ✓ fuses with the inferior ramus of pubis

Landmarks

- **Ischial tuberosity**

- ✓ large posteroinferior protuberance of the ischium
- ✓ supports body during sitting

- **Ischial spine**

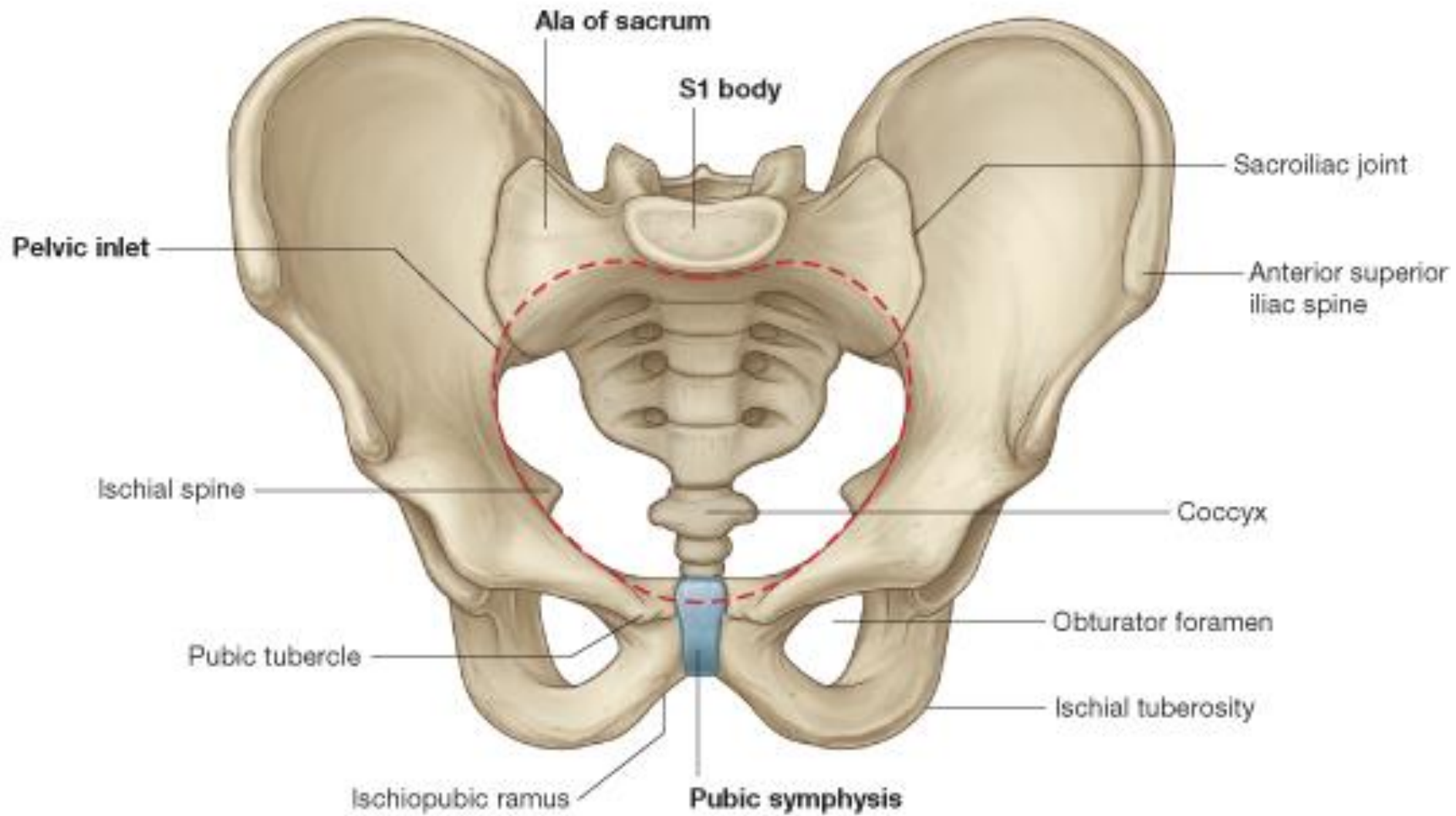
- ✓ small pointed posterior projection near the junction of the ramus and body

Pubis:

- Anteromedial part of hip bone
- Forms anterior part of the acetabulum
- Angulated bone; has two rami (inferior & superior) and body

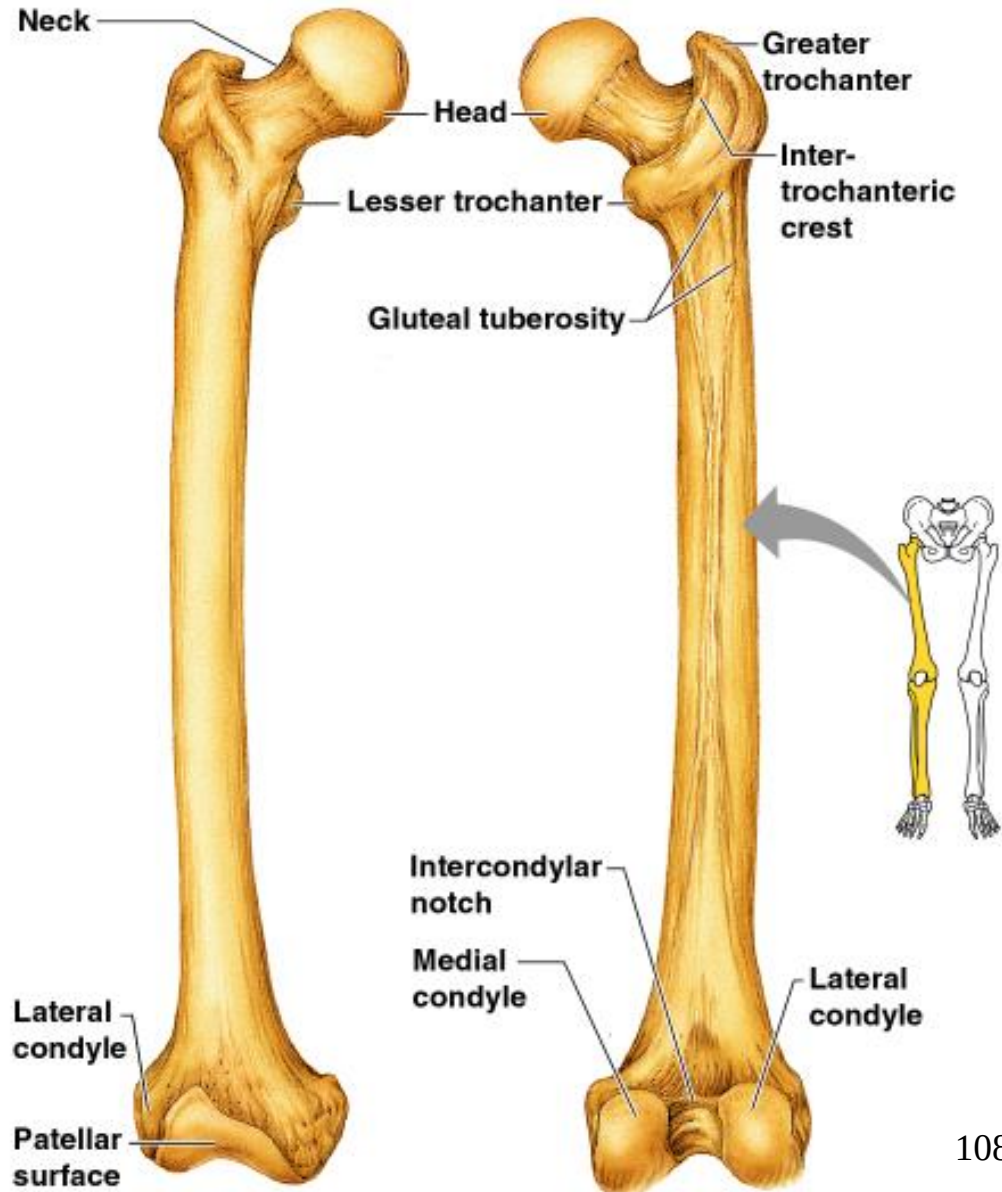
Landmarks

- pecten pubis or pectineal line
 - ➔ oblique ridge on superior pubic ramus on its superior aspect
- pubic crest
 - ➔ thickening on the anterior part of the body of the pubis
 - ➔ ends laterally as a swelling - pubic tubercle



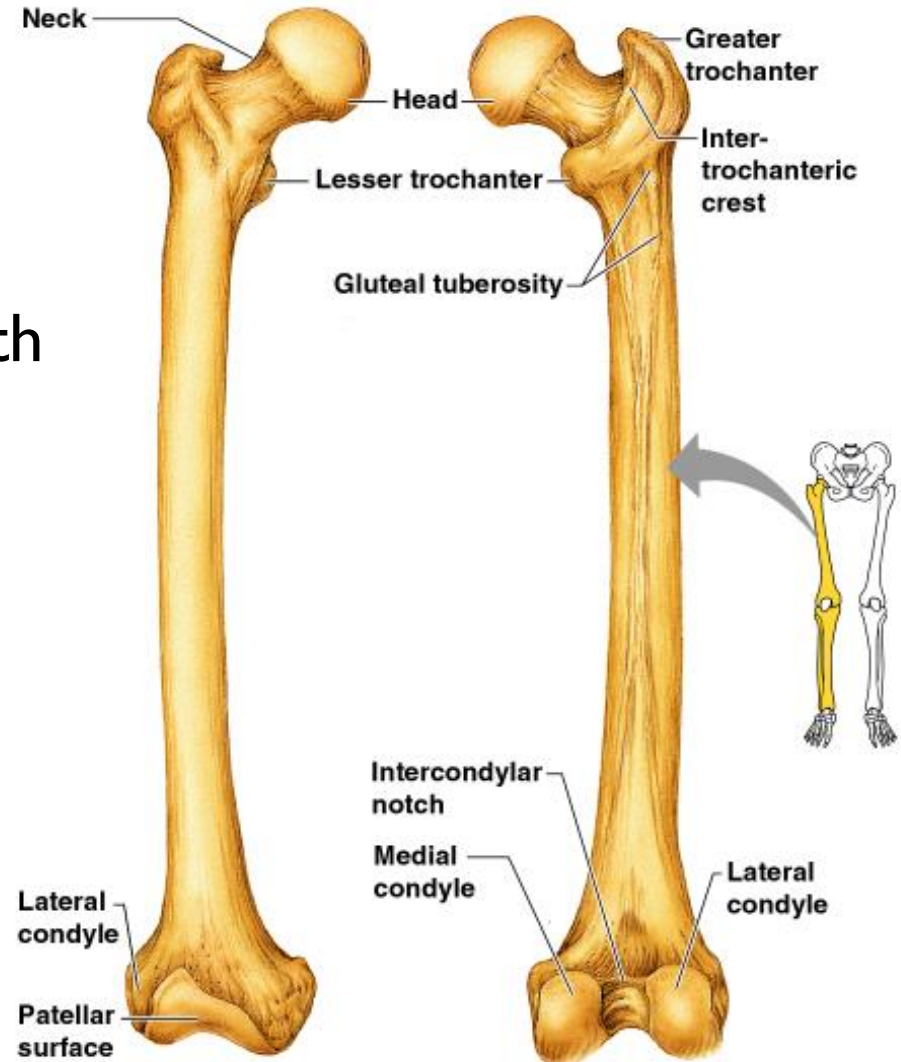
Bones of the Lower Limbs

- The thigh has one bone
 - ✓ Femur



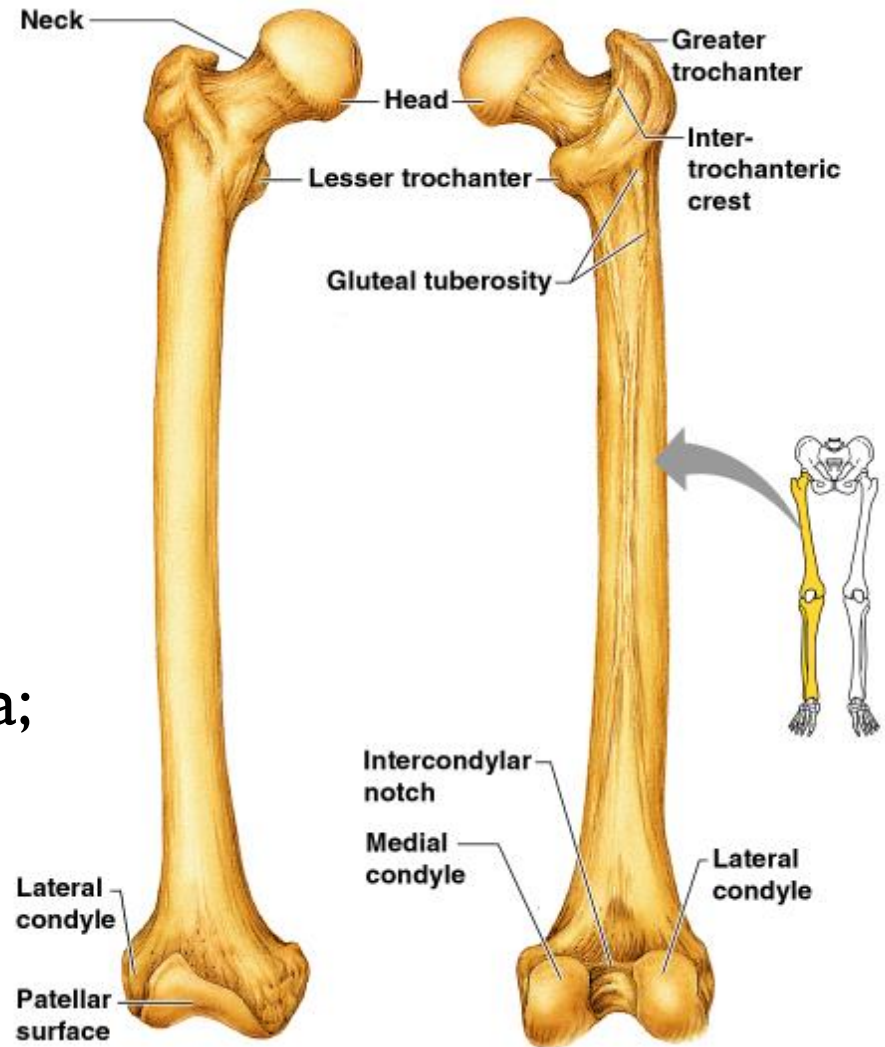
Femur

- longest, heaviest & strongest bone in the body
- proximal head articulates with acetabulum
- Fovea capitis femoris for ligament attachment
- The neck is most commonly fractured site



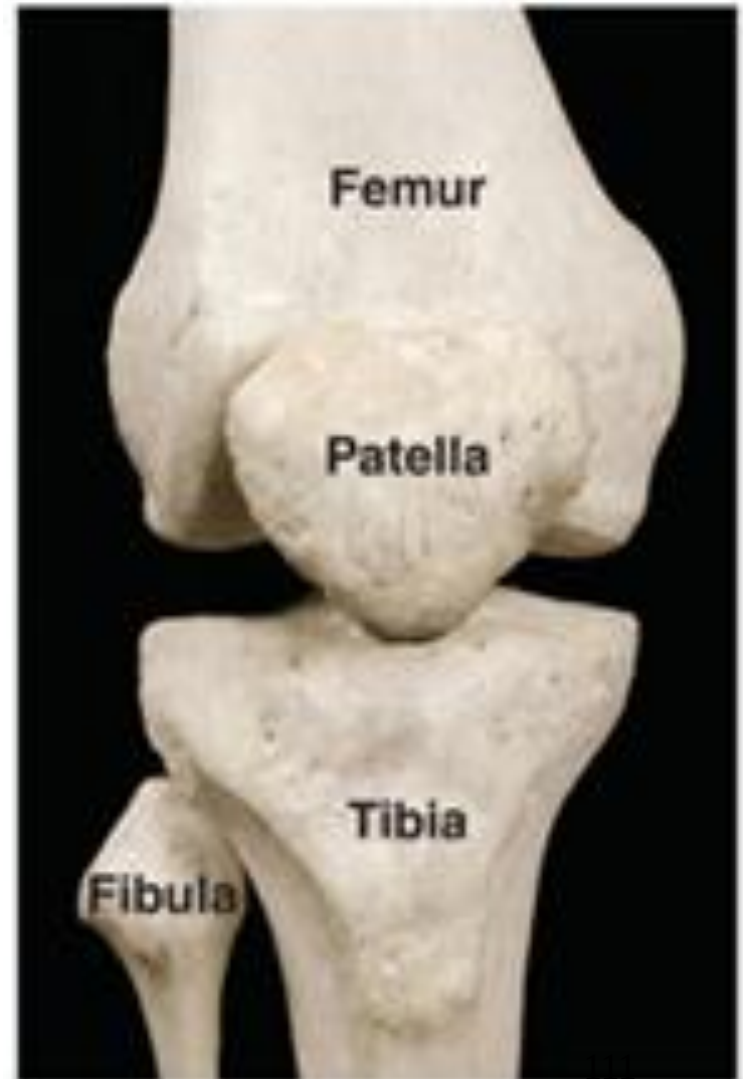
Femur

- has trochanter (greater & lesser) and intertrochanteric line in b/n
- vertical ridge posterior called **linea aspera**
- distal end articulates with tibia; and has medial & lateral condyles



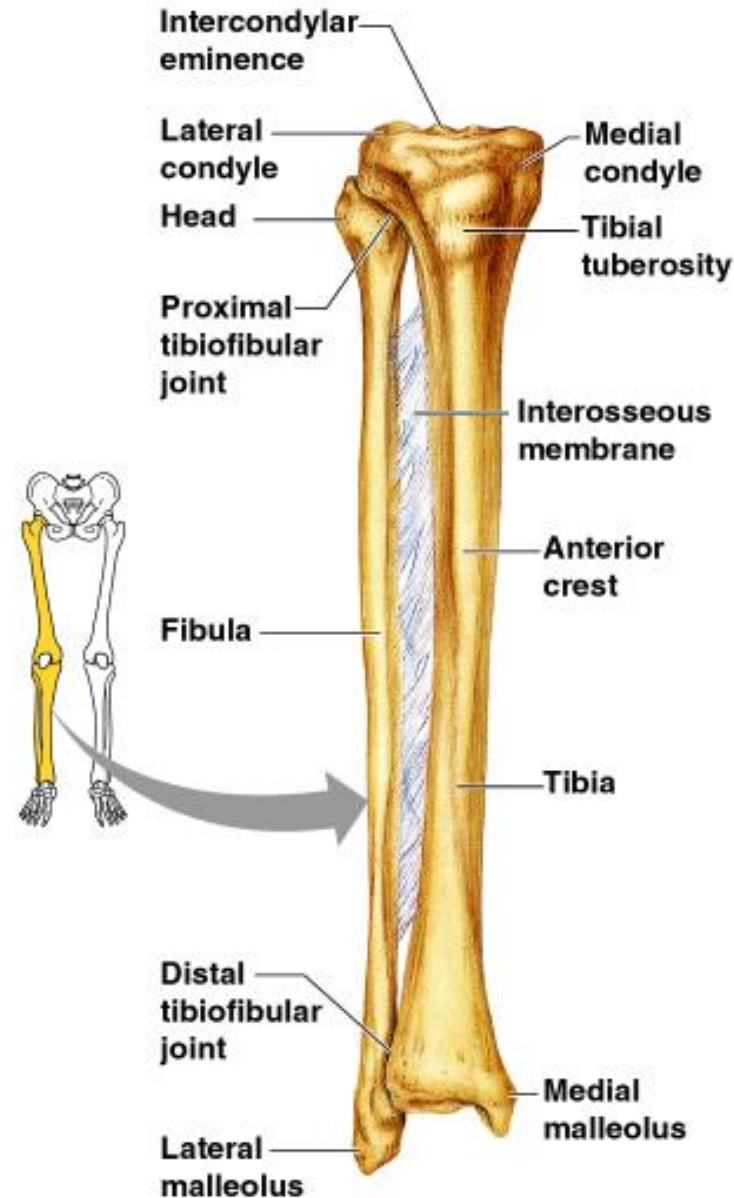
Patella

- Triangular, sesamoid bone lie on the anterior side of knee
- Has base, apex & articular facets
- **Function** to protect knee joint & strengthen the **quadriceps tendon**



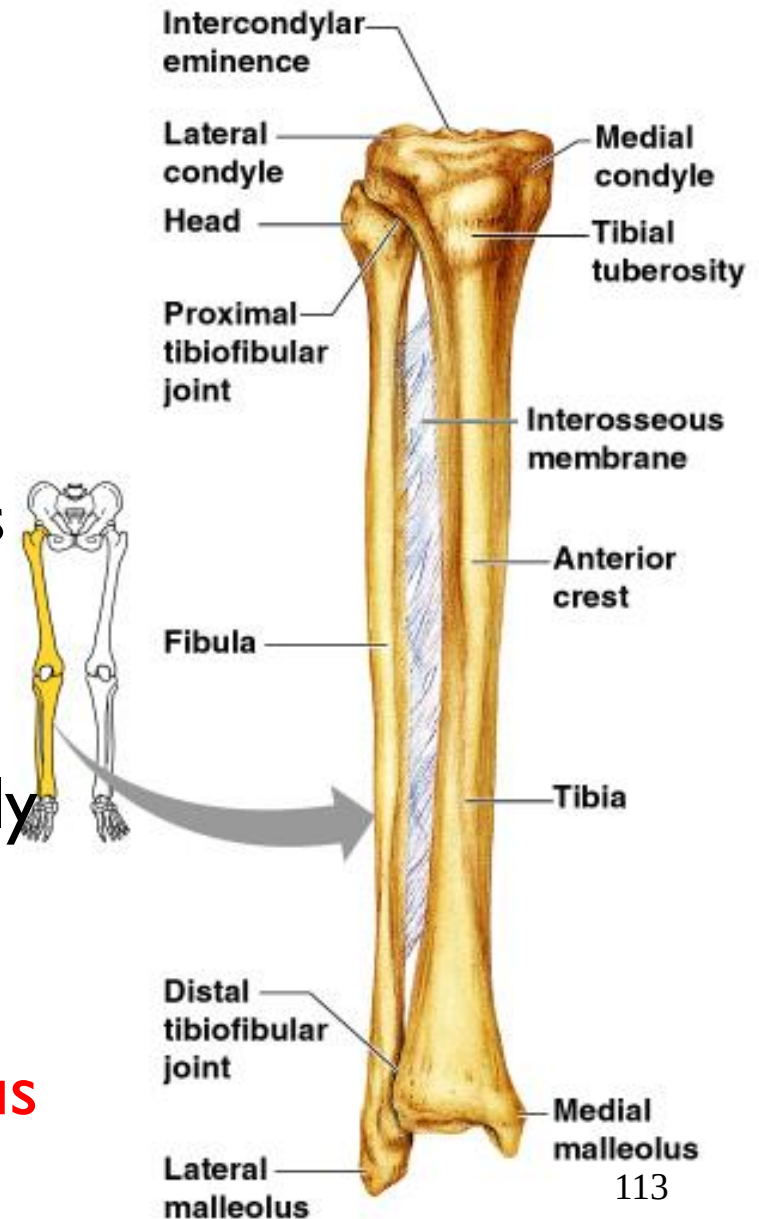
Bones of the Lower Limbs...

- The leg has two bones
 - ✓ Tibia
 - ✓ Fibula



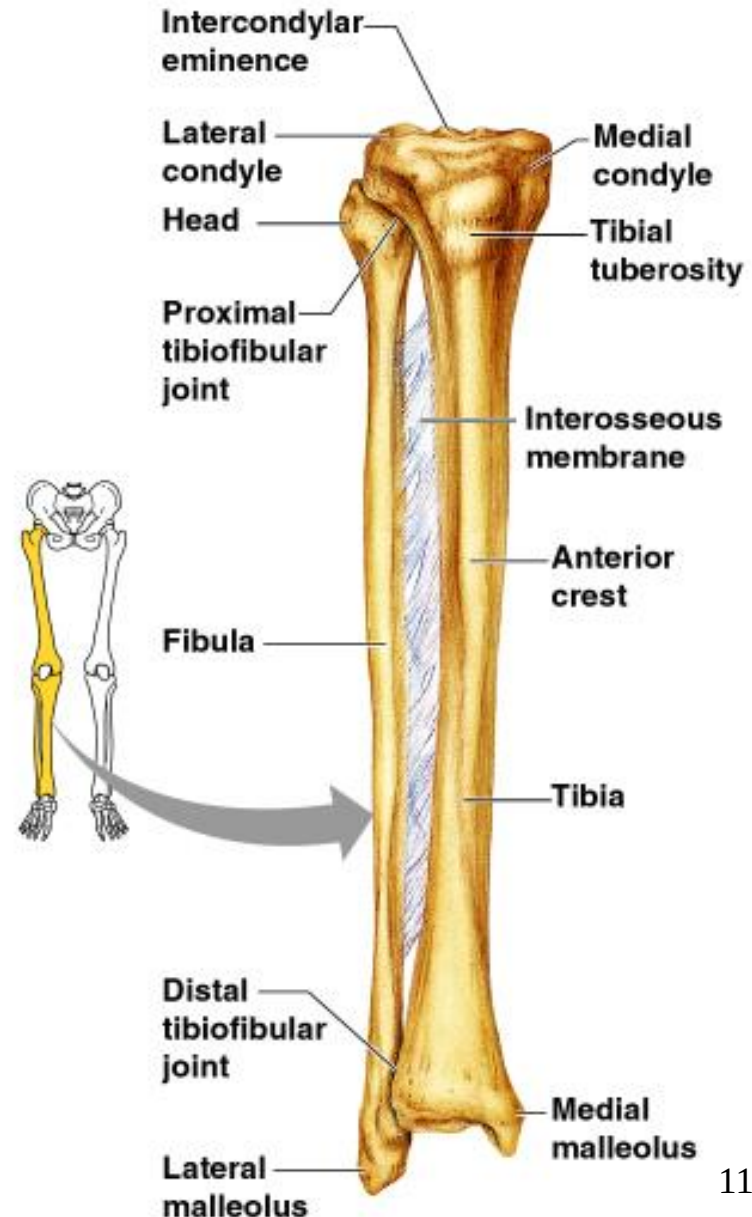
Tibia

- Articulate proximally with femur at knee joint
- Distally articulate with the talus of ankle
- Has medial & lateral condyles & intercondylar eminence in b/n
- Has **tibial tuberosity** anteriorly where patellar ligament attaches
- Distally it has **medial malleolus**



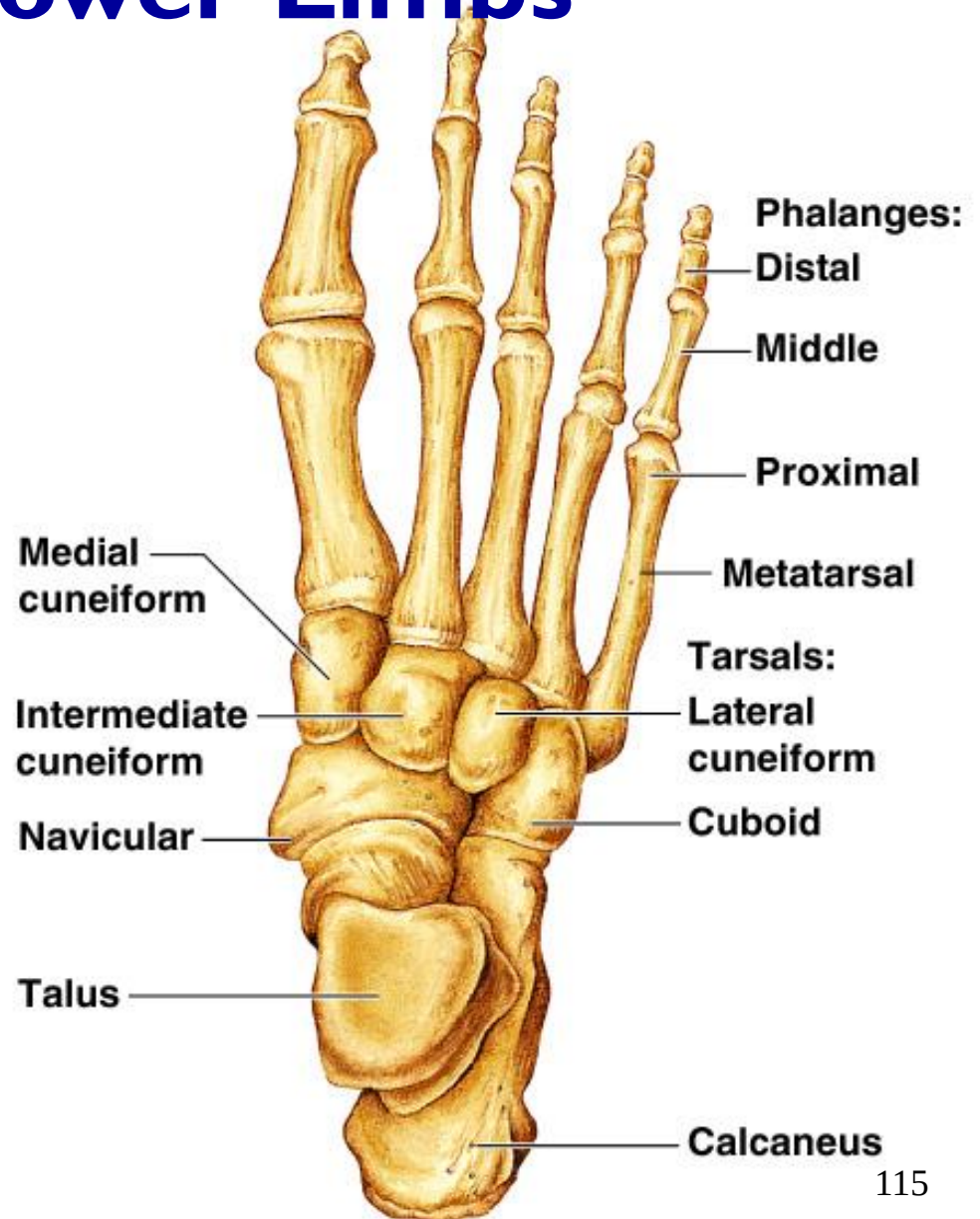
Fibula

- Long, narrow bone on the lateral side of leg
- Important for muscle attachment than support
- Has head, neck & lateral malleolus



Bones of the Lower Limbs

- The foot
 - ✓ Tarsus – ankle
 - ✓ Metatarsals – sole
 - ✓ Phalanges – toes



Tarsal Bones:- 7 bones, form ankle joint

- talus, calcaneus, navicular, 3 cuneiforms & cuboid

Metatarsal Bones:- 5 bones, form part of foot

- each bone has base, body & head
- forms tarsometatarsal & metatarsophalangeal joint

Phalanges:- 14 bones

- lateral 4 digits have proximal, middle & distal phalanx
- great toe (hallux) has only proximal & distal phalanx
- each phalanx has base, body & head

Muscular System

Muscular system

- Consists of all the muscles of the body
- Skeletal muscles
 - ✓ constitute the great majority of the named muscles.
- Other types of muscle tissue
 - ✓ constitute a few named muscles. e.g., ciliary and detrusor muscles, & arrector muscles of hairs
 - ✓ form important components of the organs of other systems, including the cardiovascular, alimentary, genitourinary, integumentary, and visual systems.

Types of Muscle (Muscle Tissue)

- Muscle cells
 - ✓ often called **muscle fibers** because they are long and narrow when relaxed
 - ✓ specialized contractile cells
 - ✓ **organized into tissues** that move body parts or temporarily alter the shape of internal organs
- Associated connective tissue
 - ✓ conveys neurovascular structures to the muscle cells as it binds them into bundles or fascicles.

Types of Muscles

- Classification of muscles based on the presence or absence of cross striations, location & functions
- Based on striations muscles could be:
Striated or Non-striated
- Based on location can be classified as:
Skeletal, Cardiac & Smooth muscles
- Based on function:
Voluntary VS Involuntary



Cardiac muscle cell



Skeletal muscle cell



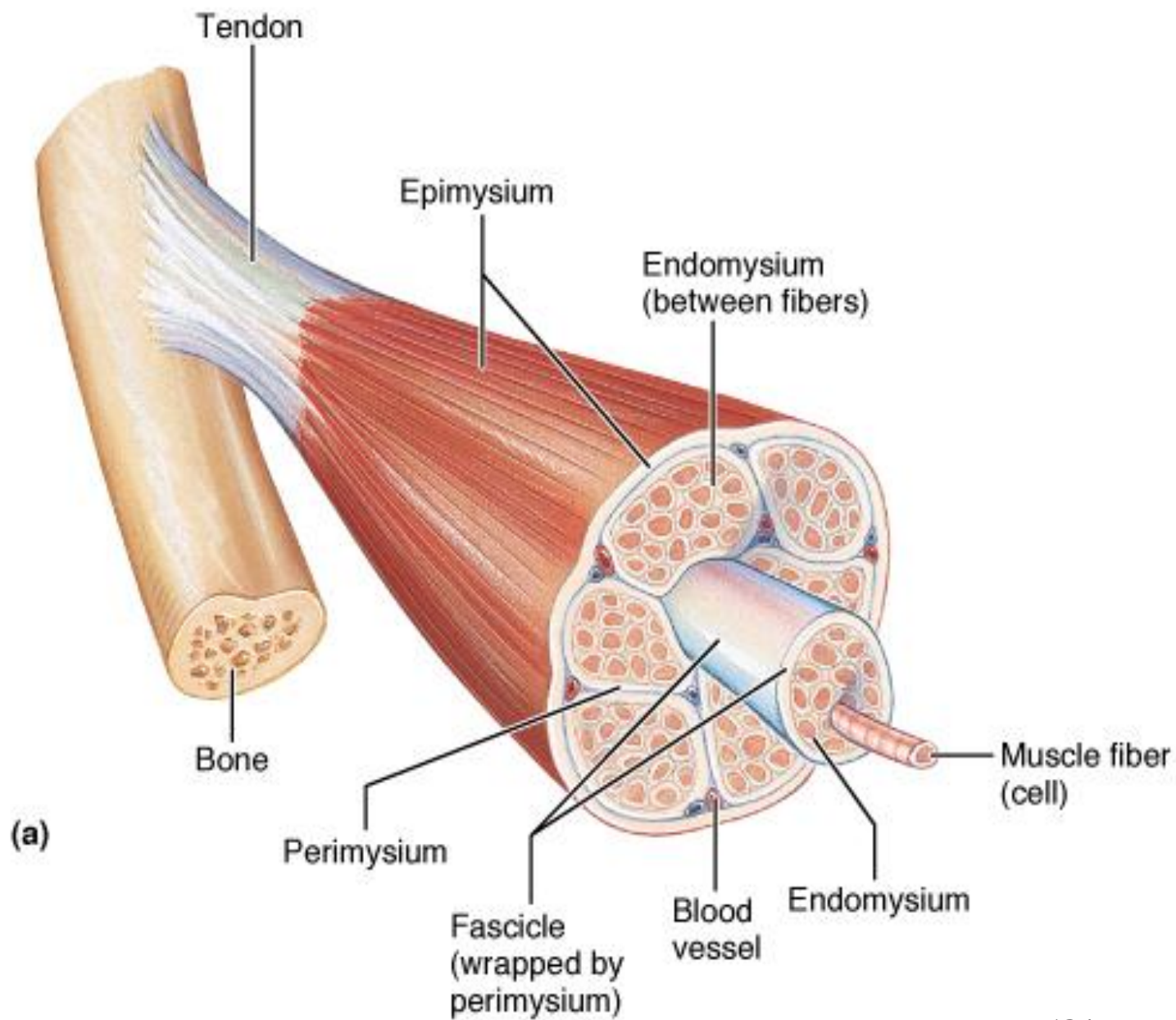
Smooth muscle cell

Skeletal Muscles

- ✓ **Features**
- ✓ **Naming of Muscles**

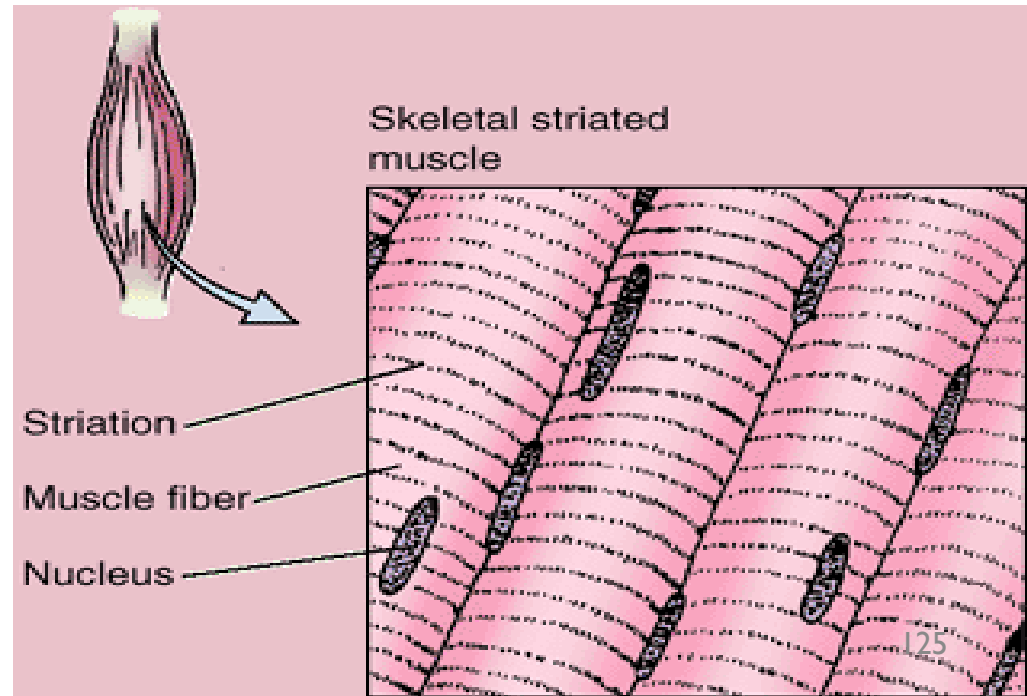
Connective Tissue Investment of Muscles

- Cells are surrounded and bundled by connective tissue (CT)
- ✓ **Endomysium**:- fine sheath of CT covering the individual muscle fibers
- ✓ **Perimysium**:- binds groups of fibers together into bundles called *fasciculi*
- ✓ **Epimysium**:- covers the entire muscle is continuous with tendon.
 - Separates the muscle from surrounding tissue
- **Fascia** - wrapping around epimysium



Skeletal Muscles

- Have fleshy, reddish contractile portions (which is striated) with white non-contractile portions (tendons)
 - ✓ Tendons composed mainly of organized collagen bundles, that provide a means of attachment.
- Therefore, the length a muscle is the distance between its attachments

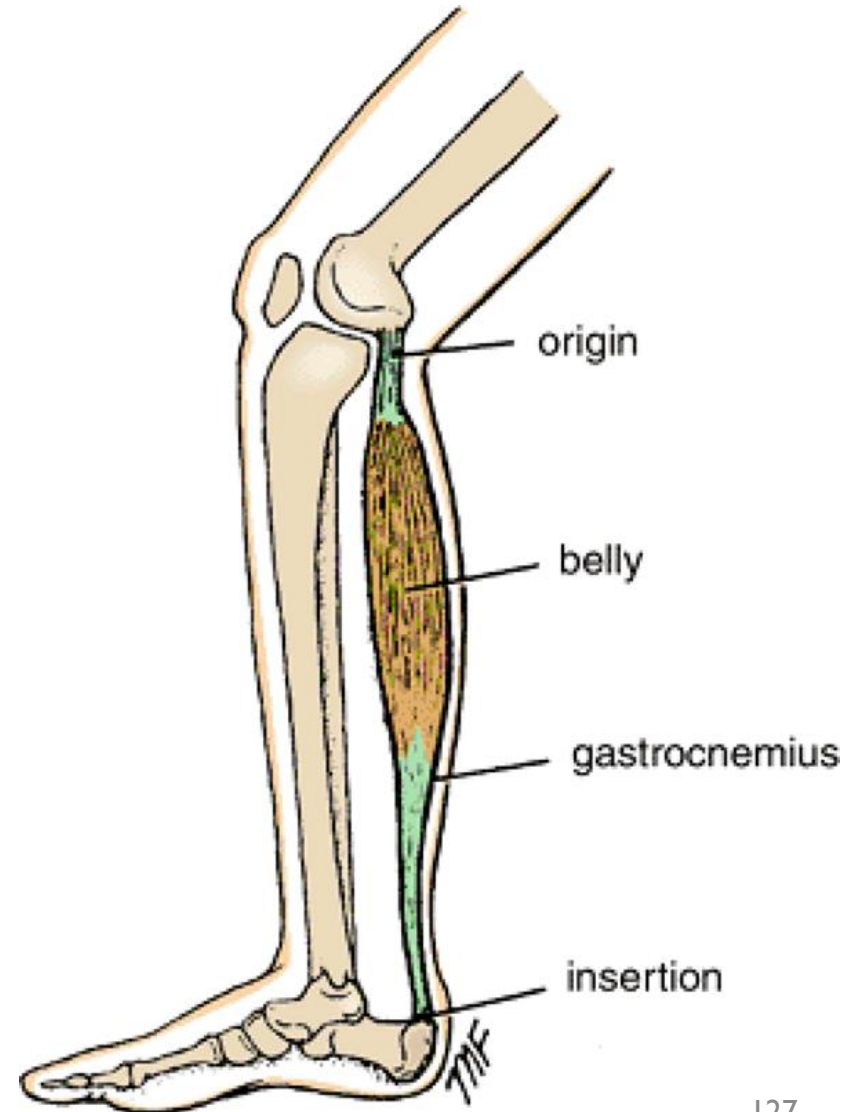


Skeletal Muscles

- Most skeletal muscles are attached directly or indirectly
 - ✓ to bones, cartilages, ligaments, or fascias or
 - ✓ to some combination of these structures
- Some muscles are attached to
 - ✓ Organs e.g. the eyeball,
 - ✓ Skin e.g. facial muscles, and
 - ✓ Mucous membranes e.g. intrinsic tongue muscles

Skeletal Muscles...

- Produce active movements because they usually have at least two attachments
 - a. The more fixed (that moves least) = **proximal attachment** = **origin** of a muscle
 - b. The more movable (that moves most) = **distal attachment** = **insertion** of a muscle



Skeletal Muscles...

- Attached to bones, cartilages, ligaments, fascia or skin by
 - I. **Rounded cords** of fibrous tissue = **tendons** (transfer force of contraction from muscles to bones)
e.g. tendon of limb muscles
 - II. **Flat sheets** of fibrous tissue = **aponeurosis**
e.g. aponeurosis of abdominal muscles
- Have fleshy part = **belly** of a muscle
e.g. belly of biceps brachii muscle

Naming of Skeletal Muscles

- Names incorporate **appearance**, **location**, **function**, **orientation**, and **unusual features**
 - ➔ Names of muscles provide clues to location, orientation, or action (function)
- **Location of the muscle**
 - e.g. Intercostal, Temporalis*
- **Shape of the muscle**
 - e.g. Deltoid; Trapezius*
- **Relative size of the muscle**
 - e.g. Maximus, medius, minimus*
- **Direction of muscle fibers**
 - e.g. Rectus (straight), transverse; oblique*

- **Number of origins**

e.g. Biceps; triceps; quadriceps

- **Attachment (origin and insertion)**

e.g. Sternocleidomastoid muscle,

- **Action (function) of the muscle**

e.g. Flexor, extensor, adductor, abductor

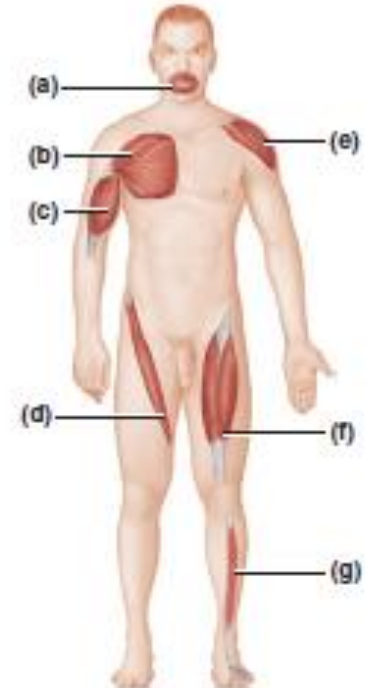
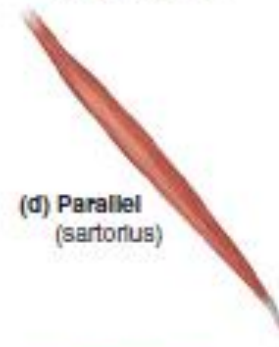
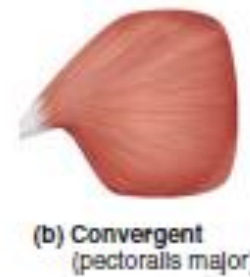
✓ The abductor digiti minimi muscle, for example, abducts the little finger

- **Relative position**

e.g. lateral, medial, anterior, & posterior

Muscle Architecture

- The architecture and shape of muscles vary
- **Fusiform muscles**
 - thick in middle & tapered at ends
 - biceps brachii m.
- **Convergent muscle**
 - broad at origin and tapering to a narrower insertion
- **Parallel muscles**
 - Parallel fascicles
 - E.g. Sartorius, Rectus abdominis m.



Muscle Architecture

- **Pennate** - are feather-like (L. pennatus, feather) in the arrangement of their fascicles
- Occur as

Unipennate: Fascicles are arranged on only one side of the tendon

→ e.g. *Extensor digitorum longus muscle (EDL)*

Bipennate → Fascicles are arranged on both sides of centrally positioned tendons

→ E.g. *Rectus femoris muscle*

Multipennate: → Fascicles attach obliquely from many directions to several tendons. E.g. *Deltoid muscle*

Three pennate muscles

a Extensor digitorum muscle



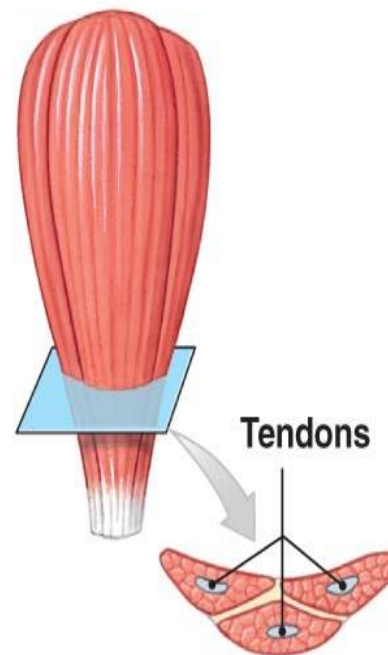
If all the muscle fibers are on the same side of the tendon, the pennate muscle is **unipennate**.

b Rectus femoris muscle

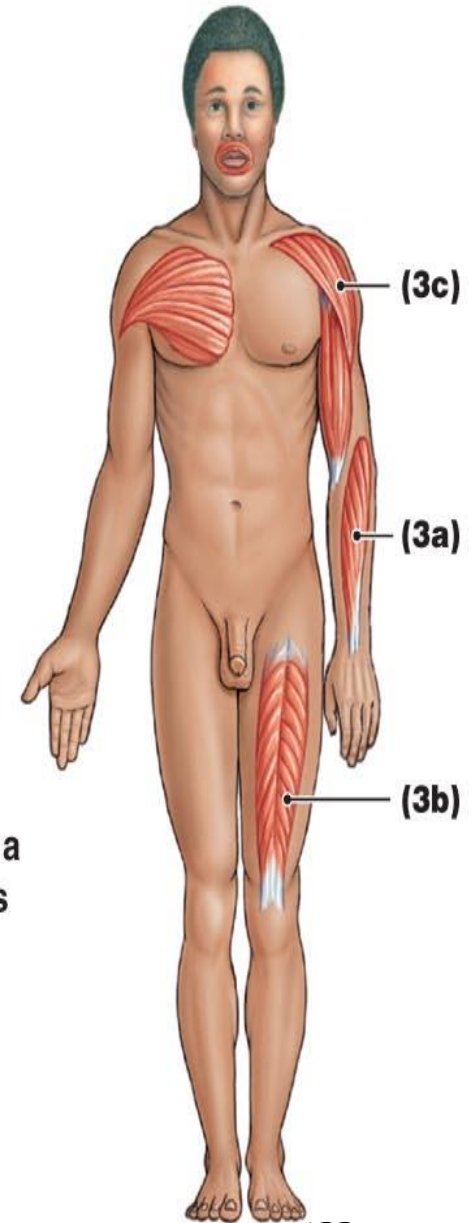


If a pennate muscle has fibers on both sides of the tendon, it is called **bipennate**.

c Deltoid muscle



If the tendon branches within a pennate muscle, the muscle is said to be **multipennate**.



Muscle Architecture

■ Circular(sphincter) muscles

- act as sphincters
- ring around body opening

E.g; Orbicularis oris
Orbicularis oculi



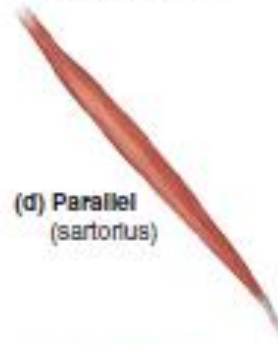
(b) Convergent
(pectoralis major)



(a) Circular
(orbicularis oris)



(c) Fusiform
(biceps brachii)



(d) Parallel
(sartorius)



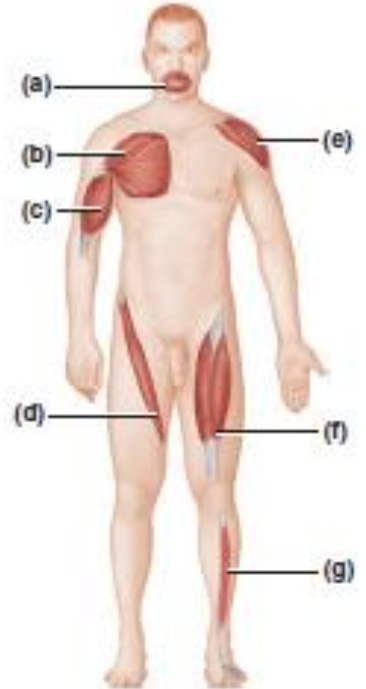
(e) Multipennate
(deltoid)



(f) Bipennate
(rectus femoris)



(g) Unipennate
(extensor digitorum longus)



Contraction of Muscles

- A muscle fiber is the structural unit of a muscle
- A motor unit, consisting of a motor neuron & the muscle fibers it controls, is the functional unit of a muscle.

✓ When a *motor neuron* is stimulated, it initiates an impulse that causes simultaneous contraction of the muscle fibers.

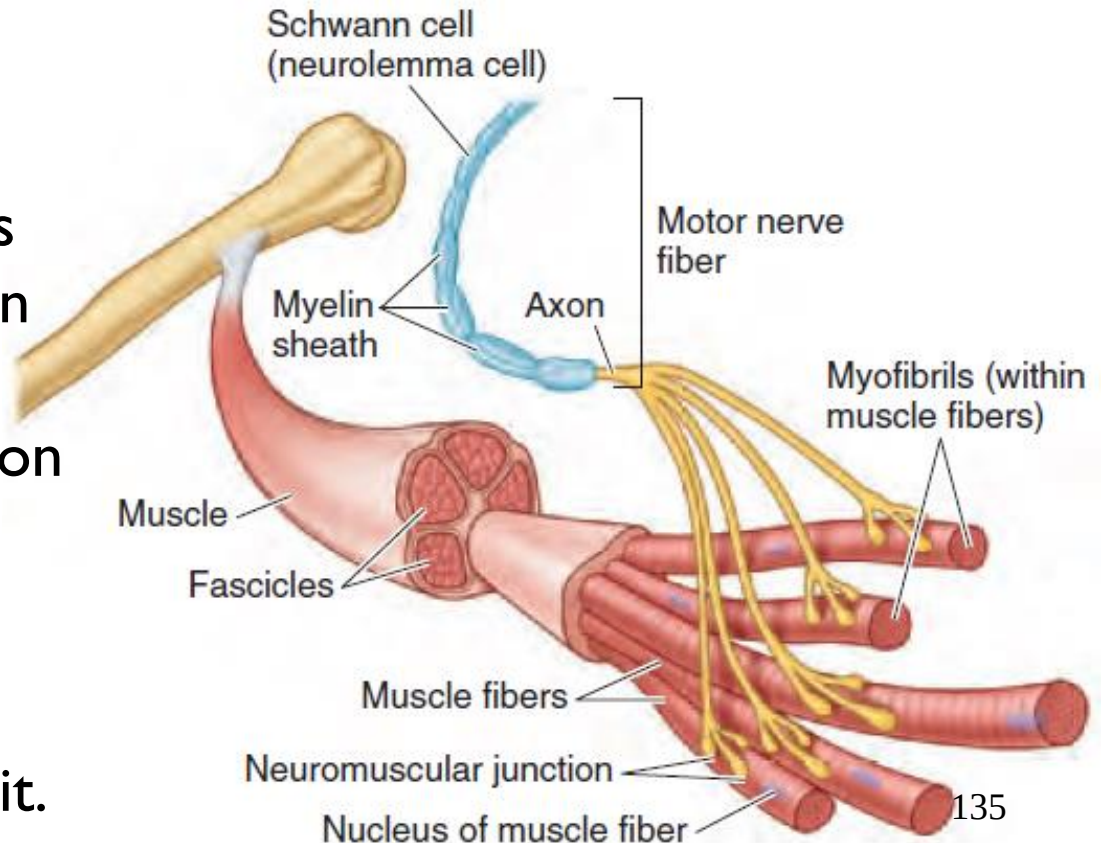


Fig. Motor unit.

Contraction of Muscles

❖ There are 3 types of contractions

- **Reflexive Contraction**

- **Tonic Contraction**

- **Phasic Contraction**

1. **Isotonic contractions**

- ✓ Centric contraction

- ✓ Eccentric contraction

2. **Isometric contractions**

Contraction of Muscles

Reflexive Contraction

- automatic and not voluntarily controlled

E.g. respiratory movements of the diaphragm & the myotatic reflex (e.g. knee jerk reflex)

Tonic Contraction

- slight contraction even in relaxed state, is called muscle tone
- does not produce movement or active resistance (as phasic contraction does)
- gives the muscle a certain firmness, assisting the stability of joints & the maintenance of posture
- Muscle tone is usually absent only when unconscious (as during deep sleep or under general anesthesia) or after nerve lesion result in paralysis

Contraction of Muscles

Phasic Contraction

- There are two main types of phasic (active) muscle contractions
 1. Isotonic contractions
 2. Isometric contractions

Contraction of Muscles

I. Isotonic contractions → the muscle changes length in relationship to the production of movement

There are 2 types of isotonic contractions

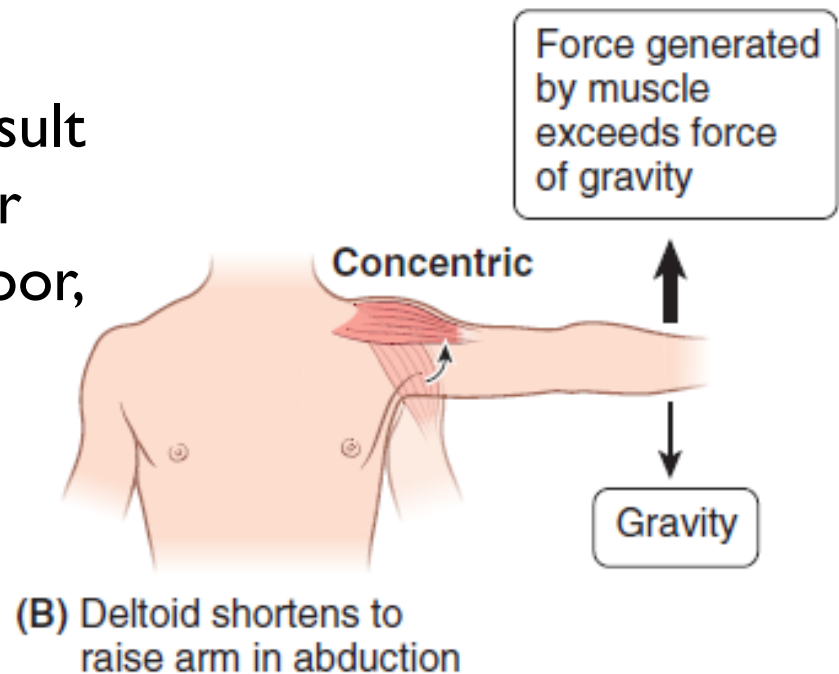
- ✓ **Concentric contraction**
- ✓ **Eccentric contraction**

Contraction of Muscles

Types of isotonic contractions

- **Concentric contraction**

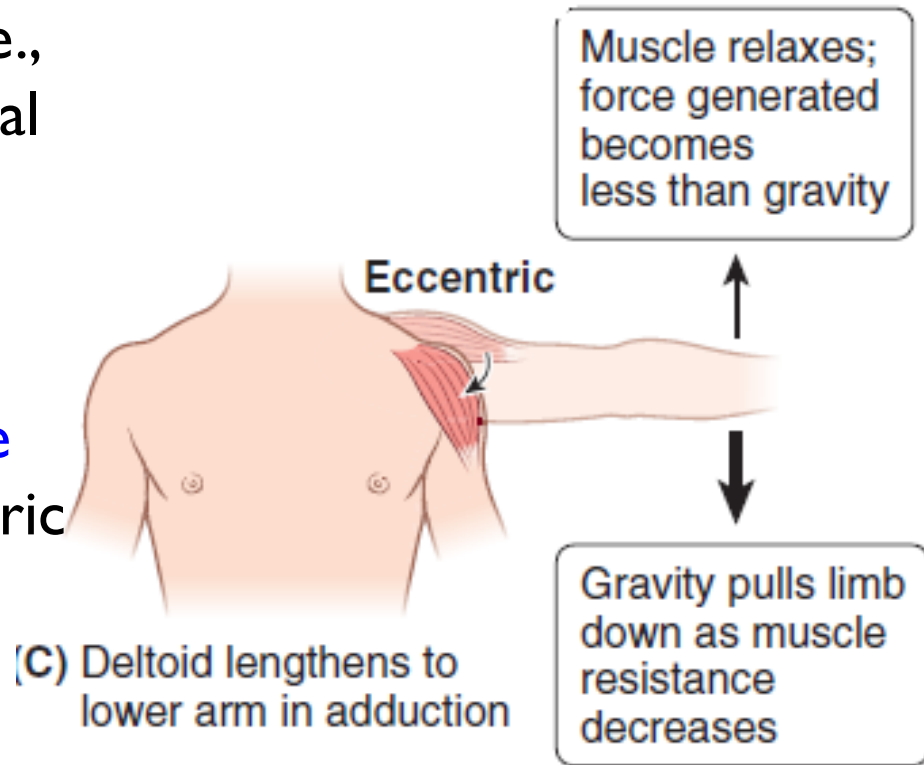
→ movement occurs as a result of the muscle shortening, for e.g. lifting a cup, pushing a door, striking a blow



Contraction of Muscles

Types of isotonic contractions...

- **Eccentric contraction** → a contracting muscle lengthens, i.e., undergoes a controlled & gradual relaxation while continually exerting a (diminishing) force.
- ✓ when the main muscle of a particular movement (the **prime mover**) is undergoing a concentric contraction, its antagonist is undergoing a coordinated eccentric contraction
- ✓ Eccentric contractions require less energy



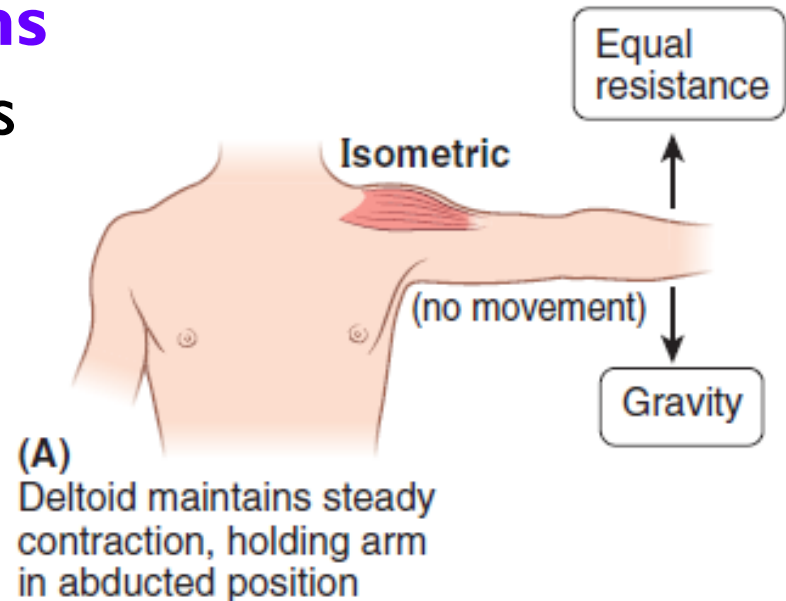
Contraction of Muscles

Phasic Contraction...

2. Isometric contractions

→ muscle length remains the same, no movement occurs, but the force (muscle tension) is increased above tonic levels

- ✓ ***important in maintaining upright posture & when muscles act as fixators or shunt muscles***

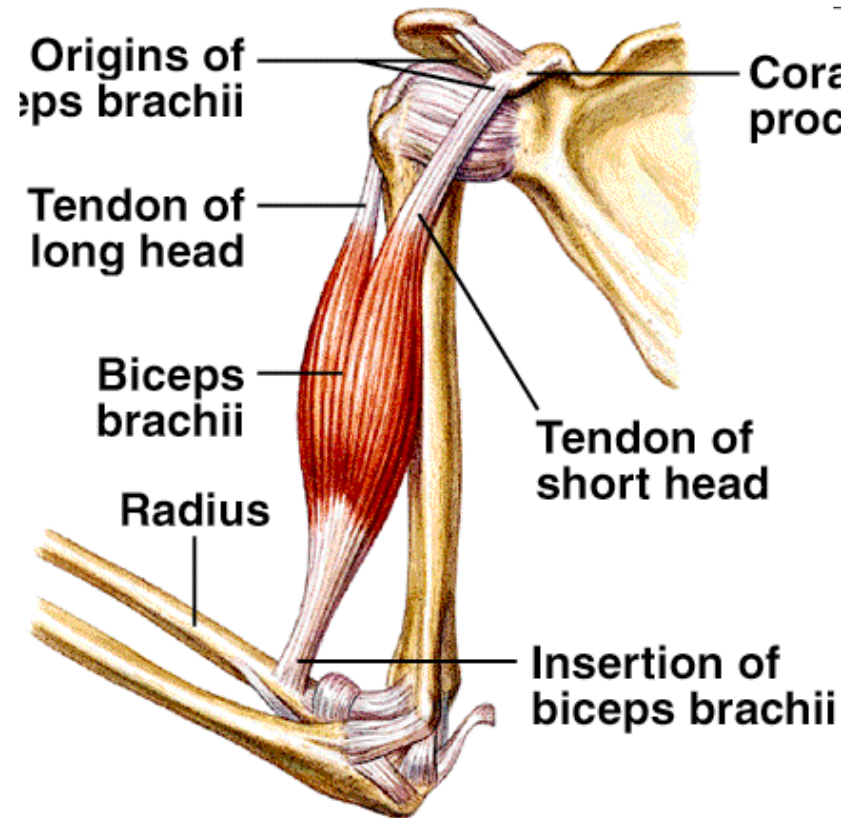


Functions of Muscles

- All movements are the result of the coordinated action of many muscles

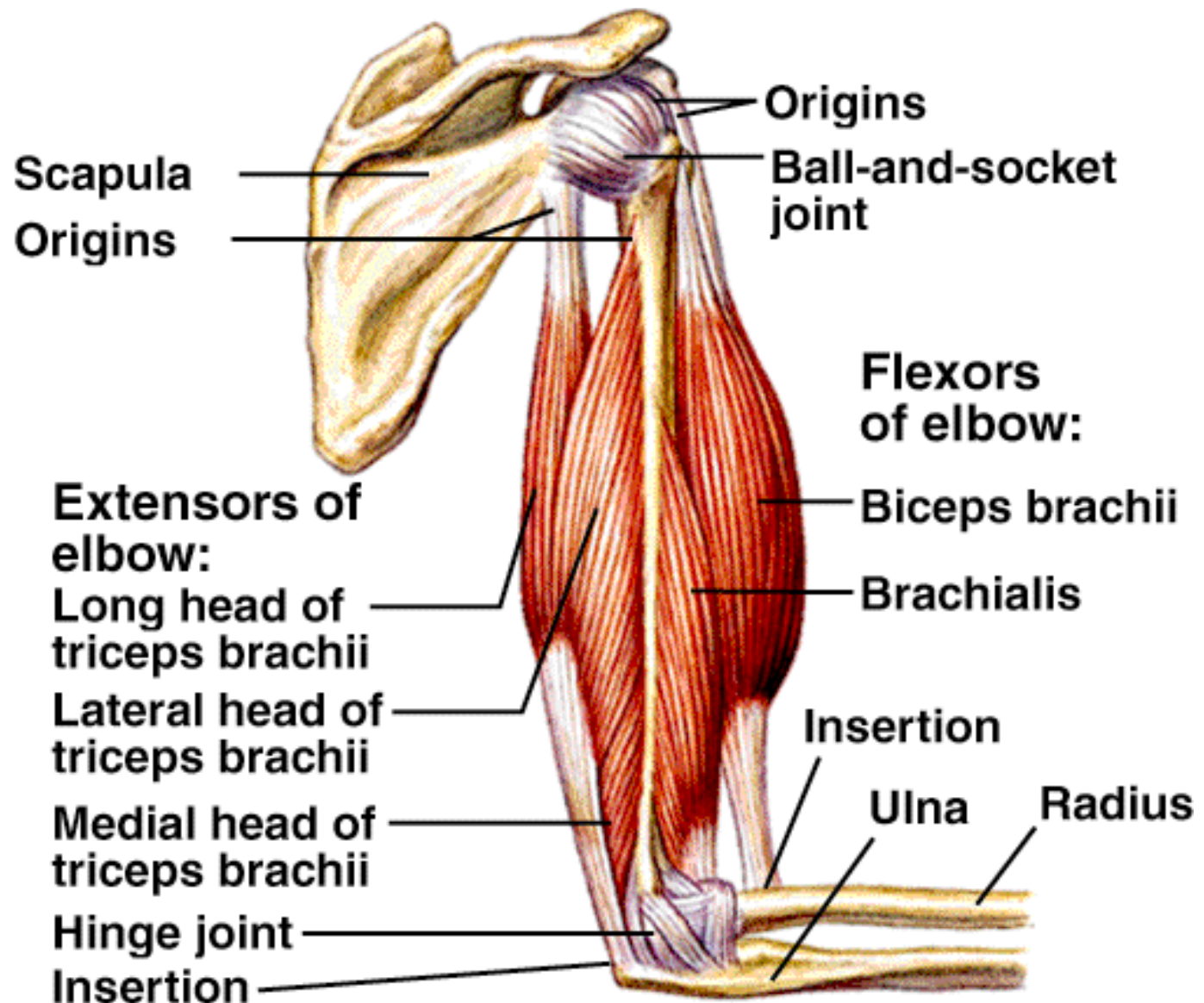
A muscle may work in the following four ways:

1. **Prime mover:** A muscle is a prime mover when it is the chief muscle or member of a chief group of muscles responsible for a particular movement
- *For example; Biceps brachii is prime mover of elbow flexion*



Functions of Muscles

- 2. Antagonist:** Muscles that oppose the action of another muscle
- ✓ A primary antagonist directly opposes the prime mover, but synergists may also be opposed by secondary antagonists.
 - ✓ As the active movers concentrically contract to produce a movement, antagonists eccentrically contract, relaxing progressively in coordination to produce a smooth movement.



Functions of Muscles

- 3. Fixator:** A fixator contracts isometrically (i.e., contraction increases the tone but does not in itself produce movement) to stabilize the origin of the prime mover so that it can act efficiently.
- ✓ (i.e. movements are occur in distal parts by prime mover)

Functions of Muscles

4. Synergist: complements the action of a prime mover

- ✓ It may directly assist a prime mover, providing a weaker component of the same movement, or indirectly, by serving as a fixator of an intervening joint when prime mover passes over more than one joint

Thank you